The Value of Lucky Hires: Match Quality Innovations and Firm Dynamics

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Abstract

Match productivity plays a central role in search-and-matching models of job creation, job destruction, and worker turnover. Yet match quality, the component of productivity specific to a worker-firm pair, remains difficult to measure empirically. This paper develops a new measure of match quality from a matched worker-employer panel built from LinkedIn profiles compiled by Revelio Labs (2007–2025). A hidden Markov model classifies employment spells into latent match states based on spell duration, transition outcomes, and the worker's employment history. For each spell, the match innovation measures the deviation of the realized match-state distribution from its predicted distribution based on information available at the time of match formation. These innovations provide an empirical analogue of unexpected match quality and are aggregated to the firm level to examine their relationship with subsequent firm outcomes. Among Compustat firms, conditional on firm controls, a 10-percentage-point shift in match innovations from "churning" matches prone to separation toward "internal growth" tracks predicts 3.9 percentage points higher employment growth, a 1.5-percentage-point higher probability of survival, and 4.8 percent higher sales at the firm level. These magnitudes are comparable to prior estimates of the effects of a one-standard-deviation increase in TFP across U.S. manufacturing establishments, suggesting that match quality is an important component of the firm-level heterogeneity underlying employment growth and survival.

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1 Introduction

Matching workers to firms is a noisy process. A central challenge is finding the right worker for the right firm, as the productivity of an employment relationship depends not only on the characteristics of the worker and the firm, but also on how well they fit together. This match-specific component of productivity, commonly referred to as match quality, is an important element in theories of labor market dynamics, shaping the formation and continuation of employment relationships (Mortensen and Pissarides, 1994; Jovanovic, 1979). Yet despite its theoretical importance, much less is known about the empirical relationship between match quality and firm performance. Measuring match quality is empirically challenging: It is inherently unobserved, match-level output is rarely recorded, and any candidate proxy is confounded by unobserved worker and firm heterogeneity. In this paper, I construct a residual-based measure of match quality at the job-spell level using a matched employer-employee panel built from LinkedIn profiles and standardized by Revelio Labs. Aggregating this measure to the firm level, I show that firms with more favorable match quality experience faster employment and sales growth and are more likely to survive.

I use the term match quality to refer to the match-specific component of a worker-firm pair that shapes how long the spell lasts and how it ends, including, for example, promotion, a move to another employer, or a transition to nonemployment. This component may evolve over time and does not transfer with the worker to a new employer. It is also distinct from firm-level technical efficiency, which raises productivity regardless of whom the firm hires. Conceptually, match quality captures forces such as complementarities between a worker and the incumbent workforce; a worker may be especially productive on a team whose members she works well with. It may also reflect labor-augmenting technology specific to the pair. For example, a salesperson may develop expertise in a particular firm's products that makes her especially effective at selling them.

Existing papers typically proxy for match quality with job duration or wages, on the premise that better matches last longer and pay more (Bowlus, 1995; Mühlbauer and Weber, 2024).[1] While these proxies are informative about underlying match quality, they have two limitations. First, no single match outcome is likely to capture all dimensions of match quality. Second, and more importantly, realized match outcomes also reflect unobserved worker and firm heterogeneity, making it important to distinguish these ef-

[1] An exception is Jackson (2013), who uses student achievement to estimate the productivity of specific teacher-school pairings separately from persistent teacher and school effects. Defining match quality as the time-invariant productivity of a given pairing, he shows that a sizable share of what would otherwise be attributed to teacher quality is in fact school-specific.

fects from the match-specific component. Using individual LinkedIn profiles compiled by Revelio Labs, I construct a new measure of match quality that addresses both limitations. The measure combines multiple dimensions of realized match outcomes and nets out the component of these outcomes that was predictable before the match formed.

Data from resume profiles are particularly well suited for measuring match quality because they provide detailed information on workers' employment histories at the position level. I observe not only transitions across firms, but also the positions they hold within each firm, including job titles, seniority rankings, and the timing of each transition. This allows me to distinguish promotions, lateral moves, and demotions within firms, as well as changes in job rank associated with moves across employers. These detailed career transitions provide multiple signals about how a worker-firm match evolves and ultimately ends, capturing dimensions of the employment relationship that traditional employer-employee records alone cannot provide.

I construct a residual-based measure of match innovation that captures the "surprise" in match outcomes in three steps. Throughout, I use *match quality* to refer to the structural, match-specific productivity of a worker-firm pair, while *match state* refers to the latent classification of realized match outcomes, which may also reflect worker- and firm-specific factors. *Match innovation* refers to the residual component of the match-state distribution after removing the part that is predictable before the match is formed. To the extent that match innovations reflect match quality rather than residual worker or firm factors, their conditional relationship with firm outcomes is informative about the link between match quality and firm performance.

In the first step, I classify each spell outcome into one of seven latent match states using a multichannel Hidden Markov model. I choose the number of states based on the transition dynamics and give them economic meaning by examining the joint distribution of spell duration and transition type, which includes internal promotions, demotions, lateral transfers, movements to higher- or lower-ranking positions at other firms, relocation, transitions into education or nonemployment, and ongoing employment. Each state is characterized by its probabilities of generating these observable outcomes and by its transition probabilities, which describe how workers move between latent states across successive employment spells. Because the model does not use worker demographics, firm characteristics, or wages to define the states, it provides a fixed classification based only on the observed evolution of employment relationships.

The seven states range from short, separation-prone positions to stable jobs with long durations and opportunities for internal advancement. A definitive ranking of the seven states is not always meaningful, since their relative desirability may depend on the

outcome of interest and on the stage of a worker's career. However, separation-prone states are generally viewed as less favorable outcomes than states associated with internal growth and stability. I assess the economic content of this classification using wages, which are excluded from the estimation. Entry wages generally rise as states move from temporary, separation-prone positions toward more stable positions with opportunities for internal advancement.

In the second step, I compute, for each position, a posterior probability of belonging to each of the seven states and examine which worker and firm observables systematically correlate with the classification. Rather than assigning a position to a single state, I retain the full posterior distribution to account for classification uncertainty. I then estimate spell-level regressions relating these posterior probabilities to observable worker and firm characteristics, including workers' age, gender, ethnicity, and educational attainment, as well as firms' size, age, and sector. Observable characteristics predict the classification in intuitive ways. For example, younger workers are more likely to hold shorter-duration, separation-prone positions, and larger and older firms are more likely to offer positions associated with stability and internal growth. These patterns indicate that characteristics observed prior to match formation systematically predict realized match outcomes, motivating the construction of match innovations that net out this predictable component.

In the third step, I construct the match innovation by comparing a position's realized posterior distribution across match states with the distribution predicted using all information available before the current match is formed. I first re-estimate the initial-state and transition equations using a richer set of predetermined worker characteristics, including the worker's employment history and demographic characteristics. These equations generate a predicted distribution across match states for each spell. The residual between the realized and predicted distributions is my measure of match innovation, capturing the component of the match realization left unexplained by systematic differences in expected outcomes across workers.

To assess the quantitative importance of match innovations for firm outcomes, I aggregate spell-level innovations to the firm level and examine how firms' exposure to different match innovations relates to subsequent employment growth and survival, conditional on firm fixed effects, lagged firm outcomes, and industry-by-time fixed effects. I find that greater exposure to Churning matches, spells lasting less than 15 months that typically end in nonemployment, relocation, or a job-to-job move to a lower-ranking position at another firm, is associated with the largest decline in employment growth. By contrast, innovations toward internal-growth states, which represent jobs more likely to end in lateral transfers or promotions, are associated with the largest increases in employ-

ment growth. A 10 percentage point shift in a firm's new-hire composition from Churning to Maturing Growth matches is associated with a 2.85 percentage point increase in subsequent employment growth and a 0.25 percentage point increase in survival probability. Within the Compustat sample, a 10 percentage point shift is associated with a 3.3–3.9 percentage point increase in employment growth, a 1.5 percentage point increase in survival probability, and 2.4-4.8 percent higher sales. To put this magnitude in context, Churning innovations fall by more than 10 percentage points while Maturing Growth innovations rise by more than 10 percentage points in 3.8% of consecutive-year firm observations in the full sample and 1.2% among Compustat firms. The magnitude of the employment-growth association is comparable to prior estimates of the effect of a one-standard-deviation increase in firm-level TFP among U.S. manufacturing establishments ((Blackwood et al., 2021)).

Importantly, the relationship between match innovations and firm outcomes is not confined to the year immediately following hiring. Tracing employment growth, survival, and sales over the subsequent three years, I find that for many match states and outcomes the associations remain economically meaningful two to three years later and, in some cases, grow stronger over time. For example, shifting 10 percentage points of new hires from Churning to Early Growth matches is associated with 4.7 percent higher sales in the following year and 9.1 percent higher sales three years later. This persistence indicates that the relationship reflects more than short-run turnover or post-hire adjustment. In addition, shifts toward growth-oriented matches predict both higher total factor productivity and faster productivity growth, beyond what is captured by firm size, age, and the predicted composition of firms' matches. Taken together, the magnitude and persistence of these patterns suggest that match quality is an important source of firm-level productivity heterogeneity.

1.1 Literature

This paper contributes to the firm dynamics literature on the determinants of employment growth and survival. A large empirical literature shows that firm-level fundamentals are central to firms' growth and survival (Syverson, 2011; Sterk et al., 2021; Decker et al., 2020). For example, (Blackwood et al., 2021) show that, across alternative measures of firm productivity, a one-standard-deviation increase in TFP is associated with roughly a 4.3–4.7 log-point increase in employment growth and a 1.4 percentage-point decline in the probability of exit. On the theory side, models of persistent firm heterogeneity and time-varying firm-specific shocks use these forces to generate labor adjustment as well

as firms' entry and exit decisions (Hopenhayn and Rogerson, 1993; Elsby and Michaels, 2013; Cooper et al., 2007). Yet the underlying sources of the time-varying firm-level heterogeneity that drives these dynamics remain less well understood. I show that firms' exposure to unexpectedly favorable worker-firm matches is one such source: firms with greater exposure to favorable match quality subsequently experience higher employment growth and are more likely to survive.

This paper also contributes to the empirical literature on measuring worker-firm match quality. Because match quality is not directly observed, existing work typically relies on proxies such as wages and job tenure (Bowlus, 1995; van Ours and Vodopivec, 2008; Caliendo et al., 2013; Mühlbauer and Weber, 2024; Koenraadt and Timmermans, 2026), job satisfaction and employer performance evaluations (Ferreira and Taylor, 2011), or measures of skill fit (Cabral Vieira, 2005; Mavromaras et al., 2007). However, as Belot et al. (2024) emphasize, these measures may reflect factors other than match quality and do not always correlate closely with one another, suggesting that they capture different underlying objects of the employment relationship. Rather than selecting a single observed outcome as a proxy for match quality, I infer a latent match state jointly from multiple outcomes over the worker's employment history. Spell duration and the way a spell ends provide complementary signals about the underlying state of the match, allowing otherwise similar employment spells to be distinguished by their transition dynamics.

My empirical approach is related to a growing literature that uses hidden Markov and latent-state models to recover persistent unobserved labor market heterogeneity from workers' employment histories. Shibata (2019) estimates a hidden Markov model in which latent labor market states summarize workers' prior histories and predict subsequent transitions. Ahn et al. (2023) similarly use a hidden Markov model to uncover segments of the U.S. labor market that differ in employment stability, unemployment dynamics, and labor force attachment. More recently, Hong et al. (2026) allow workers' latent wage types to evolve according to a hidden Markov process that depends on the type of firm employing them. I focus on a different latent object: match states of worker-firm pairs at the spell level, rather than latent worker states. Methodologically, I develop a three-step procedure that uses the model's transition structure to separate the unexpected component of a realized match state from the predictable part using information available before the match is formed.

2 Data

The primary data sources are the Individual Profile Data and Company References from Revelio Labs. Revelio aggregates and standardizes information from LinkedIn professional profiles on a weekly basis, and has become increasingly used in academic research in recent years ((Bloom et al., 2026; Chang et al., 2026; Gulati et al., 2026; Hampole et al., 2025). The resume data provide detailed information on workers' current and historical employment positions, educational backgrounds, skills, and demographic characteristics. The data enable us to directly observe firm-to-firm transitions, internal moves and promotions within firms, as well as the timing of each position change. Employers are assigned persistent identifiers, which allow employment positions to be linked across workers and over time and enable us to construct longitudinal worker–firm employment histories as well as firm-level workforce panels.

Definition of firms. Revelio provides an employer identifier, *ultimate_parent_rcid*, which links related employer entities to their ultimate parent company. This definition more closely corresponds to the enterprise-level concept of a firm in the firm-dynamics literature and may combine multiple subsidiaries, operating units, or establishments located in different geographic areas under common ownership. I aggregate employment histories across all constituent *rcid*s belonging to the same ultimate parent when constructing firm-level measures of employment, hiring, separations, growth, and survival.

Constructing worker–firm–position spells. The unit of analysis is the *worker–firm employment spell*, defined as a contiguous period during which a worker remains employed at the same firm without changing positions. Each reported employment spell is expanded into monthly observations, with a single primary position retained for each worker-month. When a worker holds multiple positions simultaneously, priority is given to a position with the same ultimate-parent employer (*ultimate_parent_rcid*) as in the preceding month to preserve employment continuity. If no such position exists, the overlapping position with the highest reported pay is retained. Finally, months that overlap with an education spell are classified as internships rather than regular employment.

Estimation Sample. The data contain a snapshot of LinkedIn profiles as of April 2026. The estimation sample spans 2007–2025, a period over which Revelio provides broad coverage of the U.S. labor market. The following restrictions are applied to construct the

estimation sample. First, I exclude positions that ended before January 2007, as Revelio's coverage of historical employment records improves substantially thereafter. Second, I retain workers with at least one year of education history after age 16, where age is inferred from the reported education records.[2] Third, I keep workers with at least three employment spells, where a new spell may reflect either a change in position within the same firm or a job-to-job move. Requiring at least three spells provides sufficient within-worker employment dynamics for estimation. For computational purposes, I cap the number of employment spells per worker at 20; more than 99.5% of individuals have 20 or fewer spells in the estimation sample. After imposing these restrictions, my baseline sample contains 22,874,918 workers, 3,896,164 unique firms, and 113,560,440 worker–firm matches.

Revelio provides broad coverage of the professional labor market. Appendix Tables 10–13 report the number and share of unique workers and employers by sector at an annual frequency and compare them with the corresponding statistics from the Quarterly Census of Employment and Wages (QCEW). Coverage of individual employment positions is broad across sectors and relatively stable over time, while gradually improving over the sample period as online professional networking becomes more widespread. Because the data are constructed from online professional profiles, representation is stronger among more-educated workers and those in professional occupations, such as managers and other professional workers, while lower-wage, less-educated, and informal workers are less well represented. Across industries, information and professional services, financial services, and education are relatively overrepresented. Larger and more established firms are also better represented than younger and smaller firms, as documented in Chang et al. (2026).

Comparisons with independent data sources suggest that Revelio-based employment measures are informative about both aggregate and firm-level labor market dynamics. Hampole et al. (2025), for example, compare firm employment measured in Revelio with employment reported in Compustat and find correlations of 0.76 in employment levels and 0.56 in five-year employment growth. Appendix Figure 11 provides an additional validation at the aggregate level by comparing hiring and separation rates constructed from Revelio with the corresponding series from the Job Openings and Labor Turnover Survey (JOLTS). The employment histories are also self-reported and may be updated retrospectively, so some short employment spells or periods of nonemployment may be

[2] I exclude all workers without education records since education is used both to infer workers' ages and to distinguish employment gaps associated with education from periods of nonemployment. In addition, I exclude workers with a single education spell lasting more than 10 years, as such records are more likely to reflect profiles that have not been updated.

less completely recorded.

2.1 Measurement of spell transition types

A unique feature of the dataset is that it allows worker movements up and down the job ladder to be measured during both job-to-job transitions and internal transfers. A move to a higher-seniority position within a worker's employment history is interpreted as an improvement in the worker–firm match. These career movements therefore provide information about whether the match-specific component of the worker–firm relationship improves or deteriorates over time. The analysis first introduces the concept of a seniority ladder, which is central to the classification of career progression, and then describes how movements along this ladder are measured.

Seniority ladder. I use Revelio Labs' seniority variable to infer whether an individual is advancing in their career when they experience a position change. Revelio Labs constructs an ordinal seniority metric for each position using three inputs: (1) job title, employer, and industry; (2) the individual's employment history, including prior positions and their associated seniority levels; and (3) the individual's age. These three measures are averaged and benchmarked against the population distribution to produce a metric comprising seven levels, ranging from 1 (entry level) to 7 (senior executive).[3]

This seniority ladder, defined over a worker's relative hierarchical position across all positions, is conceptually distinct from the firm-level job ladder studied in the literature. The firm job ladder describes the tendency of workers to move toward more productive and higher-paying firms (Burdett and Mortensen, 1998; Moscarini and Postel-Vinay, 2012, 2016). Under the firm-ladder definition, all positions within the same employer occupy the same rung. However, workers at the same firm can hold positions at very different seniority levels depending on their roles and responsibilities, generating meaningful within-firm variation in match quality.

Classification of spell transition types. I classify the spell transition types into one of the nine categories described below:

[3]The seven seniority levels are: (1) Entry Level (e.g., Accounting Intern; Software Engineer Trainee; Paralegal); (2) Junior Level (e.g., Accounts Receivable Bookkeeper; Junior Software QA Engineer; Legal Adviser); (3) Associate Level (e.g., Senior Tax Accountant; Lead Electrical Engineer; Attorney); (4) Manager Level (e.g., Account Manager; Superintendent Engineer; Lead Lawyer); (5) Director Level (e.g., Chief of Accountants; VP of Network Engineering; Head of Legal); (6) Executive Level (e.g., Managing Director, Treasury; Director of Engineering, Backend Systems; Attorney, Partner); (7) Senior Executive Level (e.g., CFO; COO; CEO). For the full methodology, see https://www.data-dictionary.reveliolabs.com/methodology.html#seniority.

Within-firm transitions.

1. *Downgrade:* an internal move to a lower-seniority position within the same firm.
2. *Lateral transfer:* an internal move to a position at the same seniority level within the same firm.
3. *Upgrade:* an internal move to a higher-seniority position within the same firm.

Across-firm transitions.

4. *Pull separation:* a move to a different firm at a higher seniority level.
5. *Push separation:* a move to a different firm at the same or a lower seniority level.
6. *Relocation:* a move to a different firm that also involves a move to a different geographic area.[4] Relocations are classified separately from push and pull separations because they may reflect changes in preferences or other personal constraints that are less directly related to match quality.

Other outcomes.

7. *Transition to education:* the worker's next recorded activity is enrollment in an educational program. Note that this includes part-time employment overlapping with education.
8. *Transition to nonemployment:* the worker separates from the firm and does not record a subsequent position or educational enrollment within the observation window.[5]
9. *Ongoing:* the position has not ended as of the last data update.

Figure 1 shows the distribution of spell transition types across the estimation sample. Downgrades are relatively uncommon, accounting for 3.8% of spells, compared with 9.8% for upgrades and 8.5% for lateral transfers. Push separations are nearly twice as

[4]Geographic areas are defined using the BLS combined MSA classification.

[5]The spell-level share of transitions into nonemployment should not be interpreted as a layoff or discharge rate measured at a regular frequency. The former is defined over completed employment spells, with all observed job spells in the sample period serving as the denominator, whereas the latter is typically measured as the fraction of involuntary separations occurring within a given period. Transitions into nonemployment may be measured with some upward bias because workers may be less likely to report short-duration, part-time, or lower-ranking jobs on their profiles. However, such omitted transitions are also likely to reflect relatively undesirable post-separation outcomes and therefore still contain information consistent with low match quality at the previous employer. Grouping these cases with true transitions into nonemployment is thus consistent with the interpretation of this outcome as signaling a relatively poor prior match.

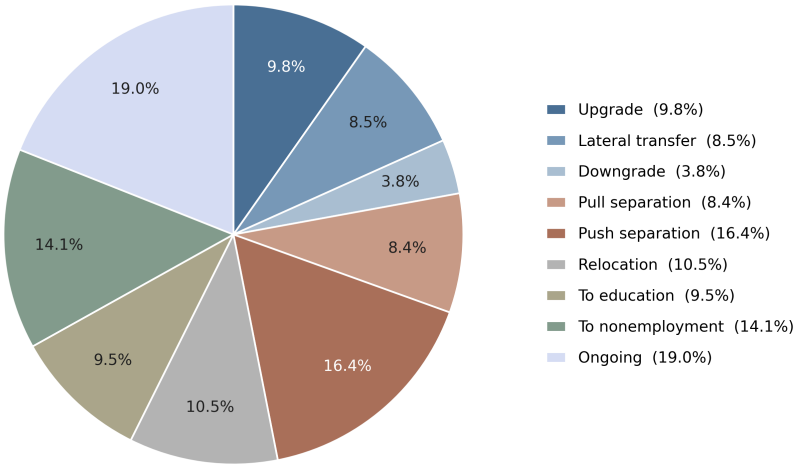


Figure 1: Distribution of Spell Transition Types

Note: The figure shows the distribution of spell transition types across the full eligible sample of workers with at least three spells. Shares are calculated over 113,560,440 spells with nonmissing exit types, including ongoing spells. Upgrade, lateral transfer, and downgrade denote transitions within the same employer.

common as pull separations, representing 16.4% and 8.4% of spells, respectively. Transitions to nonemployment account for 14.1%, while relocation and transitions to education represent 10.5% and 9.5%. The remaining 19.0% are ongoing spells.

Observed transitions to nonemployment may partly reflect users omitting jobs that could signal poor performance or low ability, potentially overstating the true frequency of transitions into nonemployment. I do not view this as a major concern for my estimates, insofar as these omitted transitions also reflect very low match quality with the previous employer, similar to genuine transitions into nonemployment.

Push separations may be more common than pull separations because workers accept lower-seniority positions in exchange for better long-run prospects, or because they disproportionately capture workers who cycle through short-lived matches while searching for a good fit. In either case, a push separation may reflect deterioration in the previous match or higher match quality with the new employer, despite lower observed seniority or wages.

Section 8.2 describes more details about the classification and 8.1 in the Appendix reports the descriptive statistics for the estimation sample.

3 Classification of Match Outcomes

In this section, I estimate a multichannel latent Markov model to recover a discrete latent match state for each position in a worker's employment history. The model captures the idea that unobserved latent match state is persistent but can evolve over a worker's employment history, and is only imperfectly observed through outcomes such as employment duration and exit type. This structure is analogous to a discrete state-space model, in which the latent-state transition process plays the role of the state equation and the emission process plays the role of the measurement equation. The latent state is allowed to vary within a worker–firm pair, as changes in positions within the same firm may signal changes in the underlying match state.

The primary purpose of this step is classification: selecting the number of latent states and characterizing the distinct types of worker–firm matches that they represent. I deliberately limit the covariates at this stage to construct a fixed definition of latent match states based only on employment outcomes and career dynamics, rather than worker's demographic characteristics.

3.1 A Multi-channel Hidden Markov Model

I model latent match outcomes at the position level as a first-order Markov process. The economic interpretation of this assumption is that, conditional on the current latent state and the gap between spells, earlier match states contain no additional information about the state of the subsequent match. Thus, the current state serves as a parsimonious summary of past employment history that are relevant for future match formation. The first-order Markov structure allows the latent match states to feature persistence and history dependence while keeping the worker's employment history tractable.

Figure 2 illustrates the structure of the model. Time is discrete. Let $n = 1, 2, \ldots, N$ index the worker's employment positions in chronological order, where the number of observed spells N_i may vary across workers. Let $U_n \in \{1, \ldots, K\}$ denote the latent match state associated with spell n. To simplify notation, I suppress the worker index i throughout this subsection unless it is needed for clarity.

Initial State Distribution. At the first observed position for each worker, the worker is assigned to latent match state k with probability $I(v_i) = P(U_{i,1} = k \mid v_i)$, where v_i denotes the birth cohort of individual i. I allow the initial distribution of latent match states to vary by birth cohort because the first observed positions of workers in earlier cohorts typically occur in mid-career, whereas those of workers in later cohorts are more

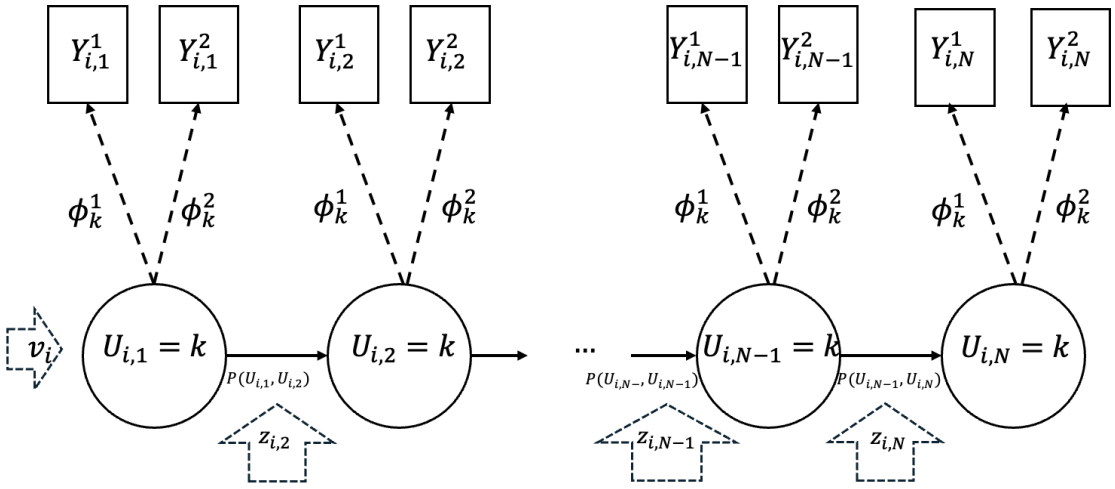


Figure 2: Latent Markov Model Structure

Note: This figure illustrates the structure of the model. $U_{i,n}$ denotes the latent match state for worker i in position n, while $Y^m_{i,n}$ denotes emission m observed for worker i in position n. The parameter ϕ^m_k denotes the emission probability for outcome m conditional on latent state k. The term $P(U_{i,n+1} \mid U_{i,n})$ denotes the transition probability between latent states across consecutive positions. The covariate v_i enters the initial-state distribution and corresponds to the worker's birth cohort, while $z_{i,n}$ enters the transition probabilities and captures the nonemployment gap preceding the new position.

likely to represent their first jobs.[6] I parameterize the initial-state probabilities using a multinomial logit model:

$$\log\left(\frac{I_{v_i}(k)}{I_{v_i}(k=4)}\right) = \alpha_k + \mathbf{v}_i'\boldsymbol{\beta}_k, \qquad k \neq 4, \tag{1}$$

where $I_{v_i}(k) = P(U_{i,1} = k \mid v_i)$ and $\mathbf{v}_i$ contains birth-cohort indicators, with the pre-1982 cohort as the omitted category. Transitional Mismatch ($k = 4$) is the reference latent state, so $\alpha_4 = 0$ and $\boldsymbol{\beta}_4 = \mathbf{0}$.

The implied initial-state probabilities are

$$I_{v_i}(k) = \frac{\exp\left(\alpha_k + \mathbf{v}_i'\boldsymbol{\beta}_k\right)}{\sum_{\ell=1}^{K} \exp\left(\alpha_\ell + \mathbf{v}_i'\boldsymbol{\beta}_\ell\right)}, \qquad k = 1, \ldots, K. \tag{2}$$

[6] $\mathbf{v}_i$ contains indicators for eleven cohorts: pre-1982, for whom the first job likely occurred before the start of the sample period; 1982–1984; and subsequent three-year cohorts through 2006–2008. An additional category is included for workers with missing birth-year information.

Transition Probabilities. After the initial spell, the latent match state evolves across successive employment spells. Define

$$Q_{z_{i,n}}(k,k') = P(U_{i,n} = k' \mid U_{i,n-1} = k, z_{i,n}), \qquad n = 2,\ldots,N_i, \tag{3}$$

where $z_{i,n}$ denotes worker i's nonemployment-gap category preceding spell n. $\mathbf{z}_{i,n}$ contains three indicators: a gap of one to six months, a gap of seven or more months associated with education, and a gap of seven or more months out of both work and school. No intervening gap is the omitted category.

For each originating state k, I specify a multinomial logit model with remaining in the same state as the reference outcome:

$$\log\left(\frac{Q_{z_{i,n}}(k,k')}{Q_{z_{i,n}}(k,k)}\right) = a_{kk'} + \mathbf{z}'_{i,n}\boldsymbol{\gamma}_{kk'}, \qquad k' \neq k. \tag{4}$$

The intercepts and gap coefficients vary across origin–destination pairs. Normalizing $a_{kk} = 0$ and $\boldsymbol{\gamma}_{kk} = \mathbf{0}$, the transition probabilities are

$$Q_{z_{i,n}}(k,k') = \frac{\exp\left(a_{kk'} + \mathbf{z}'_{i,n}\boldsymbol{\gamma}_{kk'}\right)}{\sum_{\ell=1}^{K}\exp\left(a_{k\ell} + \mathbf{z}'_{i,n}\boldsymbol{\gamma}_{k\ell}\right)}, \qquad k,k' = 1,\ldots,K. \tag{5}$$

Emission Probabilities. The emission model describes probability of observing outcome y^m in emission channel m when the underlying match is in latent state k. I consider two emission channels: first, spell duration Y^1, which takes four values: 1–6 months, 7–15 months, 16–36 months, 37 or more months; second, job exit type Y^2: takes nine values: downgrade, equal-level internal transfer, upgrade, transition to education, transition to nonemployment, pull separation, push separation, relocation, and ongoing employment. For each channel m, the emission probabilities are

$$\phi_k^m(c) = P(Y_{i,n}^m = c \mid U_{i,n} = k), \qquad c = 1,\ldots,C_m, \quad \sum_{c=1}^{C_m}\phi_k^m(c) = 1, \tag{6}$$

where $C_1 = 4$ for spell duration and $C_2 = 9$ for exit type. These probabilities vary across latent states but are constant across workers and spell positions.

The model assumes that the latent match states capture all systematic correlation between spell duration and exit type, as well as dependence in observed outcomes across positions. These states may therefore reflect not only match-specific characteristics but

also worker- and firm-specific characteristics that shape employment outcomes. For example, a demographic characteristic associated with more stable employment may be reflected in a worker's propensity to occupy particular latent states.

Conditional on the latent-state sequence $\mathbf{U}_i$, observed outcomes are assumed to be independent across positions, with their distribution at each position depending only on the current latent state $U_{i,n}$. Within each position, the two emission channels are further assumed to be independent conditional on that state.

Likelihood and Estimation. The model has four sets of parameters: $\Gamma = \{I_v, Q_z, \Phi^1, \Phi^2\}$. I_v is a $1 \times K$ vector giving the initial distribution over latent match states for birth cohort v, and Q_z is a $K \times K$ transition matrix associated with nonemployment gap category z. The matrices Φ^1 and Φ^2 contain the emission probabilities for spell duration and job exit type, respectively, with dimensions $4 \times K$ and $9 \times K$.

Given the model parameters Γ, the likelihood of observing individual i's employment history ($\mathbf{Y}_i$) is

$$\begin{aligned} P(\mathbf{Y}_i \mid v_i, \mathbf{z}_i; \Gamma) &= \sum_{\mathbf{k} \in \{1,\ldots,K\}^{N_i}} P(U_{i,1} = k_1 \mid v_i) \prod_{n=2}^{N_i} P(U_{i,n} = k_n \mid U_{i,n-1} = k_{n-1}, z_{i,n}) \\ &\quad \times \prod_{n=1}^{N_i} P(Y^1_{i,n} \mid U_{i,n} = k_n) P(Y^2_{i,n} \mid U_{i,n} = k_n) \qquad (7) \\ &= \sum_{\mathbf{k} \in \{1,\ldots,K\}^{N_i}} I_{v_i}(k_1) \prod_{n=2}^{N_i} Q_{z_{i,n}}(k_{n-1}, k_n) \prod_{n=1}^{N_i} \phi^1_{k_n}(Y^1_{i,n})\, \phi^2_{k_n}(Y^2_{i,n}). \end{aligned}$$

where the summations integrate over all (K^{N_i}) possible sequences of latent match states. The first term gives the probability of the initial latent state, the second captures transitions between latent states across successive positions, and the third gives the probability of the observed duration and exit outcomes conditional on each latent state. The likelihood of simultaneously observing labor history of all the individuals $\{Y\}_{i \in I}$ can be written as

$$\mathcal{L}(\Gamma) = \sum_{i=1}^{I} \log P(\mathbf{Y}_i \mid v_i, \mathbf{z}_i; \Gamma). \qquad (8)$$

The model is estimated by maximizing the sample log-likelihood using a random sample of one million workers. I use the `LMest` package in R, which implements maximum likelihood estimation through the expectation–maximization (EM) algorithm. The EM

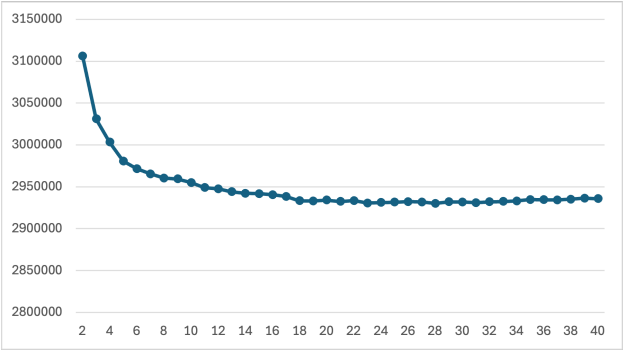


Figure 3: Bayesian Information Criterion

Notes: The figure reports the Bayesian Information Criterion (BIC) for latent Markov models estimated with $K \in [2, 40]$.

algorithm alternates between computing posterior probabilities of latent states and transitions given the observed employment histories (E-step) and updating the model parameters to maximize the expected complete-data log-likelihood (M-step). I use five independent random initializations and use the estimates with the highest log-likelihood. Appendix 9.0.1 provides further details on the estimation procedure.

3.2 Number of Latent States and Model Selection

I choose the number of latent states to balance statistical fit with the ability to distinguish stable and economically meaningful patterns in employment outcomes. Increasing the number of states mechanically allows the model to fit the data more closely, but additional states may capture small distinctions only available in the sample data and make the resulting classification more difficult to interpret.[7] I select $K = 7$ based on four considerations: information criteria, out-of-sample fit, the similarity of transition patterns across states, and the stability and economic interpretation of the estimated states.

- *Information criteria.* I estimate models with $K = 2, \ldots, 40$ latent states and compare the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Figure 3 reports the BIC, while Appendix Figure 14 reports the AIC. The BIC reaches its global minimum at $K = 28$, while the AIC continues to decline through $K = 40$. However, most of the improvement in fit occurs at relatively small values of K, and both criteria flatten substantially after $K = 5$. Relative to the BIC improvement

[7] Theorem 6 in Allman et al. (2009) provides a generic identifiability bound on the number of latent states as a function of the number of observable response combinations and repeated observations. In my setting, the implied upper bound exceeds four million latent states and is therefore non-binding for the values of K considered here.

attainable at its minimum, $K = 7$ already captures approximately 83% of the total reduction in BIC.

- *Out-of-sample fit.* I evaluate predictive performance on a separate sample of 100,000 workers. Table 20 reports the average log-likelihood per worker in the training and validation samples for selected values of K. Out-of-sample fit improves as the number of states increases, indicating that the additional states capture some genuine structure in the data rather than over-fitting to small features in the training sample. However, the improvement as K increases is very modest.

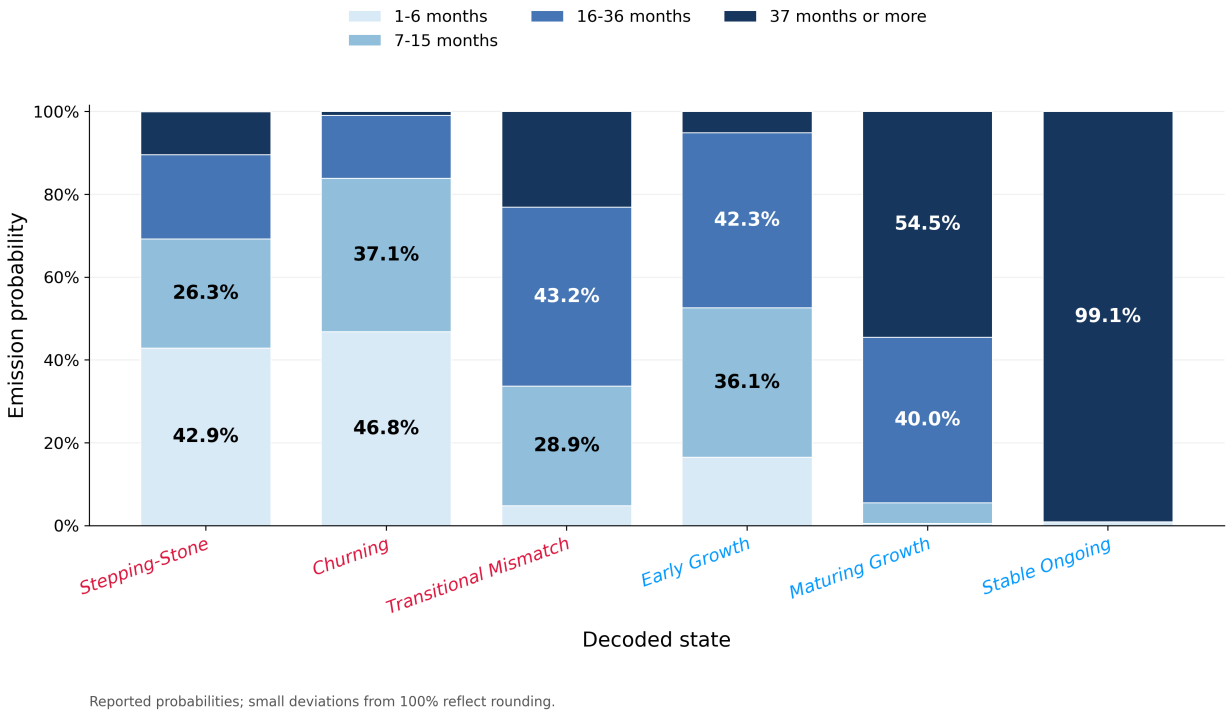
- *Separation of the transition dynamics.* With $K = 7$, the estimated transition matrices are full rank and well behaved numerically. As K increases, the transition matrices remain formally full rank but become increasingly close to singular, with several columns becoming nearly linearly dependent. This pattern indicates that some of the additional latent states have very similar transition dynamics and therefore provide limited additional distinction in workers' employment histories.

- *Parameter stability and economic interpretation.* Finally, I compare the estimated emission and transition probabilities across independently drawn samples. With larger values of K, the estimated parameters become increasingly sensitive to the sample used for estimation: states that appear in one sample do not always have a clear counterpart with similar emission and transition patterns in another. In contrast, the $K = 7$ specification produces stable parameter estimates across samples and yields a consistent set of latent states with distinct and economically interpretable employment-outcome patterns.

Taken together, larger models provide modest improvements in statistical fit, but these gains come primarily from splitting the seven broader states into more weakly differentiated clusters. The seven-state model captures most of the improvement in fit while producing transition dynamics and emission patterns that are stable across samples and readily interpretable economically.

3.3 Classification of latent match states

In this section, I characterize the seven latent match states that capture different stages of the employment trajectory, and describe their emission profiles and transition dynamics associated with each latent state



Notes: This figure reports the estimated emission probabilities for spell duration conditional on each latent match state. Spell duration is divided into four categories: 1–6 months, 7–15 months, 16–36 months, and 37 months or longer. Each bar represents the probability distribution of spell duration within a latent state.

Figure 4: Spell Duration Distribution by Latent Match State

3.3.1 Characterization of the Latent States

Table 1 summarizes the employment-duration, spell transition-type, and transition profiles of the seven latent match states. The states can be broadly classified into three groups. The first consists of three separation-prone states: *Stepping-Stone Job*, *Churning Job*, and *Transitional Mismatch*, characterized by frequent separations with the current employer. The second consists of three growth-oriented states: *Early Growth Track*, *Maturing Growth Match*, and *Stable-Ongoing Match*, characterized by internal advancement or stable employment relationships with the employer. The remaining state primarily captures the first observation for older cohorts whose earlier histories are unobserved in the sample. Figures 4 and 5 show the estimated emission probabilities for spell duration and exit type for the six economically meaningful states. The underlying distribution of these latent states, and their emission and transition probabilities are reported in Appendix 9.2.

Separation-Prone Matches. *Stepping-Stone Matches* consists primarily of short-duration employment relationships, with 43 percent lasting 1–6 months and another 26 percent lasting 7–15 months. Transitions to education and nonemployment together account for

Table 1: Summary of the Latent Match States

Labels	Employment-relationship profile	Most likely next states
Separation-Prone Matches		
Stepping-Stone	Short-duration, with transition concentrated in education (37%) and nonemployment (26%).	Stepping-Stone Job (45%); Churning Job (30%).
Churning	Short-duration, with transition concentrated in push separation (26%), nonemployment (22%), and relocation (20%).	Churning Job (35%); Stable-Ongoing Match (28%); Transitional Mismatch (16%).
Transitional Mismatch	Medium-duration and separation-dominated, with frequent push separation (33%), nonemployment (24%), and pull separation (19%).	Transitional Mismatch (45%); Stable-Ongoing Match (37%).
Growth-Oriented Matches		
Early Growth Track	Medium-duration, advancement-oriented relationships; same-level moves and upgrades account for 64% of exits.	Early Growth Track (55%); Stable-Ongoing Match (20%); Transitional Mismatch (10%).
Maturing Growth Match	Medium-long duration, advancement-oriented relationships; same-level moves and upgrades account for 50% of exits.	Stable-Ongoing Match (54%); Maturing Growth Match (28%); Early Growth Track (12%).
Stable-Ongoing Match	Long, ongoing relationships: 99% last at least 37 months and 99.5% remain Ongoing at the end of the observation window.	Stable-Ongoing Match (79%); Early Growth Track (20%).
Left-Censored Match	Long-tenure relationships, with exits spread across nonemployment (21%), push separations (20%), pull separations (12%), and relocation (11%).	Concentrated among first observed spells of older cohorts.

Notes: The table summarizes the emission and transition profiles of the seven latent states estimated by the multi-channel latent Markov model. The final column reports the most likely destination states for consecutive employment spells with no intervening nonemployment gap. Stable-ongoing primarily contains ongoing spells (over 99%), its duration and persistence partly reflect right-censoring. The Left-Censored Match state primarily captures first observed spells of older cohorts whose earlier employment histories are unobserved.

over 62% of spell transitions. This state is especially prevalent in the initial match-state distributions of later birth cohorts, whose employment histories are more likely to be observed from labor-market entry. Thus, stepping stone matches often represent temporary or early-career employment.

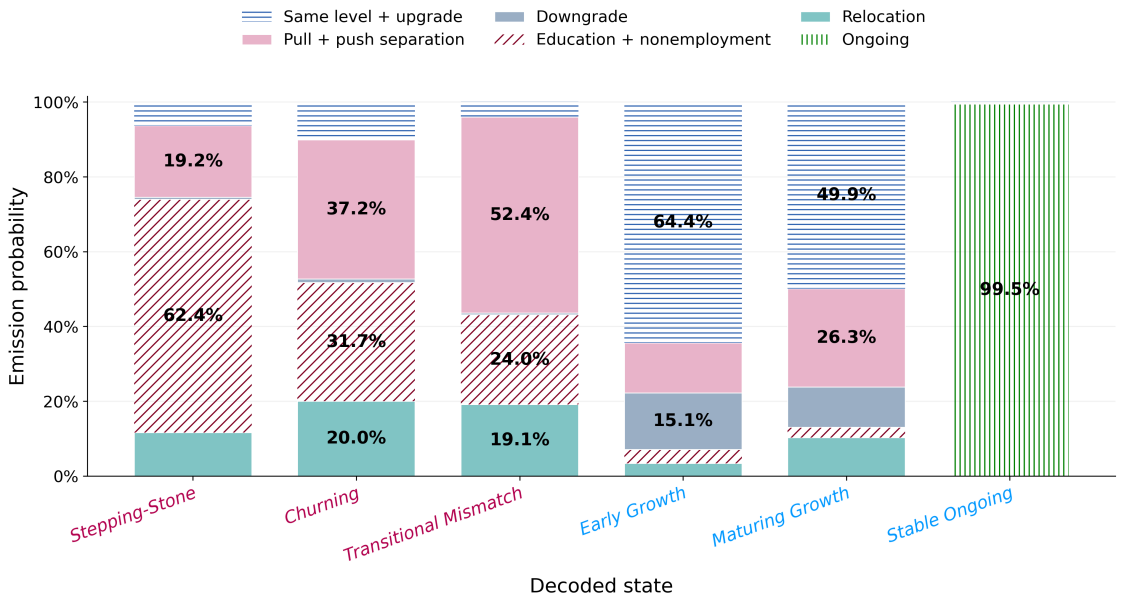
Churning Match also consists of short employment relationships, with 47 percent lasting 1–6 months and 37 percent lasting 7–15 months. However, unlike *Stepping-Stone Job*, exits from Churning matches are more concentrated in separations to other firms. These patterns suggest unstable employment relationships characterized by frequent separations rather than early-career transitions into education.

Transitional Mismatch is similarly separation-prone but has a longer duration profile. Approximately 43 percent of these relationships last 16–36 months, compared with only 17 percent for *Churning Job*. Separations to other firms account for an even larger share of exits, with push separations accounting for 33 percent, followed by transitions to nonemployment (24 percent) and pull separations (19 percent). Despite the prevalence of separations, the transition dynamics suggest that *Transitional Mismatch* is more likely than *Churning Match* and *Stepping-Stone Matches* to be followed by a *Stable-Ongoing Match*, with an estimated transition probability of 37 percent compared with 28 percent for *Churning Match* and 2 percent for *Stepping-Stone Mathces*.

Growth-Oriented Matches. *Early Growth Track* is characterized by employment relationships of medium duration and frequent internal advancement. Approximately 36 percent of relationships last 7–15 months and 42 percent last 16–36 months. Same-level internal transfers and upgrades together account for over 64 percent of spell transition outcomes, the highest share among all latent states. These relationships are also likely to be followed by another *Early Growth Track* or a *Stable-Ongoing Match*, suggesting continued employment progression.

Maturing Growth Match represents longer employment relationships with continued opportunities for internal advancement. Approximately 55 percent last at least 37 months, while another 40 percent last 16–36 months. Same-level transfers and upgrades account for approximately 50 percent of spell transition outcomes. Compared with *Early Growth Track*, *Maturing Growth Match* are more likely to be followed by *Stable-Ongoing Match*, with an estimated transition probability of 54 percent. These patterns are consistent with employment relationships becoming longer and advancement less frequent as workers progress into more mature positions.

Stable-Ongoing Match consists almost entirely of long-duration employment relationships that remain active at the end of the observation window. Approximately 99 percent



Reported probabilities; small deviations from 100% reflect rounding.

Notes: This figure reports the estimated emission probabilities for spell transition types conditional on each latent match state. Transition outcomes are further grouped into six categories: same-level transitions and upgrades within the firm, downgrades within the firm, pull and push separations, transitions to education or nonemployment, relocations, and ongoing employment spells. Each bar represents the probability distribution of exit outcomes within a latent state, with probabilities summing to one. Ongoing spells refer to positions that have not ended by the end of the sample period.

Figure 5: Spell Transition Distribution by Latent Match State

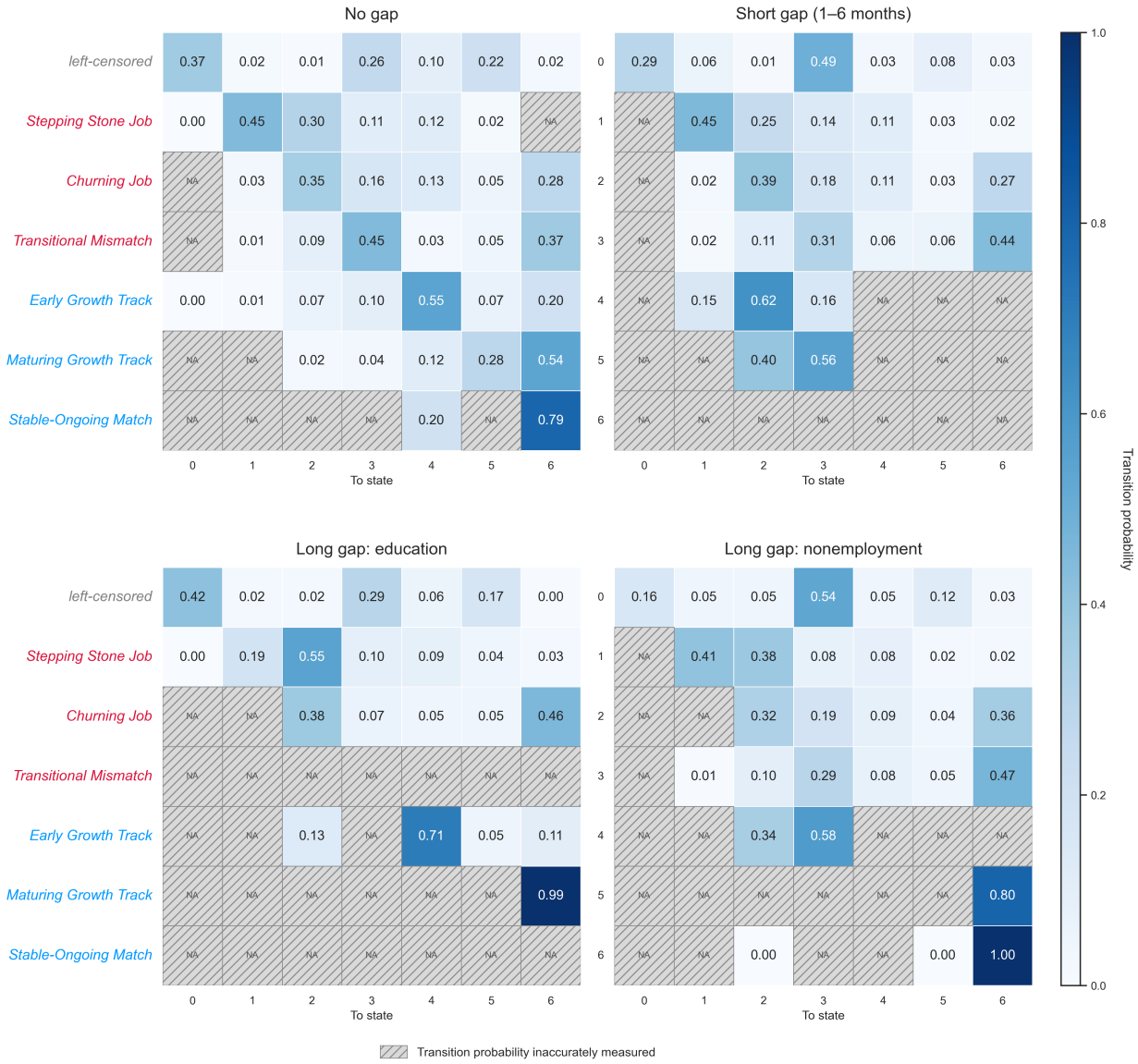
last at least 37 months, and 99.5 percent are still ongoing. This state is a common destination from several other states, particularly for *Maturing Growth Match*.

3.3.2 Transition Dynamics of the Seven States

Figure 6 reports the estimated transition matrices separately by the gap between consecutive employment spells. Rows denote the state of the previous spell, while columns denote the state of the subsequent spell. The first panel reports transitions with no intervening gap, the second reports transitions following a short gap of 1–6 months, and the final two panels report transitions following gaps of at least seven months associated with education and nonemployment, respectively. Approximately 65.7 percent of transitions involve no gap, 10.0 percent follow a short gap, 11.5 percent follow a long education-related gap, and 12.7 percent follow a long nonemployment gap. Shaded cells indicate transition probabilities that have large standard errors because the corresponding origin–destination transition have a small number of observations in the estimation sample. Imprecisely estimated transition probabilities are concentrated among transitions involving employment gaps, because some combinations of pre-employment gap categories and state-to-state transitions are observed only rarely in the data. I calculate standard errors using the delta method, accounting for the nonlinear mapping from the estimated multinomial-logit coefficients to transition probabilities. Appendix 9.4 describes the delta-method calculation and reports the standard errors for the transition matrices.

The no-gap transition matrix shows a clear concentration along the diagonal and in the upper-right portion of the matrix. The large diagonal entries indicate persistence of the latent states: workers frequently enter a subsequent employment relationship of the same latent type as their previous relationship. At the same time, transitions above the diagonal are generally more common than transitions below it, indicating that workers tend to move toward longer duration jobs with more internal advancement opportunities. The process is not monotonic, however. I also observe repeated states and movements toward moving down the career trajectory, suggesting that career progression often involves recurring or transitional steps rather than a one-directional movement up a deterministic ladder.

A short employment gap weakens the asymmetry between the upper-right and lower-left portions of the matrix. After such a gap, the two growth-oriented states transition mainly into shorter duraton, separation-prone states. *Early Growth Track* and *Maturing Growth Match* moves mainly to *Churning Match* and *Transitional Mismatch*. An employment interruption thus breaks the upward career trajectory seen in no-gap transitions. Direct transitions between growth-oriented states after a gap are often imprecisely esti-



Notes: Each panel reports the estimated probability of transitioning from the latent match state shown on the vertical axis to the state shown on the horizontal axis, conditional on the type of gap preceding the subsequent employment relationship. Shaded cells indicate imprecisely estimated transition probabilities for which the width of the 95 percent confidence interval exceeds 0.20.

Figure 6: Transition Probabilities by Pre-employment Gap

mated. Few workers in the data end a growth-oriented match with a nonemployment spell and then move immediately into another growth-oriented match, so these estimates rely on a small number of cases.

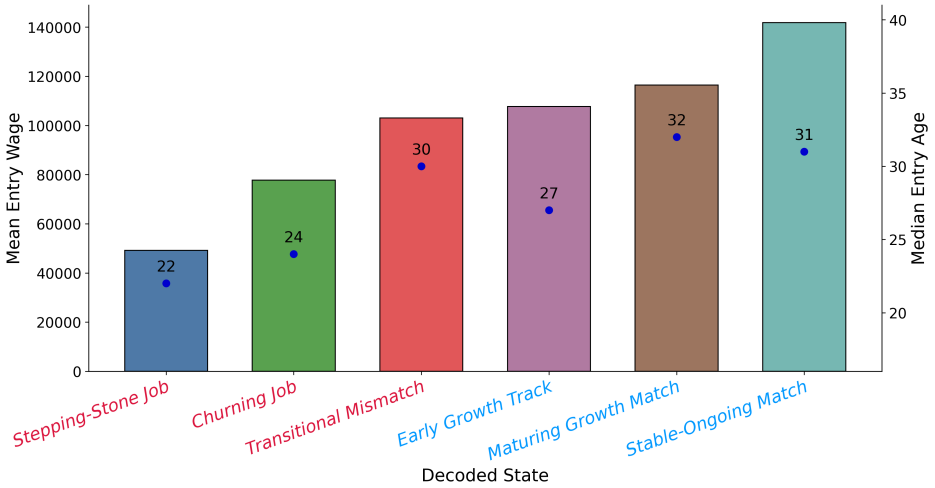
The transition matrix following a long education-related gap is estimated less precisely still, since this category is smaller and some origin-state–gap combinations are rare. Transitions following education are especially poorly measured for *Transitional Mismatch* and *Stable-ongoing Match*, because these two states are unlikely to be followed by education. Among the more precisely estimated cells, *Stable-ongoing Matches* remains a key destination. *Maturing Growth Mathches* moves to *Stable-ongoing Matches* with near-certain probability, consistent with workers using an education spell to move from a maturing growth relationship into a stable, ongoing one. *Early Growth Matches* is also highly persistent after education: workers return to it with probability 0.71, suggesting education reinforces an early growth trajectory. *Churning Matches* is another common destination after education, including from *Stepping-Stone Matches* — possibly reflecting a period of renewed search and experimentation as workers look for a suitable post-education match.

Following a long nonemployment gap, the transition distribution becomes more dispersed, and transitions from growth-oriented states are again frequently imprecisely estimated. Among the precisely estimated transitions, *Churning Jobs* and *Transitional Mismatch* emerge as common destinations, suggesting that workers returning to the labor market often pass through short-to-medium-duration, separation-prone matches before settling into more stable employment. Transitions from *Maturing Growth Matches* to *Stable-Ongoing Matches* are also prevalent, potentially reflecting recalls to previous employers.

3.3.3 Descriptive evidence on wages and age by states

Figure 7 compares the mean entry wages and median entry ages across states. The bars report mean wages and the markers report the median age of workers for that group. Note that neither wages nor worker age is used to define the latent states, so systematic differences along these dimensions provide an independent check that the estimated states capture economically meaningful differences across matches.

The patterns broadly align with a life-cycle interpretation of the latent match states. States associated with workers' initial labor-market experiences tend to have lower mean wages, whereas more mature matches are associated with higher wages. *Stepping-Stone Jobs*, *Churning Jobs*, and *Early Growth Matches* are disproportionately held by younger workers. This pattern is particularly pronounced for *Stepping-Stone Jobs*, which has a median age of 22. Its strong concentration among very young workers supports the interpretation of *Stepping-Stone Jobs* as an entry-stage employment state rather than one that is



Notes: This figure plots the mean entry wage and median starting age of workers in each latent match state. Bars show mean wages at the start of employment spells (left axis). Means are calculated across spells with nonmissing entry wages, excluding spells whose observed annualized wage-growth factor falls outside [0.5, 2]. Blue dots and numerical labels report median age at spell entry (right axis), calculated as entry year minus birth year. The age statistics cover spells beginning in 2007 or later, with entry ages between 15 and 100. The left-censored state is omitted. *Source: Revelio Labs and author's calculation.*

Figure 7: Wage and Age distribution by State

prevalent throughout workers' careers. In contrast, *Maturing Growth Matches*, *Transitional Mismatches*, and *Stable-Ongoing Positions* have median ages of 30 or above.

In the next section, I examine these relationships more systematically using spell-level multinomial regressions that relate latent-state membership to a broader set of worker and firm characteristics.

4 Characteristics of Latent Match States

In the second step, I recover the posterior distribution over latent states for each employment spell and provide descriptive evidence on how worker and firm characteristics are associated with the types of matches formed. This analysis helps identify predetermined factors that should be controlled for in the third step.

4.1 Posterior State Distribution

Taking the model parameters estimated in the first step as given, the second step recovers the posterior distribution over latent match states for each observed position. Rather than assigning each position to its most likely state, the full vector of posterior probabilities across the K latent states is retained. This soft classification preserves uncertainty in

latent-state assignments.

Two posterior distributions are considered, differing in the information used to infer the latent state. First, *local decoding* conditions on the worker's full observed employment history. For position n of worker i, the locally decoded posterior probability of latent state k is

$$\pi_{i,n,k}^{Local} \equiv P\left(U_{i,n} = k \mid \mathbf{Y}_{i,1:N_i}, \mathbf{z}_{i,1:N_i}, v_i; \widehat{\Gamma}\right), \tag{9}$$

where $\mathbf{Y}_{i,1:N_i} = (\mathbf{Y}_{i,1}, \ldots, \mathbf{Y}_{i,N_i})$ denotes the full sequence of observed employment outcomes for worker i, and $\mathbf{z}_{i,1:N_i} = (\mathbf{z}_{i,1}, \ldots, \mathbf{z}_{i,N_i})$ denotes the corresponding sequence of employment gaps.

Second, *filtered decoding* assigns the posterior probabilities conditioning only on information observed through the current position.

$$\pi_{i,n,k}^{Filtered} \equiv P\left(U_{i,n} = k \mid \mathbf{Y}_{i,1:n}, \mathbf{z}_{i,1:n}, v_i; \widehat{\Gamma}\right). \tag{10}$$

Unlike local decoding, filtered decoding does not use outcomes from positions occurring after n. It therefore represents the posterior distribution over the current latent state based only on the worker's employment history observed up to that point.

Both procedures produce a K-dimensional probability vector for each spell, $\widehat{\boldsymbol{\pi}}_{i,n}^{d} = \left(\hat{\pi}_{i,n,1}^{d}, \ldots, \hat{\pi}_{i,n,K}^{d}\right)$, where $\sum_{k=1}^{K} \hat{\pi}_{i,n,k}^{d} = 1$, where $d \in \{Local, Filtered\}$ denotes local or filtered decoding.

For the spell-level analysis in this section, I use the locally decoded posterior probabilities because the objective is to characterize the latent states as accurately as possible using the worker's complete observed employment history. In Section 5, where I construct match-state innovations, I instead use filtered posterior probabilities so that the inferred state at position n does not incorporate information from subsequent employment outcomes.

4.2 Worker and Firm Characteristics of Latent Match States

Next, I estimate a spell-level multinomial logit model to characterize how worker and firm attributes are associated with decoded match-state membership. I interpret the estimates as descriptive associations rather than causal effects of these characteristics on match quality.

Let $\widehat{U}_{i,n}$ denote the decoded latent match state for worker i in spell n, and let $\mathbf{X}_{i,n}$ denote the corresponding worker and firm characteristics. Taking Transitional Mismatch

(state 4) as the reference outcome, I specify

$$\log\left[\frac{P(\widehat{U}_{i,n}=k \mid \mathbf{X}_{i,n})}{P(\widehat{U}_{i,n}=4 \mid \mathbf{X}_{i,n})}\right] = \theta_0^k + \mathbf{X}'_{i,n}\boldsymbol{\theta}^k = \theta_0^k + \sum_{r=1}^{R} X_{i,n,r}\theta_r^k, \qquad k \neq 4, \tag{11}$$

where $\mathbf{X}_{i,n}$ includes indicators for worker age bins, highest educational attainment at the start of the match, worker ethnicity, firm industry, firm size bins, firm age bins, and year-quarter fixed effects.

For a covariate r, the coefficient θ_r^k gives the change in the log-odds of membership in state k relative to Transitional Mismatch. The corresponding odds ratio is $\exp(\theta_r^k)$. Tables 23–29 in Appendix 9.5 report the estimated coefficients and odds ratios.

Figures 8 and 9 report the estimated odds ratios from the multinomial logit regressions, organized by covariate and latent match state. I focus on four economically interpretable, non-censored states—Stepping-Stone Jobs (S1), Churning Jobs (S3), Early Growth Track (S5), and Maturing Growth Track (S6)—and omit the two states that are strongly affected by censoring. The baseline outcome category in the multinomial logit is Transitional Mismatch (S4). Appendix Tables 23–29 report the corresponding coefficient estimates and standard errors.

The first row of Figure 8 reports odds ratios by worker age, using workers between ages 30 and 35 as the reference group. Age is strongly associated with the type of match formed. Younger workers are substantially more likely to be observed in Stepping-Stone Jobs, Churning Jobs, and Early Growth Track positions relative to Transitional Mismatch. The age gradient is particularly pronounced for *Stepping-Stone Matches*: workers under age 20 have much higher odds of belonging to this state, and the odds decline sharply with age. This pattern is consistent with Stepping-Stone Jobs disproportionately capturing positions held early in workers' careers. *Churning Matches* display a similar, though less pronounced, age gradient. *Early Growth Track* positions are also more common among younger workers, while workers above age 35 are progressively less likely to be observed in this state. In contrast, the *Maturing Growth Track* is concentrated more heavily among prime-age workers: workers below age 30 and workers above age 45 are less likely to be observed in this state than workers ages 30–35.

The second row reports odds ratios by workers' highest degree at the time of hire, with a bachelor's degree as the reference category. Educational attainment is also strongly related to the distribution of workers across latent match states. Workers with only a high-school or associate degree are substantially less likely to be observed in the two growth-track states, particularly the *Maturing Growth Track*. By contrast, workers with graduate degrees are generally more likely to appear in several of the non-censored states. Mas-

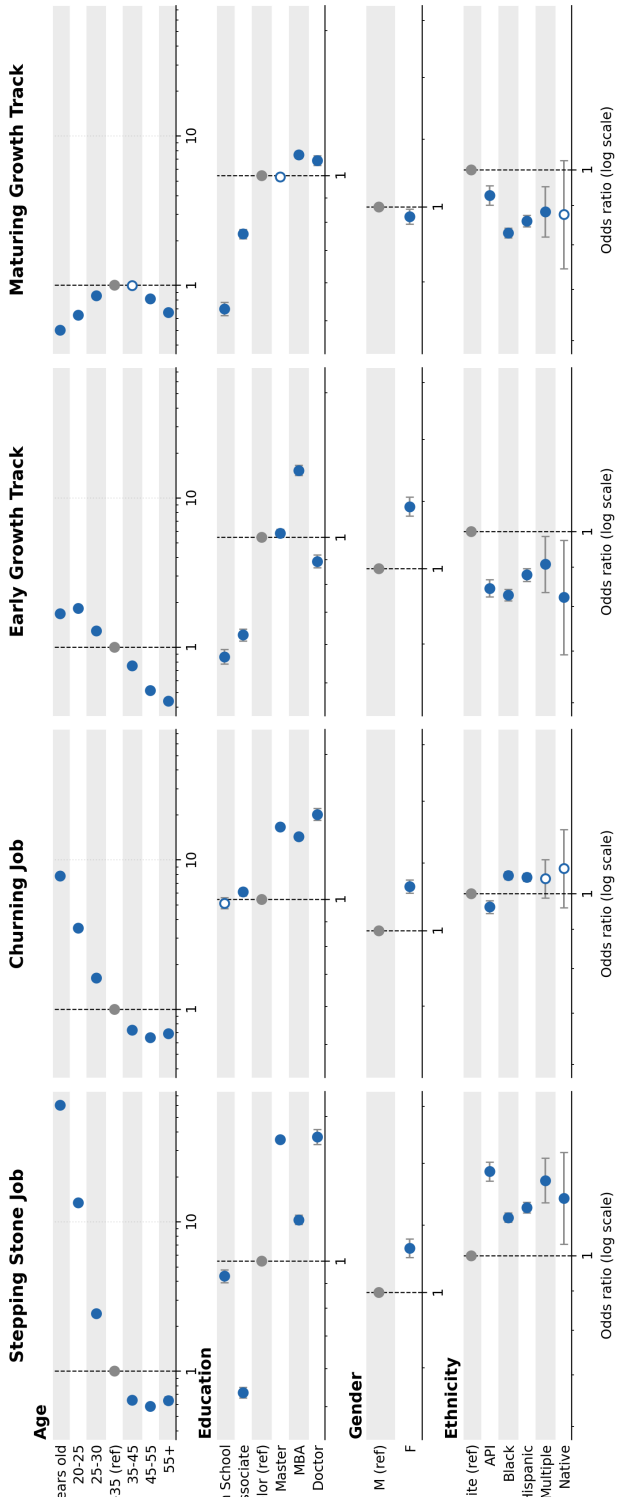


Figure 8: Demographic Characteristics

ter's, MBA, and doctoral degree holders have particularly high odds of being observed in *Stepping-Stone* and *Churning Jobs* relative to workers with a bachelor's degree. MBA holders are also substantially more likely to be observed in the *Early Growth Track*, while the differences between bachelor's and graduate-degree holders are smaller for the *Maturing Growth Track*.

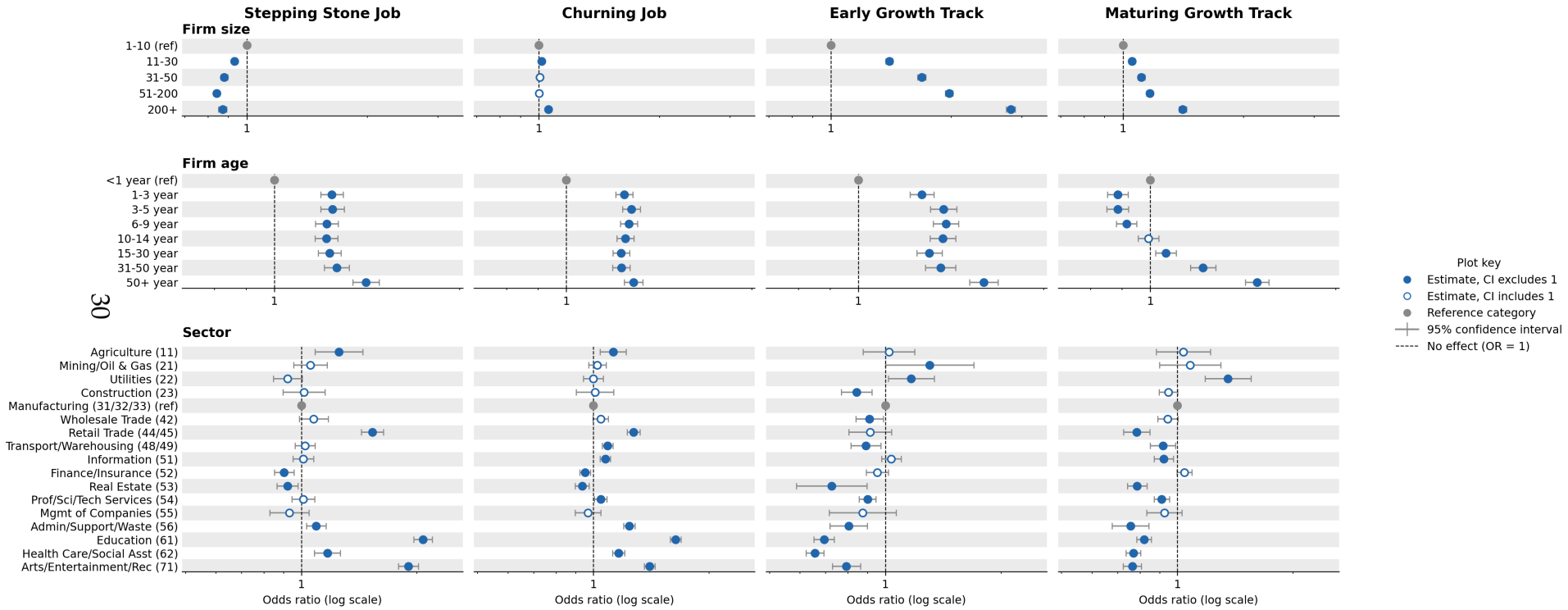
The third row reports odds ratios by gender, with male workers as the reference group. Female workers are more likely than male workers to be observed in *Stepping-Stone Matches*, *Churning Jobs*, and *Early Growth Track positions* relative to *Transitional Mismatch*. The difference is largest for *Stepping-Stone Matches* and *Early Growth Track* positions. In contrast, female workers are slightly less likely to be observed in the Maturing Growth Track.

The final row reports odds ratios by ethnicity, with White workers as the reference group. The associations differ considerably across latent states. API, Black, and Hispanic workers are more likely than White workers to be observed in *Stepping-Stone Matches*, with particularly large differences for API workers. Black and Hispanic workers are also somewhat more likely to be observed in *Churning Jobs*. In contrast, these groups are less likely than White workers to be observed in either the *Early Growth Track* or the *Maturing Growth Track*. The estimates for workers classified as Multiple or Native ethnicity are less precise in several specifications, as reflected in their wider confidence intervals.

Figure 9 reports the estimated odds ratios for firm characteristics, including firm size, firm age, and sector, using *Transitional Mismatch* as the omitted match-state category. Thus, the coefficients describe how firm characteristics are associated with the odds of belonging to each reported state relative to *Transitional Mismatch*.

The first row reports odds ratios by firm size, with firms employing 1–10 workers as the reference group. Firm size is strongly associated with the distribution of latent match states, particularly for the two growth-oriented states. The odds of being observed in the *Early Growth Track* increase monotonically with firm size. The relationship is even more pronounced for the *Maturing Growth Track*, where the odds rise more distinctly across firm-size categories, with the largest firms substantially more likely to form these matches than firms with 1–10 employees. In contrast, larger firms are less likely to form *Stepping-Stone Matches*, while the relationship between firm size and *Churning Matches* is comparatively weak. These patterns suggest that larger firms are disproportionately associated with matches characterized by internal advancement and longer-term career progression.

The second row reports odds ratios by firm age, with firms younger than one year serving as the reference group. Firm age is positively associated with several match states, but the pattern differs across states. Older firms are more likely to form *Stepping-Stone* and


Stepping Stone Job
Churning Job
Early Growth Track
Maturing Growth Track
Firm size
1-10 (ref)
11-30
31-50
51-200
200+
Firm age
<1 year (ref)
1-3 year
3-5 year
6-9 year
10-14 year
15-30 year
31-50 year
50+ year
Sector
Agriculture (11)
Mining/Oil & Gas (21)
Utilities (22)
Construction (23)
Manufacturing (31/32/33) (ref)
Wholesale Trade (42)
Retail Trade (44/45)
Transport/Warehousing (48/49)
Information (51)
Finance/Insurance (52)
Real Estate (53)
Prof/Sci/Tech Services (54)
Mgmt of Companies (55)
Admin/Support/Waste (56)
Education (61)
Health Care/Social Asst (62)
Arts/Entertainment/Rec (71)
1
Odds ratio (log scale)
Plot key
Estimate, CI excludes 1
Estimate, CI includes 1
Reference category
95% confidence interval
No effect (OR = 1)

Churning Jobs relative to very young firms, with the association remaining fairly stable across much of the firm-age distribution. The *Early Growth Track* also becomes more likely as firms age, particularly among the oldest firms. The strongest age gradient appears for the *Maturing Growth Track*. Firms in the intermediate age categories are initially less likely than newly established firms to be associated with this state, but the relationship reverses among more established firms: firms aged 31–50 years and especially firms older than 50 years are substantially more likely to form *Maturing Growth Tracks*. Importantly, these associations are estimated conditional on firm size, indicating that the relationship between firm maturity and growth-oriented matches is not driven solely by older firms being larger.

The final row reports odds ratios across sectors, with Manufacturing as the reference category. There is substantial heterogeneity in the distribution of latent match states across industries. Educational Services and Arts, Entertainment, and Recreation are relatively more likely to be associated with *Stepping-Stone* and *Churning Jobs*, while being less likely to form either *Early or Maturing Growth Track Matches*. Retail Trade shows a similar, though less pronounced pattern, with relatively greater representation in the shorter-duration states and lower odds of appearing in the growth-oriented states.

By contrast, Utilities stands out for its relatively high representation in the growth-oriented states, particularly the Maturing Growth Track. Mining, Quarrying, and Oil and Gas Extraction is also more likely to be associated with the *Early Growth Track*, although its association with the *Maturing Growth Track* is less precisely estimated. Several service sectors, including Real Estate, Administrative and Support Services, and Health Care and Social Assistance, also have relatively low odds of appearing in the growth-oriented states. These sector-level estimates should be interpreted with some caution because Revelio have relatively poor coverage in these sectors compared to their representations in the economy.

LinkedIn coverage is particularly high in Professional, Scientific, and Technical Services and Information, reflecting the platform's strong representation of knowledge-intensive and highly skilled labor markets. These sectors exhibit less pronounced differences across latent match states than sectors such as Education, Arts and Entertainment, or Retail Trade, although their state distributions are still not uniform. Information sector is relatively more concentrated in *Churning Jobs* and less represented in the *Maturing Growth Track*, while Professional, Scientific, and Technical Services is somewhat less likely to be associated with either growth-oriented state. These patterns are consistent with the relatively high employment turnover commonly observed in these sectors.

5 Match Innovations

In the third step, I construct a spell-level innovation to isolate the match-specific component of a worker–firm relationship. The latent Markov model's initial-state and transition equations are re-estimated using a richer set of predetermined worker characteristics. These estimates generate a predicted distribution over latent match states for each spell, conditional on the worker's observable characteristics and prior employment history. The match-state innovation is then constructed as the difference between the realized posterior state distribution and the predicted distribution. It therefore measures the extent to which the realized match outcome differs from what could have been predicted based on information available to the firm at the time the match was formed. Relative to the residual from a two-way fixed-effects model, this measure also allows for time-varying worker heterogeneity summarized by the worker's most recent employment spell.

5.1 Construction of Spell-level Match Innovations

Re-estimating State Distributions with Expanded Worker Covariates. I construct the match-state innovation as the difference between the realized posterior distribution over latent match states and the distribution predicted using information available prior to the current match. I fix the latent-state definitions and emission probabilities estimated in Step 1 and estimate richer covariate-dependent models for the initial-state and transition distributions. Including a large set of worker covariates makes estimation computationally demanding. Thus, to reduce computation time, I use the previously computed locally decoded posterior probabilities from Step 2 as fixed estimation weights, rather than running a full EM procedure that repeatedly updates posterior weights and model parameters.

For the first observed position, I estimate a weighted multinomial logit for the initial-state distribution. Let $\mathbf{X}_{i,1}$ denote predetermined worker characteristics observed at the first position, including gender, ethnicity, education, and year-quarter of hire, while v_i denotes the worker's birth cohort. The model yields the initial distribution

$$P(U_{i,1} = k \mid \mathbf{X}_{i,1}, v_i), \qquad k = 1, \ldots, K. \tag{12}$$

The initial-state model uses the locally decoded posterior probability $\pi_{i,1,k}^{Local}$ from Step 2 as the weight associated with state k. Thus, each first position contributes to each candidate state in proportion to its posterior probability, rather than being assigned deterministically to a single state.

For subsequent positions, I estimate a separate multinomial logit for each possible origin state. The covariate vector $\mathbf{X}_{i,n+1}$ includes worker age in place of birth cohort, together with gender, ethnicity, educational attainment, and year-quarter of hire. Let $z_{i,n+1}$ denote the pre-employment gap preceding position $n+1$. The transition model is

$$P\big(U_{i,n+1} = k' \mid U_{i,n} = k, \mathbf{X}_{i,n+1}, z_{i,n+1}\big), \qquad n = 1, \ldots, N_i - 1. \tag{13}$$

For each origin state k, estimation uses the product of the locally decoded marginal posterior probabilities for the origin and destination states:

$$w_{i,n+1,kk'} = \pi^{Local}_{i,n,k} \pi^{Local}_{i,n+1,k'}. \tag{14}$$

For computational tractability, I approximate the joint posterior probability of consecutive latent states by the product of their marginal posterior probabilities. This approximation allows the transition equations to be estimated using weighted multinomial logit regressions without re-estimating the full hidden Markov model. Each observed transition contributes to all possible origin–destination state pairs according to these soft-assignment weights.

Constructing Match Innovations with Filtered State Probabilities. Then, I use the filtered decoded posterior probabilities to construct the match-state innovation at the spell-level. This ensures that both the predicted and realized state distributions do not incorporate information from employment positions happening after the current match.

For the first observed position, the predicted state distribution is given directly by the estimated initial-state model:

$$\widehat{p}_{i,1,k} = \widehat{P}\Big(U_{i,1} = k \mid v_i, \mathbf{X}^I_{i,1}\Big). \tag{15}$$

For positions $n \geq 2$, I combine the filtered posterior distribution over the previous latent state with the estimated covariate-dependent transition probabilities. Specifically, the predicted probability that worker i enters state k in position n is

$$\widehat{p}_{i,n+1,k'} = \sum_{k=1}^{K} \pi^{Filtered}_{i,n,k} \, \widehat{P}\big(U_{i,n+1} = k' \mid U_{i,n} = k, \mathbf{X}_{i,n+1}, z_{i,n+1}\big), \qquad n = 1, \ldots, N_i - 1. \tag{16}$$

Let $\widehat{\mathbf{p}}_{i,n} = (\widehat{p}_{i,n,1}, \ldots, \widehat{p}_{i,n,K})$ denote the vector of the one-step-ahead predicted distribution over latent match states, and let $\boldsymbol{\pi}^{Filtered}_{i,n} = (\pi^{Filtered}_{i,n,1}, ..., \pi^{Filtered}_{i,n,K})$ denote the vector of filtered posterior probabilities over the realized matches. I define the match-state innova-

tion as

$$\boldsymbol{\varepsilon}_{i,n} = \boldsymbol{\pi}_{i,n}^{Filtered} - \widehat{\mathbf{p}}_{i,n}. \tag{17}$$

Because both $\boldsymbol{\pi}_{i,n}^{Filtered}$ and $\widehat{\mathbf{p}}_{i,n}$ are probability distributions,

$$\sum_{k=1}^{K} \varepsilon_{i,n,k} = 0. \tag{18}$$

The distinction between local and filtered decoding is important for the construction. Local posterior probabilities are used when re-estimating the richer initial and transition models because they provide the most informative ex-post classification of each position. On the other hand, filtered probabilities are used when constructing the predicted and realized state distributions so that the innovation does not incorporate information from future employment outcomes. The resulting innovation measures the component of the current match-state distribution that cannot be predicted from the worker's observable characteristics and employment history available before the current match outcomes are formed.

Table 2 reports summary statistics for the seven components of the spell-level match innovation. The mean innovation is close to zero for every state, indicating that the predicted state distributions are well aligned with the realized filtered distributions on average. At the same time, the innovations exhibit substantial dispersion at the spell level. The standard deviation ranges from 0.094 for Stable-Stagnant Matches (S2) to 0.317 for Stable-Ongoing Matches (S7), while the mean absolute innovation ranges from 0.046 to 0.203 across states.

Part of this difference in dispersion reflects the way the two censored states are identified in the data. Stable-Stagnant Matches (S2) is associated with left-censored employment histories and is concentrated among the first observed positions of workers whose earlier employment histories are not observed. Because this censoring status is almost entirely defined by worker's demographic characteristics, assignment to this state is comparatively easier to predict from information available prior to the current match. Stable-Ongoing Matches (S7), by contrast, captures matches that remain ongoing at the end of the sample. Whether a currently observed match will ultimately remain ongoing through the sample endpoint is inherently more difficult to predict in advance. As a result, realized posterior probabilities for S7 may differ more from their predicted probabilities, contributing to the larger dispersion of its match innovation.

A remaining concern is that the innovation may still contain unobserved worker-level

Table 2: Summary Statistics of Spell-Level Match Innovations

	Mean	Std. Dev.	Mean Absolute Value
inn_1 (Stepping-stone)	0.002	0.141	0.083
inn_2 (Stable-Stagnant)	0.001	0.094	0.046
inn_3 (Churning)	-0.001	0.176	0.120
inn_4 (Transitional)	-0.002	0.211	0.169
inn_5 (Early Growth)	0.000	0.224	0.161
inn_6 (Maturing Growth)	-0.002	0.130	0.079
inn_7 (Stable Ongoing)	0.002	0.317	0.203

Notes: The table reports summary statistics for the seven components of the spell-level match innovation, $\boldsymbol{\varepsilon}_{i,n} = \boldsymbol{\pi}_{i,n}^{Filtered} - \widehat{\mathbf{p}}_{i,n}$. Statistics are calculated over 86,108,301 employment spells in the sample.

information that the prediction step does not capture. Because such information travels with the worker, it would make a worker's innovations similar across consecutive spells. I therefore use within-worker persistence as a diagnostic, reported in Appendix 9.6. First, I compare the cosine similarity of a worker's consecutive spells with a baseline that pairs each spell with an observably comparable spell of an unrelated worker and firm. Relative to this baseline, the excess similarity of match innovations is about 55 percent lower than that of the underlying state probabilities for moves across firms, and about 57 percent lower for position changes within a firm (Table 30). Second, an unrestricted vector autoregression, in which each component of the previous spell's distribution may predict each component of the next spell's distribution, yields much smaller coefficients for innovations than for state probabilities (Tables 31 and 32). Both checks indicate that the prediction procedure removes most of the worker-level component of latent match states. Some residual persistence is nonetheless expected. As pointed out by Borovičková and Shimer (2024), only realized spells are observed, so the measured innovations are already a selected sample of potential match innovations: a match forms only when its match-specific shock clears an acceptance threshold, and endogenous separations further select which matches survive. Because the same worker's acceptance threshold governs selection in each of their spells, realized innovations can be weakly correlated across spells even when the underlying potential innovations are independent.

6 Firm Growth and Survival

Having constructed spell-level measures of match-state innovations, I next aggregate them to the firm level and examine whether firms' exposure to unexpectedly favorable

or unfavorable matches predicts their subsequent performance. The main analysis focuses on innovations among newly formed positions; results based on the firm's stock of workers and on completed spells are reported in the Appendix. The analysis proceeds in four steps. First, using the full Revelio sample, I relate match innovations to firms' subsequent separation rates, hiring rates, employment growth, and survival. Second, I link firms to Compustat and show that the same patterns hold among publicly listed firms, both for outcomes measured in Revelio and for employment, survival, and sales measured independently in Compustat. Third, I trace these relationships over the three years following hiring. Finally, I examine how match innovations are associated with firm-level TFP using Compustat data.

6.1 Firm-level Exposure to Match Innovations

I construct firms' exposure to match innovations at the annual frequency in three ways. Although the underlying data allows measuring changes in firms' exposure to match innovations at the monthly frequency, I aggregate these measures to the annual level. This reduces the influence of short-run fluctuations in hiring and worker turnover and allows the firm-level measures to capture more persistent changes in the composition and quality of a firm's worker matches.

For each firm j in year t, its match innovation for state k is

$$\bar{\varepsilon}^{m}_{j,t,k} = \frac{1}{N^{m}_{jt}} \sum_{(i,n)\in\mathcal{S}^{m}_{jt}} \varepsilon_{i,n,k}, \quad m \in \{New\,spells,\ Stock,\ Completed\,spells\} \tag{19}$$

For the new-position measure, $\mathcal{S}^{New}_{jt}$ consists of spells beginning at firm j during year t, so $\bar{\varepsilon}^{New}_{jtk}$ captures the average match innovation among new positions over the year. For the stock measure, $\mathcal{S}^{Stock}_{jt}$ consists of employment spells active at firm j in the fourth quarter of year t; $\bar{\varepsilon}^{Stock}_{jtk}$ therefore summarizes the average unexpected component of match quality among the firm's year-end workforce. An increase in the innovation toward state k indicates greater exposure to that state among the current workforce.

For the completed-spell measure, $\mathcal{S}^{Completed}_{jt}$ consists of employment spells ending at firm j during year t. This measure addresses a timing concern with the new-position and stock measures: the spell durations and transition types used to infer match quality may not yet be realized when exposure is measured. Relating innovations among spells completed in year $t-1$ to firm outcomes in year t ensures that these signals are observed before the outcome year. The interpretation also differs: a higher innovation toward state k indicates greater exposure among spells ending at that firm. Such completions may

reduce current exposure to state k, although the net change also depends on new and continuing spells. I present results for the new-hire measure in this section and report results for the stock and completed-spell measures in the Appendix.

Since the spell-level innovation vector sums to zero across states, the firm-level averages satisfy the same adding-up restriction:

$$\sum_{k=1}^{K} \bar{\varepsilon}^{m}_{j,t,k} = 0. \tag{20}$$

Because the seven innovation components sum to zero, they cannot all be included in a regression simultaneously. I omit Transitional Mismatch throughout, so each included coefficient is interpreted relative to Transitional Mismatch: an increase in the innovation toward state k, holding the other included innovations fixed, corresponds to an equal reduction in the innovation toward Transitional Mismatch. Coefficients thus capture reallocations of unexpected match-state probability rather than independent increases in individual components.

Table 3 reports the frequency of changes in firms' average new-hire match innovations between consecutive years, using firms with observed innovations in both years. Each cell gives the fraction of these firm-year pairs in which the innovation for the row state decreases by at least 10 percentage points while that for the column state increases by at least 10 percentage points. Panel A reports the full sample, and Panel B restricts the sample to firms matched to Compustat in the starting year. These events are not mutually exclusive: for example, a 30 percentage point decline in Churning innovations could coincide with a 10 percentage point increase in Early Growth innovations and a 20 percentage point increase in Maturing Growth innovations, contributing to two cells. Consequently, the table as a whole need not sum to one.

In the full sample, declines in Transitional Mismatch innovations most frequently coincide with increases in Stable Ongoing innovations, with this combination occurring in 11.6% of consecutive-year firm observations. Declines in Transitional Mismatch also coincide with increases in Churning and Early Growth innovations in 7.6% and 5.4% of observations, respectively. Declines in Churning innovations coincide with increases in Early Growth and Maturing Growth innovations in 3.9% and 3.8% of observations, respectively. All reported combinations are less frequent among Compustat firms. Nevertheless, declining Transitional Mismatch paired with rising Stable Ongoing remains the most frequent combination, occurring in 4.0% of observations. The corresponding frequencies for declining Churning paired with rising Early Growth or Maturing Growth are 1.3% and 1.2%.

Table 3: Changes in New-Hire State Innovations of at Least 10 Percentage Points

	Increasing state						
Decreasing state	L-censored	Stepping	Churning	Transitional	Early Gr.	Maturing Gr.	Ongoing
Panel A: Full sample (N = 4,927,388)							
L-censored	0.0000	0.0396	0.0222	0.0221	0.0122	0.0085	0.0169
Step.	0.0393	0.0000	0.0302	0.0606	0.0324	0.0199	0.0395
Churn.	0.0208	0.0279	0.0000	0.0690	0.0387	0.0375	0.0931
Trans.	0.0205	0.0637	0.0758	0.0000	0.0541	0.0290	0.1163
Early Gr.	0.0122	0.0385	0.0452	0.0629	0.0000	0.0147	0.0592
Maturing Gr.	0.0083	0.0225	0.0405	0.0328	0.0134	0.0000	0.0345
Ongoing	0.0169	0.0439	0.0990	0.1225	0.0600	0.0345	0.0000
Panel B: Compustat sample (N = 71,239)							
L-censored	0.0000	0.0105	0.0053	0.0071	0.0052	0.0034	0.0052
Step.	0.0104	0.0000	0.0045	0.0129	0.0092	0.0062	0.0081
Churn.	0.0058	0.0042	0.0000	0.0158	0.0135	0.0119	0.0263
Trans.	0.0064	0.0139	0.0193	0.0000	0.0262	0.0143	0.0400
Early Gr.	0.0049	0.0097	0.0140	0.0262	0.0000	0.0057	0.0286
Maturing Gr.	0.0033	0.0065	0.0118	0.0152	0.0060	0.0000	0.0142
Ongoing	0.0055	0.0084	0.0254	0.0402	0.0254	0.0129	0.0000

Notes: Table 3 reports the frequency of changes in firms' average new-hire match innovations between consecutive years, using firms with observed innovations in both years. Each cell gives the fraction of these firm-year pairs in which the innovation for the row state decreases by at least 10 percentage points while that for the column state increases by at least 10 percentage points. Each pair receives equal weight, and the denominator includes all eligible pairs, denoted by N. A pair may satisfy the conditions for multiple cells, so row sums need not equal one. *Source: Author's calculation.*

Appendix Tables 33 and 34 report the corresponding frequencies of changes in firms' average match innovations among incumbent workers and completed spells, respectively.

6.2 Empirical Methods

I estimate the following regression at the firm level:

$$O_{jt} = \sum_{k \in \{1,2,3,5,6,7\}} \delta_k \, \bar{\varepsilon}_{j,t-1,k} + C_{j,t-1} + u_{jt}, \tag{21}$$

where O_{jt} measures the activity of firm j in year t and $\bar{\varepsilon}_{j,t-1,k}$ is the firm's realized match innovation for outcome k in year $t-1$. In the full Revelio sample, I consider four outcomes: (1) firm's annual separation rate, (2) firm's annual hiring rate, (3) the firm's annual

employment growth rate, following Davis et al. (2013), and (4) an indicator for whether the firm survives from $t-1$ to t. [8] $C_{j,t-1}$ contains lagged firm-level controls that vary across specifications. I estimate the survival regression as a linear probability model, and I cluster standard errors at the firm level throughout.

The identification rests on two assumptions. First, the predicted probability distribution of employment outcomes captures all unobserved worker heterogeneity, including, for example, persistent differences in ability and the accumulation of human capital over time. Because the hidden Markov model conditions on workers' employment histories and a rich set of worker covariates, including worker's birth cohorts, education attainments, gender, ethnicity, employment gap before the current position, and the time of the hire, the realized match innovations are plausibly exogenous with respect to worker-specific effects. The persistence exercise in Appendix 9.6 also provides supporting evidence that the procedure substantially reduces worker-specific persistence.

Even if the assumption that match innovations are independent of unobserved worker heterogeneity does not fully hold, the results provide new evidence on how unexpected match outcomes relate to subsequent firm performance. At the time of hiring, firms may infer workers' otherwise unobserved characteristics from their observable attributes and employment histories. To the extent that these characteristics are captured by the information used to construct the predicted match-state distribution, their influence is removed from the match innovation. The resulting measure therefore captures deviations from expected match outcomes conditional on this information. The findings thus shed light on the economic value of unexpectedly favorable or unfavorable matches to firms.

Second, the identification assumes that conditional on $C_{j,t-1}$, realized match innovations are orthogonal to other time-varying firm-level determinants of subsequent outcomes. I need suitable firm-level controls because the distribution of exposure to match innovations across industries and business characteristics is not random, as shown in section 4. For example, employment in retail is typically much less stable than in manufacturing, and large firms are more likely than small startups to offer internal job ladders. Because the predicted probabilities contain no information about the destination firm, controlling for firm characteristics is important when estimating effects on firm outcomes.

I consider the following set of controls. I include firm fixed effects to absorb persistent

[8] The separation and hiring rates are defined as the number of separations and new hires during the year, respectively, divided by employment at the beginning of the year. Employment growth is measured using the DHS growth rate, $g_{jt} = \frac{N_{jt}-N_{j,t-1}}{\frac{1}{2}(N_{jt}+N_{j,t-1})}$. The survival indicator equals one if the firm has positive employment in year t. Because employment is measured using LinkedIn profiles, the disappearance of a firm's observed workforce may indicate a substantial contraction or cessation of activity, but does not necessarily imply firm exit.

differences in firm-level demand and productivity, which prior work identifies as important determinants of firms' growth paths (Sterk et al., 2021). I also include industry-by-year fixed effects, with industries defined at the six-digit NAICS level, incorporating more than 1,000 industry cells per year. These fixed effects capture time-varying industry-level demand and technology shocks that may jointly affect match outcomes and firm performance. In addition, I control for lagged firm outcomes to account for transitory firm-level shocks that may influence subsequent firm activity. In a second set of regressions, I additionally control for workers' predicted match-state distributions aggregated to the firm level. These controls capture the observable composition of a firm's new matches, which may be correlated with realized match innovations through sorting and may independently affect subsequent firm outcomes. Reassuringly, all specifications yield similar magnitudes and identical rankings of the effects across match-outcome states.

The empirical specification relies on the assumption that realized match innovations are unrelated to other time-varying firm-level shocks that independently predict subsequent firm outcomes. This would fail if, for example, a positive firm-specific technology shock in year t both generated better match outcomes and raised subsequent employment growth, biasing my estimates upward. The firm-level controls capture an important part of demand and productivity heterogeneity. To further assess the role of match innovations, I estimate the effects over longer horizons. If the estimates were driven only by transitory demand or productivity shocks, the effects may attenuate as the horizon lengthens. Instead, they remain statistically significant and, for some match states, the effect become stronger. I therefore interpret the estimates as economically informative about the effect of match-quality innovations on subsequent firm outcomes.

6.3 Match Innovations and Firm Outcomes

Table 4 reports the conditional correlation of match innovations on firm separation rates, hiring rate, growth rate and survival. Columns (1)-(4) controls for industry-by-year fixed effects, firm fixed effects and lagged firm outcomes; columns (5)-(8) additionally control for lagged predicted state probabilities, aggregated to the firm-level in the same way as match innovations. Columns (5)-(8) are my preferred estimates. Table 5 reports the corresponding results restricted to Compustat firms.

Effects in the following year using the full sample. The clearest contrast is between Churning and the two growth-track states. In the full Revelio sample, a 10 percentage point shift toward Churning matches is associated with a 0.88 percentage point

higher separation rate, while the same shift toward Early Growth and Maturing Growth matches is associated with declines in the separation rate of 0.78 and 2.27 percentage points, respectively. It is perhaps unsurprising that greater exposure to shorter-duration, separation-prone matches predicts higher subsequent separation rates than exposure to growth-oriented matches. However, the differences in estimates across states with similar spell-duration distributions are particularly interesting. Although Stepping-stone and Churning matches have similar spell-duration distributions, a shift from Transitional Mismatch toward these two states has opposite effects on firms' subsequent separation rates: greater exposure to Stepping-stone matches predicts lower separations, while greater exposure to Churning matches predicts higher separations. Furthermore, Early Growth matches have substantially longer durations than Stepping-stone matches, and their transition outcomes are characterized primarily by internal advancement rather than separation. Nevertheless, greater exposure to Stepping-stone matches predicts a larger decline in subsequent separation rates than greater exposure to Early Growth matches. Taken together, these findings suggest that the estimated effects on subsequent separations are not simply a mechanical reflection of the spell durations and exit outcomes used to infer the latent match states. Instead, differences in firms' exposure to these states contain additional information about subsequent workforce dynamics beyond the observed characteristics used to classify individual matches.

In terms of hiring rates, Stepping-stone and Early Growth matches have the largest positive associations with subsequent hiring. A 10 percentage point shift from Transitional Mismatch toward Stepping-stone matches is associated with a 0.25 percentage point higher hiring rate, while an equivalent shift toward Early Growth matches is associated with a 0.24 percentage point higher hiring rate. By contrast, the same shift toward Maturing Growth matches is associated with a 0.33 percentage point lower hiring rate.

Together, these results suggest that match innovations are important predictors of subsequent net employment growth and firm survival. A 10 percentage point shift toward Churning matches is associated with 1.03 percentage point lower employment growth and a 0.14 percentage point lower probability of firm survival. In contrast, shifts of the same magnitude toward Early Growth and Maturing Growth matches are associated with 1.01 and 1.82 percentage points higher employment growth, respectively, and 0.09 and 0.11 percentage points higher survival probabilities. Taken together, a 10 percentage point shift in match composition from Churning toward either of the two growth-track states predicts 2.0–2.9 percentage points higher employment growth and a 0.23–0.25 percentage point higher probability of firm survival.

Table 4: New-Hire Match Innovations and Firm Outcomes

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
inn_1 (Stepping-st)	−0.1002*	0.0240*	0.1154*	0.0153*	−0.1057*	0.0246*	0.1208*	0.0154*
	(0.0010)	(0.0010)	(0.0013)	(0.0004)	(0.0010)	(0.0010)	(0.0013)	(0.0004)
inn_3 (Churning)	0.0874*	0.0116*	−0.1039*	−0.0136*	0.0877*	0.0108*	−0.1029*	−0.0135*
	(0.0010)	(0.0009)	(0.0012)	(0.0003)	(0.0010)	(0.0009)	(0.0012)	(0.0003)
inn_5 (Early Gr)	−0.0803*	0.0217*	0.1058*	0.0093*	−0.0777*	0.0235*	0.1010*	0.0094*
	(0.0008)	(0.0009)	(0.0011)	(0.0003)	(0.0008)	(0.0009)	(0.0011)	(0.0002)
inn_6 (Maturing Gr)	−0.2128*	−0.0327*	0.1665*	0.0106*	−0.2265*	−0.0329*	0.1819*	0.0113*
	(0.0012)	(0.0013)	(0.0016)	(0.0003)	(0.0011)	(0.0013)	(0.0016)	(0.0003)
inn_7 (Stable Ongoing)	−0.1015*	−0.0039*	0.0897*	0.0041*	−0.1049*	−0.0037*	0.0933*	0.0044*
	(0.0005)	(0.0006)	(0.0007)	(0.0002)	(0.0005)	(0.0006)	(0.0007)	(0.0002)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0346	0.0117	0.0168	0.0092	0.0381	0.0118	0.0191	0.0097
N	7,476,741	7,476,741	7,509,821	7,576,673	7,476,741	7,476,741	7,509,821	7,576,673

Notes: This table reports estimates of lagged match-state innovations of new positions to firm outcomes using the full sample. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival. All models include firm and 6-digit NAICS industry-by-year fixed effects, firm fixed effects, and lagged firm outcomes. Separation and hiring regressions control for their lagged outcome; growth and survival models control for lagged employment growth. All lags are one year. Columns (5)–(8) also include lagged predicted state probabilities. Transitional Mismatch is the reference for innovations and probabilities. Left-censored state is included in the regression but not reported. I report the within R-square and standard errors are clustered at the firm level. * $p < 0.10$ *Source: Revelio Labs.*

Table 5: New-Hire Match Innovations and Firm Outcomes I (Compustat Firms)

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
inn_1 (Stepping-st)	−0.1646*	0.0268	0.1861*	−0.0079*	−0.1718*	0.0348*	0.2029*	−0.0076*
	(0.0157)	(0.0171)	(0.0232)	(0.0038)	(0.0160)	(0.0173)	(0.0236)	(0.0038)
inn_3 (Churning)	0.1248*	−0.0931*	−0.2348*	−0.0079*	0.1238*	−0.0917*	−0.2340*	−0.0081*
	(0.0152)	(0.0150)	(0.0232)	(0.0035)	(0.0152)	(0.0150)	(0.0231)	(0.0036)
inn_5 (Early Gr)	−0.1376*	0.0969*	0.2295*	−0.0014	−0.1383*	0.0973*	0.2308*	−0.0013
	(0.0107)	(0.0122)	(0.0168)	(0.0020)	(0.0107)	(0.0122)	(0.0168)	(0.0019)
inn_6 (Maturing Gr)	−0.3043*	−0.0474*	0.2393*	0.0003	−0.3060*	−0.0484*	0.2394*	0.0004
	(0.0150)	(0.0170)	(0.0239)	(0.0030)	(0.0149)	(0.0170)	(0.0236)	(0.0029)
inn_7 (Stable Ongoing)	−0.1569*	0.0161*	0.1648*	−0.0027*	−0.1592*	0.0180*	0.1692*	−0.0026
	(0.0080)	(0.0090)	(0.0125)	(0.0016)	(0.0081)	(0.0090)	(0.0125)	(0.0016)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0771	0.0729	0.0978	0.0031	0.0800	0.0740	0.1005	0.0036
N	72,815	72,815	72,831	72,869	72,815	72,815	72,831	72,869

Notes: This table reports estimates of lagged match-state innovations of new positions to firm outcomes using Compustat firms. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival, all measured using Revelio Labs data. All models include firm and 6-digit NAICS industry-by-year fixed effects, and lagged firm outcomes. Separation and hiring regressions control for their lagged outcome; growth and survival models control for lagged employment growth. All lags are one year. Columns (5)–(8) also include lagged predicted state probabilities. Transitional Mismatch is the reference for innovations and probabilities. Left-censored state is included in the regression but not reported. I report the within R-square and standard errors are clustered at the firm level. * $p < 0.10$ *Source: Revelio Labs, Compustat.*

Effects in the following year using Compustat firms. When restricting the sample to Compustat firms, the conditional association with subsequent firm performances is even stronger. Table 5 reports the corresponding results, where separation, hires, growth and survival are all measured using Revelio data. A 10 percentage point shift from Transitional Mismatch toward Churning matches is associated with a 1.24 percentage point higher separation rate, while shifts of the same magnitude toward Early Growth and Maturing Growth matches are associated with 1.38 and 3.06 percentage point lower separation rates, respectively. In terms of hiring, Stepping-stone and Early Growth matches have the largest positive associations, with a 10 percentage point shift toward these states predicting 0.35 and 0.97 percentage point higher hiring rates, respectively. By contrast, the same shift toward Maturing Growth matches predicts a 0.48 percentage point lower hiring rate.

Together, these differences translate into substantial variation in subsequent employment growth and firm survival. A 10 percentage point shift toward Churning matches is associated with 2.34 percentage points lower employment growth and a 0.08 percent-

Table 6: Match Innovations and Firm Outcomes II (Compustat Firms)

	Emp. growth (1)	Survival (2)	Log sales (3)	Emp. growth (4)	Survival (5)	Log sales (6)
inn_1 (Stepping-st)	0.1172*	0.0514*	0.0400	0.1341*	0.0536*	0.0684
	(0.0365)	(0.0180)	(0.0691)	(0.0367)	(0.0181)	(0.0709)
inn_3 (Churning)	−0.1675*	−0.0988*	−0.2407*	−0.1586*	−0.0973*	−0.2341*
	(0.0327)	(0.0195)	(0.0771)	(0.0328)	(0.0195)	(0.0779)
inn_5 (Early Gr)	0.2324*	0.0513*	0.2360*	0.2327*	0.0517*	0.2420*
	(0.0266)	(0.0137)	(0.0695)	(0.0268)	(0.0138)	(0.0705)
inn_6 (Maturing Gr)	0.1719*	0.0475*	0.0002	0.1721*	0.0473*	0.0030
	(0.0332)	(0.0165)	(0.0844)	(0.0334)	(0.0164)	(0.0843)
inn_7 (Stable Ongoing)	0.1376*	0.0554*	0.0468	0.1419*	0.0560*	0.0565
	(0.0189)	(0.0105)	(0.0447)	(0.0190)	(0.0105)	(0.0448)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	Yes	Yes	Yes
Within R^2	0.0117	0.0114	0.4673	0.0136	0.0116	0.4677
N	59,908	61,691	53,831	59,908	61,691	53,831

Notes: This table reports estimates of lagged match-state innovations of new positions to firm outcomes using Compustat firms. Outcomes are employment growth following Davis et al. (2013), firm survival and log of sales , all measured using Compustat annual data. All models include firm and 6-digit NAICS industry-by-year fixed effects, and lagged firm outcomes. Separation and hiring regressions control for their lagged outcome; growth and survival models control for lagged employment growth. All lags are one year. Columns (4)–(6) also include lagged predicted state probabilities. Transitional Mismatch is the reference for innovations and probabilities. Left-censored state is included in the regression but not reported. I report the within R-square and standard errors are clustered at the firm level. * $p < 0.10$ *Source: Revelio Labs, Compustat.*

age point lower probability of firm survival. In contrast, shifts of the same magnitude toward Early Growth and Maturing Growth matches predict 2.31 and 2.39 percentage points higher employment growth, respectively, and −0.01 and 0.004 percentage point higher survival probabilities. Taken together, a 10 percentage point shift in match composition from Churning toward either of the two growth-track states predicts 4.65–4.73 percentage points higher employment growth and a 0.07–0.09 percentage point higher probability of firm survival.

Table 6 reports the effects on employment growth, firm survival, and log sales using Compustat annual data (SP Global Market Intelligence, 2026). The Compustat data provide an independent check on the effects estimated above, as the employment and sales measures come from a separate data source. A 10-percentage-point shift in exposure from Churning toward either of the two growth-track states predicts 3.31–3.91 percentage points higher employment growth, a 1.45–1.49 percentage point higher probability of firm survival, and a 2.37–4.76% increase in sales.

Effects over the next three years using Compustat firms. Next, I trace the effects of match innovations on firm outcomes over the subsequent three years. For many states and outcomes, the effects remain economically meaningful two or three years later and, in some cases, become stronger over time. Figure 10 reports the effects of a 10-percentage-point shift in new-hire match innovations from Transitional Mismatch toward each of the indicated states on annual employment growth, firm survival (left axis), and log sales (right axis), using Compustat annual data. All specifications control for the aggregated predicted state probabilities. The figures also report 90% confidence intervals.

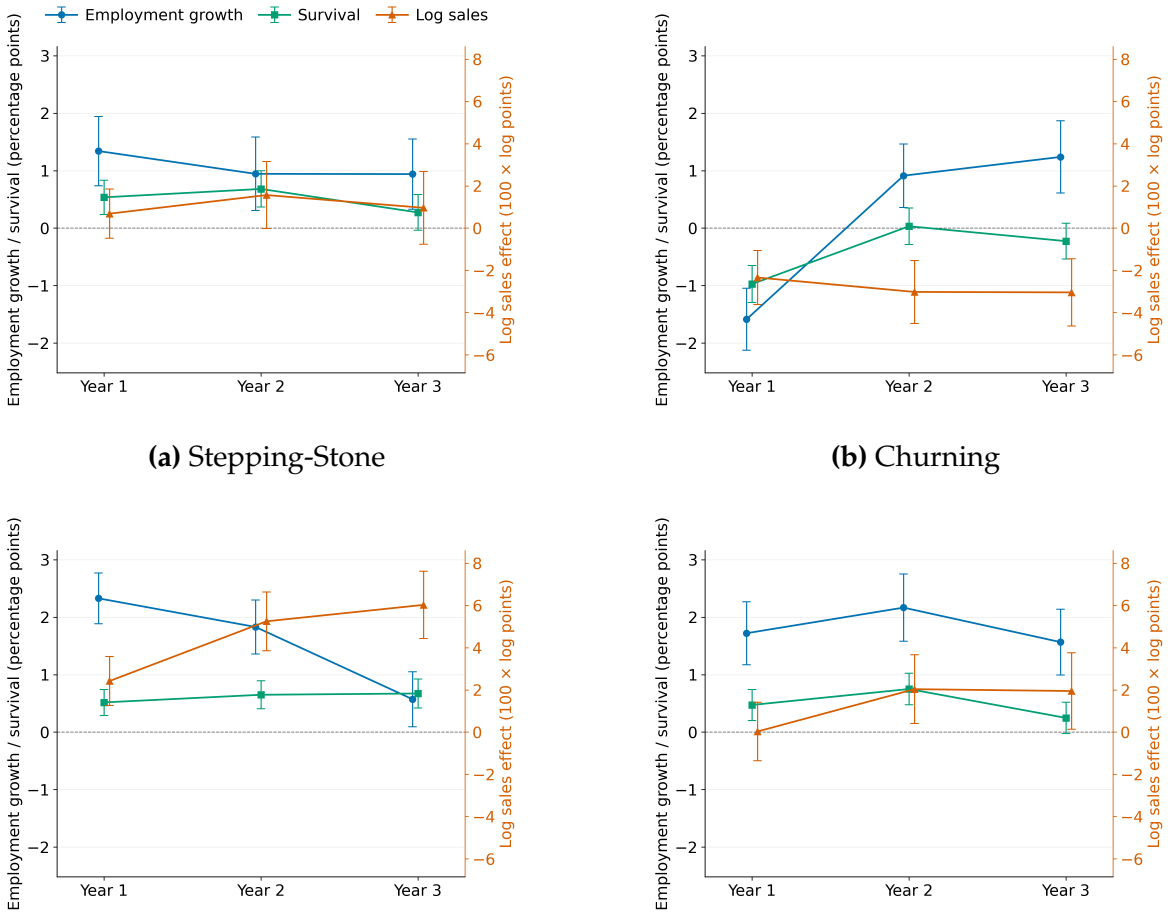
The most striking result is the strengthening positive association between Early Growth matches and firm sales. A 10-percentage-point shift in new positions from Transitional Mismatch toward Early Growth predicts approximately 2.4% higher sales in the first year. The difference increases to over 5% in year $t+2$ and becomes even larger in year $t+3$, suggesting that favorable matches are associated with persistent and growing positive effects on firm sales levels several years after hiring. In contrast, the negative association between Churning matches and sales remains relatively stable over time, potentially because these unfavorable matches have limited lasting effects on firm sales as the workers involved leave the firm.

Turning to employment growth, a shift toward Stepping-Stone, Early Growth, or Maturing Growth matches predicts higher annual employment growth even in the third year after hiring. For Maturing Growth matches, the conditional association with employment growth during the second year is even stronger than during the first year. Interestingly, while Churning matches initially predict lower employment growth, they predict higher annual employment growth in the second year, with an even stronger positive association in the third year. One possible explanation is that firms gradually replace these unfavorable matches with new workers in subsequent years, leading to increased hiring and a recovery in employment growth as firms rebuild their workforce.

I report the remaining results on hiring and separation rates, employment growth, and firm survival using the full sample, as well as results for Compustat firms based on Revelio data, in Appendices 9.9 and 9.10.

6.4 Match Innovations and Firm Productivity

I next examine how match innovation relates to firm-level TFP. Table 7 reports the conditional correlation between firm-level exposure to match innovations and TFP among Compustat firms. TFP is estimated following İmrohoroğlu and Tüzel (2014), who adapt the Olley and Pakes (1992) estimator, with investment as the proxy for unobserved pro-



(a) Stepping-Stone

(b) Churning

(c) Early Growth

(d) Maturing Growth

Notes: This table reports estimates of firm outcomes on lagged match-state innovations of new positions for Compustat firms. Outcomes are employment growth following Davis et al. (2013), firm survival, and log sales, all measured using Compustat annual data. All models include firm fixed effects, 6-digit NAICS industry-by-year fixed effects, and lagged firm outcomes. Employment growth and survival regressions control for lagged employment growth, and log-sales regressions control for lagged log sales. All lags are one year. Columns (4)–(6) also include lagged predicted state probabilities. Transitional Mismatch is the reference state for innovations and probabilities. Within R^2 is reported, and standard errors are clustered at the firm level. * $p < 0.10$ *Source: Revelio Labs, Compustat.*
Source: Revelio Labs and Compustat.

Figure 10: Effects of Match Innovations on Firm Outcomes Over Three Years

ductivity, to Compustat data; Appendix 9.11.1 describes the estimation in detail. All columns include 2-digit NAICS industry-by-year fixed effects and control for the predicted state probabilities of the same population, and columns (2) and (4) additionally control for firm size and age bins. Transitional Mismatch is the omitted reference state, so each coefficient measures the change in the outcome associated with shifting innovation mass from Transitional Mismatch toward the listed state. Appendix Table 45 reports the same regressions with 6-digit industry-by-year fixed effects.

Columns (1) and (2) relate log TFP in year t to the match innovations of the firm's stock of workers in year $t-1$. In both specifications, shifts toward Early Growth, Maturing Growth, and Stable Ongoing matches are associated with significantly higher productivity, whereas shifts toward Churning matches are not. Shifts toward Stepping Stone matches are positively associated with TFP in column (1), but the association disappears once controlling for firm size and age. The coefficients on the two growth-track states, Early Growth and Maturing Growth, are the largest among all states and are precisely estimated ($p < 0.01$) in both columns. This ranking is consistent with the earlier results on firm performance, in which shifts toward these two states predict the largest increases in employment and sales growth. The magnitudes are economically meaningful. Using the estimates in column (2), a 10 percentage point shift in the stock's innovation exposure from Transitional Mismatch is associated with 0.137 log points (14.7 percent) higher TFP in the following year when the shift is toward Early Growth, 0.141 log points (15.2 percent) when it is toward Maturing Growth, and 0.088 log points (9.2 percent) when it is toward Stable Ongoing. Controlling for size and age reduces the coefficients by roughly 40 percent relative to column (1), indicating that part of the raw association reflects the fact that larger firms are both more productive and more exposed to growth-oriented matches. These results are consistent with the interpretation that exposure to growth-oriented match innovations is associated with higher productivity.

Columns (3) and (4) relate the growth of log TFP between t and $t+1$ to the match innovations of workers hired in year t. Because the outcome is the within-firm change in TFP, these specifications abstract from persistent differences in productivity levels across firms. Firms whose new hires turn out to be better matched than predicted experience faster subsequent productivity growth. Using the estimates in column (4), shifting 10 percentage points of new hires' innovation exposure away from Transitional Mismatch is associated with faster TFP growth over the following year: by 1.6 percentage points toward Early Growth, 3.2 percentage points toward Maturing Growth, and 2.0 percentage points toward Stable Ongoing. Controlling for firm size and age leaves the estimates essentially unchanged, consistent with differencing removing the persistent, size-related

component of TFP levels. Together, these results indicate that match innovations carry meaningful information about firm productivity beyond what is captured by firm size, age, and the predicted composition of the firm's matches.

7 Conclusion

This paper develops a new measure of worker-firm match quality and examines how unexpected variation in match quality relates to subsequent firm performance. Using a matched worker-employer panel from Revelio Labs, employment spells are classified into seven latent match states through a multichannel Hidden Markov model. Match innovations are then constructed as the difference between realized and predicted state probabilities, capturing the component of match outcomes that is not predictable from information available before match formation.

Aggregating these innovations among new hires to the firm level, I show that firms' exposure to different types of match quality predicts subsequent employment growth, survival, and sales, even after controlling for firm fixed effects, lagged firm outcomes, and industry-by-year fixed effects. Greater exposure to Churning matches, characterized by short durations and separation-prone outcomes, is associated with weaker firm performance. In contrast, greater exposure to internal-growth matches, characterized by opportunities for promotions and lateral transfers, predicts faster employment growth and higher survival. Among Compustat firms, shifting 10 percentage points of new hires' match innovations from Churning toward either Early Growth or Maturing Growth matches is associated with 3.3–3.9 percentage points higher employment growth, a 1.4–1.5 percentage point higher probability of survival, and 2.3–4.8 percent higher sales, with outcomes measured independently in Compustat. These associations are persistent, remaining economically meaningful up to three years after hiring. Finally, match quality is also related to firm productivity: shifts toward growth-oriented matches predict both higher total factor productivity and faster productivity growth, beyond what is captured by firm size, age, and the predicted composition of firms' matches.

These findings suggest that the realized quality of worker–firm matches is an important dimension of firm dynamics, distinct from persistent firm heterogeneity and conventional firm-level shocks. They also raise the question of why some firms systematically form better matches than others. Future research should investigate the sources of these differences, including firms' recruiting and screening constraints, management and organizational structure, the design of jobs and internal labor markets, and the information available to firms and workers at the time of hiring. Understanding these determinants

Table 7: Match Innovations and Firm Productivity(Compustat, Olley–Pakes TFP)

	Log TFP_t Stock innovations, $t-1$		Δ log $TFP_{t,t+1}$ New-hire innovations, t	
	(1)	(2)	(3)	(4)
Stepping Stone	0.7587**	0.2514	0.1765*	0.1850*
	(0.3460)	(0.3346)	(0.1056)	(0.1053)
Churning	0.2602	0.0346	0.1761*	0.1730*
	(0.3796)	(0.3697)	(0.1019)	(0.1019)
Early Growth	2.3558***	1.3688***	0.1699**	0.1601**
	(0.2708)	(0.2384)	(0.0781)	(0.0808)
Maturing Growth	2.1917***	1.4110***	0.3256***	0.3178***
	(0.2807)	(0.2308)	(0.1017)	(0.1029)
Stable Ongoing	1.2419***	0.8817***	0.1957***	0.1972***
	(0.1562)	(0.1502)	(0.0601)	(0.0601)
NAICS2-year FE	Yes	Yes	Yes	Yes
Size & age bins	No	Yes	No	Yes
Predicted	Yes	Yes	Yes	Yes
Within R^2	0.0923	0.1821	0.0023	0.0041
N	29,469	29,469	25,243	25,243

Notes: This table reports the conditional correlation between firm-level total factor productivity (TFP) and average filtered match-state innovations. TFP is estimated from Compustat following İmrohoroğlu and Tüzel (2014), using the Olley and Pakes (1992) estimator with investment as the proxy: value added is sales minus materials, where materials equal total expenses net of an imputed wage bill (Compustat employment times the BLS QCEW national private-sector wage per employee in the firm's NAICS industry); capital is gross property, plant, and equipment deflated at its average vintage and measured at the beginning of the year. The outcome is log TFP for col (1) and (2), and Δ log TFP for col (3) and (4); all models include 2-digit NAICS industry-by-year fixed effects. In columns (1)–(2), the outcome is log TFP in year t and the exposures are innovations of the stock of all workers in year $t-1$. In columns (3)–(4), the outcome is the change in log TFP of the same firm from t to $t+1$ and the exposures are innovations of workers hired in year t. All models control for predicted state probabilities of the same population measured in the exposure year. Columns (2) and (4) add employment-size and firm-age bins. Industry (NAICS), employment-size bins, and firm-age bins are from Revelio Labs and measured in the exposure year. Firms are linked to Compustat through GVKEY, ticker, or CUSIP using Compustat membership in the exposure year; firm-years matched to multiple parents are excluded. Transitional Mismatch is the reference state for innovations and probabilities. Within R^2 is reported; firm-clustered standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Source: Compustat, Revelio Labs, BLS QCEW.*

would clarify whether match quality reflects a firm capability that can be developed, and how policies that improve the matching process might ultimately raise firm growth and productivity.

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8 Data Appendix

8.1 Sample Composition and Validation

This section describes the composition of the Revelio Labs estimation sample and assesses how well it represents the U.S. workforce and firm population. I first compare worker and firm characteristics with national benchmarks from the Current Population Survey (CPS), the Quarterly Census of Employment and Wages (QCEW), the Statistics of U.S. Businesses (SUSB), and the Business Dynamics Statistics (BDS). I then validate the employment dynamics in Revelio against two external sources: aggregate hires and separations from the Job Openings and Labor Turnover Survey (JOLTS), and firm-level employment from Compustat. Finally, I describe the employment spells that form the basis of the analysis.

8.1.1 Worker Characteristics

For each worker I observe highest level of educational attainment (high school or below, some college, bachelor's degree, master's degree, professional or doctoral degree), which is drawn from the Revelio Labs education file. Age at the time of hire is inferred from the year of the worker's earliest recorded educational enrollment. Gender is predicted from the worker's name using a classification algorithm, and predicted ethnicity (White, Black, Hispanic, Asian, or other) is similarly inferred from name and, where available, location signals.

Table 8 compares the racial and ethnic composition of the Revelio estimation sample with that of employed individuals ages 25 and older in the CPS, averaged over 2007–2025. Relative to the CPS, the Revelio sample has a higher share of White workers (68.40% versus 64.86%) and lower shares of Black workers (8.60% versus 11.04%) and Hispanic workers (7.75% versus 15.95%). The underrepresentation is largest for Hispanic workers, whose share in Revelio is roughly half of that in the CPS. Because 7.74% of Revelio workers have missing ethnicity, these gaps should be read as approximate.

Table 9 reports the corresponding comparison for educational attainment. Highly educated workers are heavily overrepresented in Revelio: 83.71% of workers hold a bachelor's degree or higher, compared with 39.95% in the CPS. Conversely, workers with a high school education or less account for 2.76% of the Revelio sample versus 33.43% in the CPS, and associate degree holders account for 5.81% versus 10.90%. This pattern is expected, as online professional profiles are more common among college-educated workers in professional occupations. The results of this paper should therefore be interpreted

Table 8: Distribution of Race and Ethnicity

Category	Revelio (%)	CPS 2007–2025 (%)
White	68.40	64.86
Black	8.60	11.04
Hispanic	7.75	15.95
Asian and Pacific Islander (API)	7.18	–
Asian	–	6.03
Multiple	0.27	–
Native	0.06	–
Other non-Hispanic races	–	2.13
Missing	7.74	–

Notes: This table compares the racial and ethnic composition of the Revelio estimation sample with that of employed individuals ages 25 and older in the U.S. civilian noninstitutional population. CPS percentages are simple averages of annual employment shares over 2007–2025. In the CPS, the White, Black, and Asian categories exclude Hispanic individuals, and the Hispanic category includes individuals of any race. Dashes indicate categories that are not reported separately in that source. *Sources:* Revelio Labs; BLS, Current Population Survey. https://www.bls.gov/cps/

as applying primarily to the college-educated workforce.

8.1.2 Firm Characteristics

Industry composition. Tables 10 through 13 report the number and sectoral distribution of firms and workers in Revelio alongside QCEW benchmarks. Revelio covers a broad range of industries but places greater weight on sectors associated with professional employment. Averaged over the sample period, professional services account for 16.9% of Revelio firms, compared with 12.5% of QCEW establishments, and manufacturing and information also account for larger shares of Revelio firms. Among workers, Revelio has higher employment shares in information (6.8% versus 2.2%), finance (7.2% versus 4.6%), and professional services (9.0% versus 6.8%), and lower shares in retail, health care, and accommodation and food services. Manufacturing shares are similar across the two sources (9.9% versus 9.7%). These comparisons are approximate, because Revelio records firms whereas the QCEW records establishments.

These patterns of sectoral representation are persistent. Between 2007 and 2024, some gaps relative to the QCEW narrow, but substantial differences remain in information, retail, health care, and accommodation and food services. The tables therefore point to modest improvement in sectoral alignment rather than general convergence toward the QCEW distribution. Changes over time may reflect both shifts in the economy and

Table 9: Distribution of Educational Attainment

Education	Revelio (%)	CPS 2007–2025 (%)
High school and below	2.76	33.43
Some college, no degree	–	15.72
Associate degree	5.81	10.90
Bachelor's degree and higher	83.71	39.95
Bachelor's degree	46.39	–
Graduate or professional degree (Master's, MBA, doctorate)	37.32	–
Missing	7.71	–

Notes: This table compares the educational attainment of the Revelio estimation sample with that of employed individuals ages 25 and older in the U.S. civilian noninstitutional population. CPS percentages are simple averages of annual employment shares over 2007–2025. Bachelor's and higher degrees are combined for comparability over the full period; indented rows show the Revelio breakdown and are not additional categories. Dashes indicate categories that are not reported separately in that source. *Sources:* Revelio Labs; BLS, Current Population Survey, annual employment by educational attainment. https://www.bls.gov/cps/cpsaat07.htm

changes in the coverage and composition of Revelio profiles.

Firm size. Table 14 compares the firm size distribution in Revelio with the Statistics of U.S. Businesses (SUSB). Revelio underrepresents the smallest firms: among firms with nonmissing size, 62.51% have fewer than 10 workers, compared with 79.17% in the SUSB. All larger size categories are correspondingly overrepresented. For example, firms with 10–49 workers account for 29.09% of Revelio firms versus 16.96% in the SUSB, and firms with 1,000 or more workers account for 0.49% versus 0.17%. This pattern is consistent with very small firms being less likely to appear in online profile data at all.

Firm age. Table 15 compares the firm age distribution in Revelio with the BDS. Young firms are strongly underrepresented: among Revelio firms with an observed founding year, only 3.17% are four years old or younger, compared with 31.93% of firms in the BDS. Revelio firms are instead concentrated among mature firms: the median firm with an observed founding year is 20 years old, the mean is 28.9 years, and more than half (52.3%) are 20 years or older. Founding year is missing for 31.00% of Revelio firms, and missingness is likely to be more common among young and small firms, so this comparison should be interpreted with caution. Taken together, the size and age comparisons indicate that the Revelio sample tilts toward larger and more established firms.

Table 10: Number of Firms and Establishments by Year and Sector

Year	Source	11 Agric.	21 Mining	22 Utilities	23 Const.	31-33 Mfg.	42 Wholesale	44-45 Retail	48-49 Transp.	51 Info.	52 Finance	53 Real Est.	54 Prof. Svcs	55 Mgmt	56 Admin	61 Educ.	62 Health	71 Arts	72 Hosp.	81 Other	92 Pub. Admin	99 Unclass.	Missing Missing
2007	Revelio	15,030	5,508	7,465	85,388	192,742	74,713	159,934	37,749	99,446	96,365	63,689	285,276	8,275	74,985	96,749	132,930	46,107	69,958	88,752	169,698	3,537	39,395
	QCEW	95,203	29,418	15,848	887,529	360,737	626,672	1,047,530	246,527	140,824	489,927	368,875	965,658	48,761	453,194	120,995	713,125	112,900	597,319	1,147,925	71,343	191,729	–
2008	Revelio	17,215	6,128	8,480	93,212	208,135	81,111	179,669	41,363	113,158	105,447	70,594	325,050	9,052	84,514	108,410	147,142	52,339	79,119	99,216	186,249	3,915	43,899
	QCEW	94,750	30,702	15,974	878,331	359,103	628,682	1,048,989	247,011	141,685	488,495	364,528	992,124	51,049	462,150	123,594	732,925	113,870	606,756	1,186,086	71,894	198,588	–
2009	Revelio	19,258	6,547	9,542	99,373	221,766	87,032	199,818	44,414	128,291	113,954	76,931	367,964	9,709	94,013	120,068	160,943	58,734	88,327	109,822	203,631	4,356	48,086
	QCEW	94,436	31,118	16,173	828,247	350,493	617,849	1,030,743	242,310	141,235	475,406	350,994	998,010	51,571	458,640	124,386	749,426	113,259	609,263	1,220,570	72,398	175,743	–
2010	Revelio	22,079	7,216	10,779	107,729	240,525	94,873	226,008	48,803	145,453	123,751	84,592	416,288	10,777	105,303	133,636	176,923	66,078	100,160	122,308	223,869	4,797	53,341
	QCEW	94,484	31,487	16,307	788,985	341,833	613,877	1,022,957	239,563	139,197	464,732	342,002	1,008,602	52,069	462,115	127,924	767,846	112,685	616,964	1,249,369	72,494	171,006	–
2011	Revelio	24,177	7,775	11,514	114,079	254,305	100,488	246,681	52,342	158,519	130,391	90,097	448,913	11,544	113,355	144,418	190,429	71,589	111,138	131,913	239,100	5,251	57,681
	QCEW	95,181	32,529	16,329	759,549	336,372	613,845	1,020,233	242,355	138,125	461,203	336,508	1,025,810	53,676	466,221	130,582	783,212	113,116	627,410	1,309,746	72,130	180,628	–
2012	Revelio	26,485	8,353	12,138	121,595	269,866	106,479	269,947	56,740	172,647	137,760	96,382	483,718	12,371	122,042	156,348	204,809	77,809	123,769	142,806	254,962	5,743	62,656
	QCEW	96,361	33,965	16,557	743,466	334,039	615,836	1,024,770	245,916	138,532	463,013	337,644	1,049,214	56,146	475,330	133,940	800,069	114,942	639,461	1,285,436	71,539	188,918	–
2013	Revelio	28,632	8,816	12,694	129,166	284,575	112,023	292,092	60,966	186,202	145,234	102,990	518,127	13,307	130,974	167,923	218,705	83,614	136,114	153,141	270,577	6,227	67,415
	QCEW	97,810	34,942	16,760	739,923	334,005	619,147	1,031,955	248,826	139,283	463,993	344,225	1,074,459	58,506	484,061	137,104	1,273,293	117,330	647,580	788,031	70,576	183,608	–
2014	Revelio	31,133	9,372	13,467	138,023	301,992	118,438	316,999	66,016	201,759	154,098	111,198	558,386	14,183	141,076	181,662	233,829	89,992	150,111	164,673	289,363	6,756	72,620
	QCEW	99,711	35,896	16,892	749,251	337,039	620,881	1,036,609	253,038	144,933	470,209	352,619	1,105,203	60,115	493,419	139,671	1,302,469	120,949	656,934	805,422	70,728	188,060	–
2015	Revelio	34,046	9,794	14,317	147,075	319,945	125,018	344,514	71,194	218,072	164,051	120,019	601,897	15,280	151,855	196,482	250,555	96,875	165,316	177,531	309,187	7,312	78,178
	QCEW	101,480	35,826	17,031	758,455	339,872	619,387	1,037,343	258,683	146,282	474,306	360,022	1,138,169	61,688	507,961	144,867	1,341,973	123,942	666,585	817,922	71,077	194,230	–
2016	Revelio	36,546	10,124	15,107	155,558	335,842	130,920	371,022	75,844	233,876	173,580	128,556	644,636	16,152	162,477	211,044	266,209	103,219	179,542	190,439	328,345	7,879	84,211
	QCEW	102,973	34,169	17,148	771,310	342,909	615,539	1,039,759	263,038	150,725	478,353	369,245	1,155,618	63,231	521,117	147,728	1,401,074	126,611	678,164	827,358	71,016	230,401	–
2017	Revelio	39,352	10,586	15,926	164,149	351,875	136,782	399,641	81,010	250,279	184,491	138,040	690,373	17,004	173,759	227,053	283,068	109,608	193,228	204,216	348,129	8,351	91,157
	QCEW	104,003	32,464	17,321	782,720	345,930	612,334	1,038,669	265,752	156,444	482,572	378,585	1,173,332	64,772	528,655	150,376	1,452,487	129,814	694,573	834,706	70,580	208,895	–
2018	Revelio	42,651	11,116	16,997	173,324	370,490	143,201	430,620	87,320	268,904	196,643	148,053	744,462	18,108	187,253	246,416	301,095	116,543	207,715	219,403	371,118	8,782	99,458
	QCEW	105,520	32,464	17,694	806,981	350,659	615,153	1,040,887	274,460	166,607	489,787	396,285	1,217,498	66,941	545,611	154,720	1,508,053	136,543	709,233	850,946	71,169	146,218	–
2019	Revelio	45,664	11,493	18,114	181,235	387,077	148,853	458,602	92,986	286,241	209,039	157,900	796,903	19,152	200,060	265,911	319,543	122,692	220,522	233,723	395,128	9,165	107,769
	QCEW	106,500	32,610	17,948	828,425	354,230	615,309	1,040,219	285,012	177,863	495,938	409,866	1,255,858	69,668	562,787	158,392	1,565,040	142,652	722,367	855,660	71,776	155,111	–
2020	Revelio	48,308	11,791	19,227	188,233	401,540	153,439	484,981	98,184	307,778	222,635	167,536	854,408	20,107	212,912	286,048	337,410	127,449	225,083	247,670	417,916	9,397	116,125
	QCEW	108,262	31,866	18,408	844,774	357,417	614,286	1,039,293	294,031	192,020	510,396	425,033	1,308,176	73,047	578,811	162,830	1,667,037	146,977	724,289	817,827	72,232	179,495	–
2021	Revelio	50,288	12,066	20,411	194,354	412,945	156,393	504,182	103,501	326,168	235,158	175,990	899,130	20,973	223,262	302,636	352,321	132,429	232,374	258,163	434,997	9,586	126,778
	QCEW	109,728	31,093	19,168	869,257	365,513	618,141	1,047,926	312,263	218,801	534,342	443,343	1,413,855	79,750	608,788	169,458	1,716,099	151,608	735,639	835,005	72,295	234,337	–
2022	Revelio	52,145	12,370	21,366	199,484	422,262	158,818	521,146	108,050	343,081	245,021	183,247	942,179	21,696	234,468	318,717	365,355	136,867	239,215	266,771	448,443	9,752	136,423
	QCEW	111,707	30,978	20,292	900,420	379,839	634,699	1,055,007	333,813	257,553	559,303	466,119	1,558,055	87,478	650,577	178,702	1,773,095	160,705	757,863	861,187	72,391	345,682	–
2023	Revelio	53,984	12,584	22,398	203,226	430,286	161,187	538,559	111,418	359,895	253,308	189,483	989,574	22,543	246,120	335,085	377,470	140,278	241,083	273,763	457,287	9,778	147,621
	QCEW	113,506	30,933	21,415	925,221	391,322	638,779	1,068,227	345,839	279,935	572,931	481,593	1,658,686	94,657	676,793	185,983	1,820,164	168,116	776,697	875,283	72,837	336,977	–
2024	Revelio	55,626	12,772	23,015	205,174	436,655	163,324	554,932	113,183	373,634	259,682	193,604	1,028,960	23,426	254,960	347,378	387,237	142,094	238,522	279,099	460,072	9,703	151,719
	QCEW	113,723	30,738	22,077	936,179	396,248	637,175	1,066,813	343,787	288,795	579,030	485,770	1,715,086	100,043	683,561	190,194	1,943,893	173,163	788,069	878,983	72,907	274,636	–
2025	Revelio	56,854	12,909	23,125	204,545	437,701	164,332	549,022	113,559	383,688	262,810	195,218	1,055,559	24,414	260,606	355,713	388,908	141,674	230,506	278,841	460,343	9,508	140,589
	QCEW	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
All	Revelio	36,814	9,859	15,583	152,891	330,554	127,233	370,967	77,086	239,847	179,654	131,269	665,884	16,214	167,052	221,142	268,152	100,842	170,095	191,697	329,917	7,358	90,796
	QCEW	102,519	32,400	17,741	822,168	354,309	620,977	1,040,996	274,568	175,491	497,441	389,625	1,211,856	66,287	534,433	148,969	1,295,071	132,177	680,843	969,303	71,743	210,237	–

Notes: This table reports the number of firms in Revelio and the number of establishments in the QCEW by year and two-digit NAICS sector. The "All" row reports average annual counts over the available years. Revelio statistics use individual position records with valid end dates. *Sources:* Revelio Labs individual position data; BLS, Quarterly Census of Employment and Wages (QCEW).

Table 11: Number of Workers by Year and Sector

Year	Source	11 Agric.	21 Mining	22 Utilities	23 Const.	31-33 Mfg.	42 Wholesale	44-45 Retail	48-49 Transp.	51 Info.	52 Finance	53 Real Est.	54 Prof. Svcs	55 Mgmt	56 Admin	61 Educ.	62 Health	71 Arts	72 Hosp.	81 Other	92 Pub. Admin	99 Unclass.	Missing Missing
2007	Revelio	59,269	213,618	185,995	542,960	4,369,894	693,058	2,021,785	616,087	2,514,070	3,098,836	638,436	3,355,802	30,926	791,443	2,254,378	1,907,212	459,646	876,697	902,890	2,790,705	19,877	8,077,434
	QCEW	1,165,065	660,278	564,101	7,470,978	13,801,244	5,987,142	15,446,908	5,038,523	3,018,755	5,965,017	2,111,695	7,536,567	1,839,616	8,315,439	8,870,932	13,945,929	1,787,026	11,310,142	4,389,416	4,444,189	216,721	–
2008	Revelio	68,266	244,345	213,229	587,838	4,657,038	758,735	2,269,640	667,299	2,729,526	3,280,733	692,982	3,693,505	35,101	890,216	2,662,457	2,223,476	539,364	1,012,239	1,055,595	3,104,190	23,223	8,988,557
	QCEW	1,166,399	713,399	574,974	7,035,736	13,349,786	5,954,822	15,244,640	5,000,400	2,973,997	5,829,985	2,070,279	7,720,772	1,895,417	7,923,164	8,979,959	14,354,052	1,811,185	11,349,733	4,434,791	4,515,167	208,318	–
2009	Revelio	76,320	253,715	228,462	601,884	4,684,464	789,935	2,461,639	680,870	2,820,297	3,328,150	726,304	3,840,229	38,168	953,312	3,063,072	2,501,425	607,817	1,113,294	1,178,280	3,392,089	25,959	9,772,801
	QCEW	1,140,124	641,021	578,042	5,866,074	11,778,517	5,561,625	14,479,860	4,668,250	2,793,101	5,590,122	1,929,876	7,379,469	1,855,139	7,084,602	8,986,657	14,645,784	1,761,776	11,007,713	4,321,928	4,522,592	173,694	–
2010	Revelio	88,967	284,799	252,317	649,073	5,007,680	868,268	2,826,663	740,724	3,104,376	3,577,662	798,188	4,223,680	43,797	1,091,889	3,575,839	2,890,049	709,204	1,294,931	1,358,067	3,744,645	30,982	10,845,514
	QCEW	1,144,585	651,364	569,343	5,409,062	11,455,600	5,466,213	14,417,382	4,575,633	2,683,163	5,455,488	1,874,874	7,355,400	1,854,778	7,331,721	8,935,733	14,908,395	1,748,010	11,029,398	4,299,245	4,529,882	152,464	–
2011	Revelio	101,428	329,882	275,530	715,074	5,447,677	964,507	3,231,500	822,988	3,401,769	3,859,873	872,853	4,636,575	48,730	1,245,273	4,055,756	3,293,838	810,549	1,500,339	1,527,005	3,984,903	36,282	11,790,746
	QCEW	1,158,995	729,776	565,878	5,393,207	11,668,416	5,545,553	14,602,592	4,656,840	2,648,137	5,475,771	1,869,031	7,568,658	1,914,543	7,645,388	8,814,066	15,189,532	1,765,456	11,297,213	4,361,236	4,407,927	173,376	–
2012	Revelio	115,080	372,953	296,406	799,763	5,862,604	1,065,399	3,642,960	911,306	3,714,665	4,160,095	962,988	5,058,217	54,402	1,410,432	4,518,111	3,720,844	927,338	1,731,137	1,718,061	4,255,088	41,898	12,816,599
	QCEW	1,188,505	797,846	565,283	5,506,227	11,872,709	5,656,442	14,801,178	4,741,656	2,643,737	5,533,414	1,902,838	7,795,401	2,003,075	7,920,502	8,751,331	15,486,986	1,814,723	11,685,068	4,513,473	4,375,428	171,660	–
2013	Revelio	127,582	404,689	312,141	895,348	6,231,600	1,154,599	4,008,676	996,273	4,016,806	4,437,999	1,074,168	5,448,302	59,225	1,554,109	4,909,798	4,117,106	1,040,070	1,939,683	1,872,174	4,490,368	47,350	13,617,283
	QCEW	1,208,781	813,087	563,840	5,739,777	11,963,258	5,738,845	15,012,242	4,813,868	2,664,089	5,594,825	1,951,372	8,020,732	2,087,081	8,202,508	8,778,877	16,242,082	1,863,739	12,081,654	4,114,555	4,347,157	165,000	–
2014	Revelio	141,832	437,657	332,806	1,004,890	6,651,997	1,253,248	4,360,468	1,110,159	4,370,818	4,729,553	1,205,418	5,888,677	64,771	1,711,422	5,295,319	4,532,786	1,156,574	2,139,282	2,043,770	4,754,959	52,999	14,367,308
	QCEW	1,228,195	841,689	565,436	6,053,318	12,124,876	5,815,746	15,283,710	4,955,722	2,686,625	5,604,138	2,001,380	8,248,497	2,154,136	8,503,186	8,806,861	16,567,260	1,924,513	12,447,720	4,199,876	4,345,431	214,677	–
2015	Revelio	156,021	422,876	355,994	1,113,324	7,063,592	1,343,504	4,678,951	1,229,305	4,762,433	5,055,875	1,342,892	6,338,888	71,507	1,862,839	5,665,731	5,014,707	1,275,645	2,339,710	2,215,352	5,030,875	59,141	15,043,331
	QCEW	1,246,155	751,894	571,723	6,340,070	12,259,494	5,874,045	15,582,161	5,167,352	2,701,773	5,703,281	2,053,504	8,522,590	2,197,652	8,718,059	8,892,883	17,004,593	1,987,762	12,856,204	4,273,143	4,352,479	239,962	–
2016	Revelio	171,296	383,092	372,633	1,210,381	7,360,802	1,410,452	4,939,558	1,337,787	5,100,116	5,339,190	1,471,388	6,710,094	78,456	1,993,221	6,019,887	5,461,811	1,392,963	2,541,489	2,392,092	5,305,539	66,044	15,701,215
	QCEW	1,257,935	613,374	571,180	6,600,554	12,264,621	5,859,324	15,765,252	5,342,733	2,740,898	5,793,444	2,088,864	8,739,889	2,230,131	8,883,037	8,936,930	17,488,144	2,061,946	13,232,468	4,350,424	4,401,656	271,704	–
2017	Revelio	188,686	387,907	393,109	1,328,130	7,726,585	1,487,235	5,223,653	1,466,082	5,455,051	5,640,626	1,618,613	7,122,049	87,009	2,137,763	6,438,302	5,908,325	1,511,314	2,750,283	2,554,870	5,580,748	73,143	16,427,799
	QCEW	1,260,212	623,706	570,695	6,860,090	12,374,561	5,898,327	15,796,495	5,528,229	2,733,504	5,875,237	2,141,670	8,893,889	2,278,042	8,993,570	8,975,323	17,902,264	2,117,797	13,522,526	4,397,367	4,406,228	219,504	–
2018	Revelio	210,655	413,924	415,715	1,463,330	8,199,255	1,575,555	5,500,539	1,619,017	5,867,863	5,970,763	1,779,174	7,637,117	100,222	2,308,964	6,925,113	6,391,789	1,637,857	2,963,632	2,756,826	5,896,416	80,659	17,153,701
	QCEW	1,262,394	673,302	569,231	7,137,234	12,615,096	5,855,321	15,731,946	5,781,483	2,749,662	5,921,583	2,201,212	9,185,159	2,347,219	9,162,003	9,039,554	18,323,260	2,181,036	13,732,954	4,464,268	4,454,650	127,754	–
2019	Revelio	233,764	423,578	436,450	1,579,312	8,548,534	1,637,742	5,699,833	1,745,847	6,288,512	6,273,804	1,926,494	8,129,512	108,395	2,456,349	7,369,161	6,855,883	1,734,522	3,097,761	2,930,148	6,208,643	86,779	17,657,232
	QCEW	1,262,034	674,835	565,841	7,390,264	12,742,690	5,883,667	15,545,740	6,063,396	2,779,183	5,995,755	2,259,847	9,437,601	2,407,183	9,208,977	9,101,102	18,742,396	2,241,807	13,922,308	4,516,136	4,492,186	130,762	–
2020	Revelio	245,392	396,679	442,245	1,635,066	8,534,843	1,630,972	5,833,403	1,812,188	6,452,933	6,409,856	2,001,329	8,301,923	106,651	2,507,628	7,361,078	7,076,971	1,610,314	2,938,781	2,984,815	6,328,966	85,166	17,998,189
	QCEW	1,240,048	540,312	560,717	7,142,252	12,050,345	5,620,210	14,701,633	6,137,336	2,638,060	6,013,102	2,130,057	9,353,431	2,318,008	8,475,759	8,597,856	18,121,463	1,594,692	10,984,979	3,907,965	4,460,959	137,503	–
2021	Revelio	265,477	374,468	468,950	1,734,196	8,949,861	1,687,395	6,095,727	1,911,169	7,096,596	6,846,418	2,131,156	8,983,746	106,615	2,677,798	7,739,262	7,576,139	1,692,784	3,053,235	3,137,754	6,614,253	92,882	18,503,192
	QCEW	1,245,817	512,935	558,681	7,359,130	12,268,879	5,696,637	15,286,515	6,554,056	2,766,823	6,115,957	2,190,486	9,793,100	2,343,063	8,985,570	8,657,168	18,448,483	1,831,183	12,068,921	4,112,963	4,424,788	193,340	–
2022	Revelio	277,051	382,196	500,192	1,838,163	9,316,300	1,741,005	6,079,505	2,009,328	7,597,749	7,142,754	2,262,899	9,525,890	108,629	2,828,121	8,232,355	7,911,241	1,835,744	3,147,660	3,298,108	6,844,323	99,126	19,008,467
	QCEW	1,256,452	555,955	565,536	7,623,199	12,724,809	5,976,699	15,425,489	6,995,217	3,012,318	6,224,488	2,314,209	10,432,594	2,503,466	9,461,397	8,917,256	18,936,325	2,143,522	13,321,329	4,370,034	4,444,010	294,731	–
2023	Revelio	277,514	385,106	516,105	1,886,378	9,302,116	1,747,836	5,889,568	2,012,024	7,411,978	7,107,350	2,310,099	9,618,358	109,773	2,853,736	8,498,382	8,057,366	1,879,032	3,097,536	3,372,524	6,968,991	98,717	19,233,699
	QCEW	1,250,147	583,984	585,230	7,887,071	12,833,958	6,101,516	15,475,648	7,019,393	2,941,470	6,257,936	2,357,858	10,625,728	2,550,472	9,266,592	9,167,033	19,812,394	2,307,915	13,864,843	4,530,861	4,548,117	274,874	–
2024	Revelio	274,481	386,043	524,847	1,921,060	9,249,796	1,746,748	5,682,017	1,982,177	7,316,256	7,136,175	2,318,276	9,695,443	110,647	2,843,216	8,629,976	8,139,888	1,874,212	2,933,602	3,423,387	7,060,606	95,235	19,286,508
	QCEW	1,252,422	586,236	599,639	8,070,019	12,698,765	6,106,539	15,429,135	7,138,441	2,869,663	6,254,839	2,373,471	10,667,147	2,613,063	9,043,581	9,334,889	20,694,655	2,414,374	13,993,845	4,630,915	4,691,582	215,661	–
2025	Revelio	263,944	376,335	525,943	1,908,079	9,011,927	1,703,574	5,315,574	1,920,336	7,177,050	7,111,566	2,284,255	9,558,253	109,905	2,765,176	8,340,899	7,958,543	1,775,656	2,650,076	3,323,507	6,875,698	87,097	18,939,504
	QCEW	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
All	Revelio	175,949	361,782	371,004	1,232,329	7,167,188	1,329,461	4,513,772	1,346,893	5,115,730	5,289,857	1,495,680	6,724,540	77,523	1,941,206	5,871,309	5,344,179	1,287,927	2,269,546	2,318,170	5,222,737	63,293	15,012,057
	QCEW	1,218,570	664,722	570,298	6,715,792	12,380,424	5,811,037	15,223,807	5,565,474	2,780,275	5,844,688	2,101,251	8,737,590	2,188,449	8,506,948	8,919,134	17,045,222	1,964,359	12,428,279	4,343,811	4,453,579	198,984	–

Notes: This table reports the number of workers in Revelio and employment in the QCEW by year and two-digit NAICS sector. The "All" row reports average annual counts over the available years. Revelio statistics use individual position records with valid end dates. *Sources:* Revelio Labs individual position data; BLS, Quarterly Census of Employment and Wages (QCEW).

Table 12: Distribution of Firms and Establishments by Year and Sector

Year	Source	11 Agric.	21 Mining	22 Utilities	23 Const.	31-33 Mfg.	42 Wholesale	44-45 Retail	48-49 Transp.	51 Info.	52 Finance	53 Real Est.	54 Prof. Svcs	55 Mgmt	56 Admin	61 Educ.	62 Health	71 Arts	72 Hosp.	81 Other	92 Pub. Admin	99 Unclass.	Missing Missing
2007	Revelio	0.8%	0.3%	0.4%	4.6%	10.4%	4.0%	8.6%	2.0%	5.4%	5.2%	3.4%	15.4%	0.4%	4.0%	5.2%	7.2%	2.5%	3.8%	4.8%	9.2%	0.2%	2.1%
	QCEW	1.1%	0.3%	0.2%	10.2%	4.1%	7.2%	12.0%	2.8%	1.6%	5.6%	4.2%	11.1%	0.6%	5.2%	1.4%	8.2%	1.3%	6.8%	13.1%	0.8%	2.2%	–
2008	Revelio	0.8%	0.3%	0.4%	4.5%	10.1%	3.9%	8.7%	2.0%	5.5%	5.1%	3.4%	15.8%	0.4%	4.1%	5.3%	7.1%	2.5%	3.8%	4.8%	9.0%	0.2%	2.1%
	QCEW	1.1%	0.3%	0.2%	9.9%	4.1%	7.1%	11.9%	2.8%	1.6%	5.5%	4.1%	11.2%	0.6%	5.2%	1.4%	8.3%	1.3%	6.9%	13.4%	0.8%	2.2%	–
2009	Revelio	0.8%	0.3%	0.4%	4.4%	9.8%	3.8%	8.8%	2.0%	5.6%	5.0%	3.4%	16.2%	0.4%	4.1%	5.3%	7.1%	2.6%	3.9%	4.8%	9.0%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	9.5%	4.0%	7.1%	11.8%	2.8%	1.6%	5.4%	4.0%	11.4%	0.6%	5.2%	1.4%	8.6%	1.3%	7.0%	13.9%	0.8%	2.0%	–
2010	Revelio	0.9%	0.3%	0.4%	4.3%	9.5%	3.8%	8.9%	1.9%	5.8%	4.9%	3.3%	16.5%	0.4%	4.2%	5.3%	7.0%	2.6%	4.0%	4.8%	8.9%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	9.0%	3.9%	7.0%	11.7%	2.7%	1.6%	5.3%	3.9%	11.5%	0.6%	5.3%	1.5%	8.8%	1.3%	7.1%	14.3%	0.8%	2.0%	–
2011	Revelio	0.9%	0.3%	0.4%	4.2%	9.4%	3.7%	9.1%	1.9%	5.8%	4.8%	3.3%	16.5%	0.4%	4.2%	5.3%	7.0%	2.6%	4.1%	4.9%	8.8%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	8.6%	3.8%	7.0%	11.6%	2.7%	1.6%	5.2%	3.8%	11.6%	0.6%	5.3%	1.5%	8.9%	1.3%	7.1%	14.9%	0.8%	2.0%	–
2012	Revelio	0.9%	0.3%	0.4%	4.2%	9.2%	3.6%	9.2%	1.9%	5.9%	4.7%	3.3%	16.5%	0.4%	4.2%	5.3%	7.0%	2.7%	4.2%	4.9%	8.7%	0.2%	2.1%
	QCEW	1.1%	0.4%	0.2%	8.4%	3.8%	6.9%	11.6%	2.8%	1.6%	5.2%	3.8%	11.8%	0.6%	5.4%	1.5%	9.0%	1.3%	7.2%	14.5%	0.8%	2.1%	–
2013	Revelio	0.9%	0.3%	0.4%	4.1%	9.1%	3.6%	9.3%	1.9%	5.9%	4.6%	3.3%	16.6%	0.4%	4.2%	5.4%	7.0%	2.7%	4.3%	4.9%	8.6%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.3%	3.8%	7.0%	11.6%	2.8%	1.6%	5.2%	3.9%	12.1%	0.7%	5.4%	1.5%	14.3%	1.3%	7.3%	8.8%	0.8%	2.1%	–
2014	Revelio	0.9%	0.3%	0.4%	4.1%	9.0%	3.5%	9.4%	2.0%	6.0%	4.6%	3.3%	16.6%	0.4%	4.2%	5.4%	6.9%	2.7%	4.5%	4.9%	8.6%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.3%	3.7%	6.9%	11.4%	2.8%	1.6%	5.2%	3.9%	12.2%	0.7%	5.4%	1.5%	14.4%	1.3%	7.3%	8.9%	0.8%	2.1%	–
2015	Revelio	0.9%	0.3%	0.4%	4.1%	8.8%	3.5%	9.5%	2.0%	6.0%	4.5%	3.3%	16.6%	0.4%	4.2%	5.4%	6.9%	2.7%	4.6%	4.9%	8.5%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.2%	3.7%	6.7%	11.3%	2.8%	1.6%	5.1%	3.9%	12.3%	0.7%	5.5%	1.6%	14.6%	1.3%	7.2%	8.9%	0.8%	2.1%	–
2016	Revelio	0.9%	0.3%	0.4%	4.0%	8.7%	3.4%	9.6%	2.0%	6.1%	4.5%	3.3%	16.7%	0.4%	4.2%	5.5%	6.9%	2.7%	4.6%	4.9%	8.5%	0.2%	2.2%
	QCEW	1.1%	0.4%	0.2%	8.2%	3.6%	6.5%	11.1%	2.8%	1.6%	5.1%	3.9%	12.3%	0.7%	5.5%	1.6%	14.9%	1.3%	7.2%	8.8%	0.8%	2.4%	–
2017	Revelio	1.0%	0.3%	0.4%	4.0%	8.5%	3.3%	9.7%	2.0%	6.1%	4.5%	3.4%	16.8%	0.4%	4.2%	5.5%	6.9%	2.7%	4.7%	5.0%	8.5%	0.2%	2.2%
	QCEW	1.1%	0.3%	0.2%	8.2%	3.6%	6.4%	10.9%	2.8%	1.6%	5.1%	4.0%	12.3%	0.7%	5.6%	1.6%	15.2%	1.4%	7.3%	8.8%	0.7%	2.2%	–
2018	Revelio	1.0%	0.3%	0.4%	3.9%	8.4%	3.2%	9.8%	2.0%	6.1%	4.5%	3.4%	16.9%	0.4%	4.2%	5.6%	6.8%	2.6%	4.7%	5.0%	8.4%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.3%	3.6%	6.3%	10.7%	2.8%	1.7%	5.0%	4.1%	12.5%	0.7%	5.6%	1.6%	15.5%	1.4%	7.3%	8.8%	0.7%	1.5%	–
2019	Revelio	1.0%	0.2%	0.4%	3.9%	8.3%	3.2%	9.8%	2.0%	6.1%	4.5%	3.4%	17.0%	0.4%	4.3%	5.7%	6.8%	2.6%	4.7%	5.0%	8.4%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.3%	3.6%	6.2%	10.5%	2.9%	1.8%	5.0%	4.1%	12.7%	0.7%	5.7%	1.6%	15.8%	1.4%	7.3%	8.6%	0.7%	1.6%	–
2020	Revelio	1.0%	0.2%	0.4%	3.8%	8.1%	3.1%	9.8%	2.0%	6.2%	4.5%	3.4%	17.2%	0.4%	4.3%	5.8%	6.8%	2.6%	4.5%	5.0%	8.4%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.3%	3.5%	6.0%	10.2%	2.9%	1.9%	5.0%	4.2%	12.9%	0.7%	5.7%	1.6%	16.4%	1.4%	7.1%	8.0%	0.7%	1.8%	–
2021	Revelio	1.0%	0.2%	0.4%	3.7%	8.0%	3.0%	9.7%	2.0%	6.3%	4.5%	3.4%	17.3%	0.4%	4.3%	5.8%	6.8%	2.6%	4.5%	5.0%	8.4%	0.2%	2.4%
	QCEW	1.0%	0.3%	0.2%	8.2%	3.5%	5.8%	9.9%	2.9%	2.1%	5.0%	4.2%	13.4%	0.8%	5.8%	1.6%	16.2%	1.4%	6.9%	7.9%	0.7%	2.2%	–
2022	Revelio	1.0%	0.2%	0.4%	3.7%	7.8%	2.9%	9.7%	2.0%	6.4%	4.5%	3.4%	17.5%	0.4%	4.4%	5.9%	6.8%	2.5%	4.4%	5.0%	8.3%	0.2%	2.5%
	QCEW	1.0%	0.3%	0.2%	8.0%	3.4%	5.7%	9.4%	3.0%	2.3%	5.0%	4.2%	13.9%	0.8%	5.8%	1.6%	15.8%	1.4%	6.8%	7.7%	0.6%	3.1%	–
2023	Revelio	1.0%	0.2%	0.4%	3.6%	7.7%	2.9%	9.7%	2.0%	6.5%	4.5%	3.4%	17.7%	0.4%	4.4%	6.0%	6.8%	2.5%	4.3%	4.9%	8.2%	0.2%	2.6%
	QCEW	1.0%	0.3%	0.2%	8.0%	3.4%	5.5%	9.3%	3.0%	2.4%	5.0%	4.2%	14.4%	0.8%	5.9%	1.6%	15.8%	1.5%	6.7%	7.6%	0.6%	2.9%	–
2024	Revelio	1.0%	0.2%	0.4%	3.6%	7.6%	2.9%	9.7%	2.0%	6.5%	4.5%	3.4%	18.0%	0.4%	4.5%	6.1%	6.8%	2.5%	4.2%	4.9%	8.1%	0.2%	2.7%
	QCEW	1.0%	0.3%	0.2%	8.0%	3.4%	5.4%	9.1%	2.9%	2.5%	4.9%	4.1%	14.6%	0.9%	5.8%	1.6%	16.6%	1.5%	6.7%	7.5%	0.6%	2.3%	–
2025	Revelio	1.0%	0.2%	0.4%	3.6%	7.6%	2.9%	9.5%	2.0%	6.7%	4.6%	3.4%	18.4%	0.4%	4.5%	6.2%	6.8%	2.5%	4.0%	4.8%	8.0%	0.2%	2.4%
	QCEW	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
All	Revelio	0.9%	0.3%	0.4%	4.0%	8.7%	3.4%	9.4%	2.0%	6.0%	4.7%	3.4%	16.9%	0.4%	4.2%	5.6%	6.9%	2.6%	4.3%	4.9%	8.6%	0.2%	2.3%
	QCEW	1.1%	0.3%	0.2%	8.6%	3.7%	6.5%	10.9%	2.8%	1.8%	5.2%	4.0%	12.5%	0.7%	5.5%	1.5%	13.2%	1.4%	7.1%	10.2%	0.8%	2.2%	–

Notes: This table reports the percentage distribution of firms in Revelio and establishments in the QCEW across two-digit NAICS sectors in each year. The "All" row reports average annual shares over the available years. Revelio statistics use individual position records with valid end dates. *Sources:* Revelio Labs individual position data; BLS, Quarterly Census of Employment and Wages (QCEW).

Table 13: Distribution of Workers by Year and Sector

Year	Source	11 Agric.	21 Mining	22 Utilities	23 Const.	31-33 Mfg.	42 Wholesale	44-45 Retail	48-49 Transp.	51 Info.	52 Finance	53 Real Est.	54 Prof. Svcs	55 Mgmt	56 Admin	61 Educ.	62 Health	71 Arts	72 Hosp.	81 Other	92 Pub. Admin	99 Unclass.	Missing Missing
2007	Revelio	0.2%	0.6%	0.5%	1.5%	12.0%	1.9%	5.6%	1.7%	6.9%	8.5%	1.8%	9.2%	0.1%	2.2%	6.2%	5.2%	1.3%	2.4%	2.5%	7.7%	0.1%	22.2%
	QCEW	0.9%	0.5%	0.5%	6.0%	11.1%	4.8%	12.5%	4.1%	2.4%	4.8%	1.7%	6.1%	1.5%	6.7%	7.2%	11.3%	1.4%	9.1%	3.5%	3.6%	0.2%	–
2008	Revelio	0.2%	0.6%	0.5%	1.5%	11.5%	1.9%	5.6%	1.7%	6.8%	8.1%	1.7%	9.1%	0.1%	2.2%	6.6%	5.5%	1.3%	2.5%	2.6%	7.7%	0.1%	22.3%
	QCEW	0.9%	0.6%	0.5%	5.7%	10.8%	4.8%	12.4%	4.1%	2.4%	4.7%	1.7%	6.3%	1.5%	6.4%	7.3%	11.7%	1.5%	9.2%	3.6%	3.7%	0.2%	–
2009	Revelio	0.2%	0.6%	0.5%	1.4%	10.9%	1.8%	5.7%	1.6%	6.5%	7.7%	1.7%	8.9%	0.1%	2.2%	7.1%	5.8%	1.4%	2.6%	2.7%	7.9%	0.1%	22.7%
	QCEW	1.0%	0.5%	0.5%	5.0%	10.1%	4.8%	12.4%	4.0%	2.4%	4.8%	1.7%	6.3%	1.6%	6.1%	7.7%	12.5%	1.5%	9.4%	3.7%	3.9%	0.1%	–
2010	Revelio	0.2%	0.6%	0.5%	1.4%	10.4%	1.8%	5.9%	1.5%	6.5%	7.5%	1.7%	8.8%	0.1%	2.3%	7.4%	6.0%	1.5%	2.7%	2.8%	7.8%	0.1%	22.6%
	QCEW	1.0%	0.6%	0.5%	4.7%	9.9%	4.7%	12.4%	3.9%	2.3%	4.7%	1.6%	6.3%	1.6%	6.3%	7.7%	12.9%	1.5%	9.5%	3.7%	3.9%	0.1%	–
2011	Revelio	0.2%	0.6%	0.5%	1.4%	10.3%	1.8%	6.1%	1.6%	6.4%	7.3%	1.6%	8.8%	0.1%	2.4%	7.7%	6.2%	1.5%	2.8%	2.9%	7.5%	0.1%	22.3%
	QCEW	1.0%	0.6%	0.5%	4.6%	9.9%	4.7%	12.4%	4.0%	2.3%	4.7%	1.6%	6.4%	1.6%	6.5%	7.5%	12.9%	1.5%	9.6%	3.7%	3.8%	0.1%	–
2012	Revelio	0.2%	0.6%	0.5%	1.4%	10.1%	1.8%	6.3%	1.6%	6.4%	7.2%	1.7%	8.7%	0.1%	2.4%	7.8%	6.4%	1.6%	3.0%	3.0%	7.3%	0.1%	22.0%
	QCEW	1.0%	0.7%	0.5%	4.6%	9.9%	4.7%	12.4%	4.0%	2.2%	4.6%	1.6%	6.5%	1.7%	6.6%	7.3%	12.9%	1.5%	9.8%	3.8%	3.7%	0.1%	–
2013	Revelio	0.2%	0.6%	0.5%	1.4%	9.9%	1.8%	6.4%	1.6%	6.4%	7.1%	1.7%	8.7%	0.1%	2.5%	7.8%	6.6%	1.7%	3.1%	3.0%	7.2%	0.1%	21.7%
	QCEW	1.0%	0.7%	0.5%	4.7%	9.8%	4.7%	12.3%	3.9%	2.2%	4.6%	1.6%	6.6%	1.7%	6.7%	7.2%	13.3%	1.5%	9.9%	3.4%	3.6%	0.1%	–
2014	Revelio	0.2%	0.6%	0.5%	1.5%	9.8%	1.9%	6.4%	1.6%	6.5%	7.0%	1.8%	8.7%	0.1%	2.5%	7.8%	6.7%	1.7%	3.2%	3.0%	7.0%	0.1%	21.3%
	QCEW	1.0%	0.7%	0.5%	4.9%	9.7%	4.7%	12.3%	4.0%	2.2%	4.5%	1.6%	6.6%	1.7%	6.8%	7.1%	13.3%	1.5%	10.0%	3.4%	3.5%	0.2%	–
2015	Revelio	0.2%	0.6%	0.5%	1.5%	9.8%	1.9%	6.5%	1.7%	6.6%	7.0%	1.9%	8.8%	0.1%	2.6%	7.8%	6.9%	1.8%	3.2%	3.1%	6.9%	0.1%	20.8%
	QCEW	1.0%	0.6%	0.4%	5.0%	9.6%	4.6%	12.2%	4.1%	2.1%	4.5%	1.6%	6.7%	1.7%	6.8%	7.0%	13.4%	1.6%	10.1%	3.4%	3.4%	0.2%	–
2016	Revelio	0.2%	0.5%	0.5%	1.6%	9.6%	1.8%	6.4%	1.7%	6.6%	7.0%	1.9%	8.7%	0.1%	2.6%	7.8%	7.1%	1.8%	3.3%	3.1%	6.9%	0.1%	20.5%
	QCEW	1.0%	0.5%	0.4%	5.1%	9.5%	4.5%	12.2%	4.1%	2.1%	4.5%	1.6%	6.7%	1.7%	6.9%	6.9%	13.5%	1.6%	10.2%	3.4%	3.4%	0.2%	–
2017	Revelio	0.2%	0.5%	0.5%	1.6%	9.5%	1.8%	6.4%	1.8%	6.7%	6.9%	2.0%	8.7%	0.1%	2.6%	7.9%	7.2%	1.9%	3.4%	3.1%	6.8%	0.1%	20.2%
	QCEW	1.0%	0.5%	0.4%	5.2%	9.4%	4.5%	12.0%	4.2%	2.1%	4.5%	1.6%	6.8%	1.7%	6.8%	6.8%	13.6%	1.6%	10.3%	3.3%	3.4%	0.2%	–
2018	Revelio	0.2%	0.5%	0.5%	1.7%	9.4%	1.8%	6.3%	1.9%	6.8%	6.9%	2.0%	8.8%	0.1%	2.7%	8.0%	7.4%	1.9%	3.4%	3.2%	6.8%	0.1%	19.7%
	QCEW	0.9%	0.5%	0.4%	5.3%	9.4%	4.4%	11.8%	4.3%	2.1%	4.4%	1.6%	6.9%	1.8%	6.9%	6.8%	13.7%	1.6%	10.3%	3.3%	3.3%	0.1%	–
2019	Revelio	0.3%	0.5%	0.5%	1.7%	9.3%	1.8%	6.2%	1.9%	6.9%	6.9%	2.1%	8.9%	0.1%	2.7%	8.1%	7.5%	1.9%	3.4%	3.2%	6.8%	0.1%	19.3%
	QCEW	0.9%	0.5%	0.4%	5.5%	9.4%	4.3%	11.5%	4.5%	2.1%	4.4%	1.7%	7.0%	1.8%	6.8%	6.7%	13.8%	1.7%	10.3%	3.3%	3.3%	0.1%	–
2020	Revelio	0.3%	0.4%	0.5%	1.8%	9.2%	1.8%	6.3%	2.0%	7.0%	6.9%	2.2%	9.0%	0.1%	2.7%	7.9%	7.6%	1.7%	3.2%	3.2%	6.8%	0.1%	19.4%
	QCEW	1.0%	0.4%	0.4%	5.6%	9.5%	4.4%	11.6%	4.8%	2.1%	4.7%	1.7%	7.4%	1.8%	6.7%	6.8%	14.3%	1.3%	8.7%	3.1%	3.5%	0.1%	–
2021	Revelio	0.3%	0.4%	0.5%	1.8%	9.2%	1.7%	6.2%	2.0%	7.3%	7.0%	2.2%	9.2%	0.1%	2.7%	7.9%	7.8%	1.7%	3.1%	3.2%	6.8%	0.1%	18.9%
	QCEW	0.9%	0.4%	0.4%	5.6%	9.3%	4.3%	11.6%	5.0%	2.1%	4.7%	1.7%	7.5%	1.8%	6.8%	6.6%	14.0%	1.4%	9.2%	3.1%	3.4%	0.1%	–
2022	Revelio	0.3%	0.4%	0.5%	1.8%	9.1%	1.7%	6.0%	2.0%	7.4%	7.0%	2.2%	9.3%	0.1%	2.8%	8.1%	7.8%	1.8%	3.1%	3.2%	6.7%	0.1%	18.6%
	QCEW	0.9%	0.4%	0.4%	5.5%	9.3%	4.3%	11.2%	5.1%	2.2%	4.5%	1.7%	7.6%	1.8%	6.9%	6.5%	13.8%	1.6%	9.7%	3.2%	3.2%	0.2%	–
2023	Revelio	0.3%	0.4%	0.5%	1.8%	9.1%	1.7%	5.7%	2.0%	7.2%	6.9%	2.3%	9.4%	0.1%	2.8%	8.3%	7.9%	1.8%	3.0%	3.3%	6.8%	0.1%	18.7%
	QCEW	0.9%	0.4%	0.4%	5.6%	9.2%	4.4%	11.0%	5.0%	2.1%	4.5%	1.7%	7.6%	1.8%	6.6%	6.5%	14.1%	1.6%	9.9%	3.2%	3.2%	0.2%	–
2024	Revelio	0.3%	0.4%	0.5%	1.9%	9.0%	1.7%	5.5%	1.9%	7.1%	7.0%	2.3%	9.4%	0.1%	2.8%	8.4%	7.9%	1.8%	2.9%	3.3%	6.9%	0.1%	18.8%
	QCEW	0.9%	0.4%	0.4%	5.7%	9.0%	4.3%	10.9%	5.0%	2.0%	4.4%	1.7%	7.5%	1.8%	6.4%	6.6%	14.6%	1.7%	9.9%	3.3%	3.3%	0.2%	–
2025	Revelio	0.3%	0.4%	0.5%	1.9%	9.0%	1.7%	5.3%	1.9%	7.2%	7.1%	2.3%	9.6%	0.1%	2.8%	8.3%	8.0%	1.8%	2.7%	3.3%	6.9%	0.1%	18.9%
	QCEW	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
All	Revelio	0.2%	0.5%	0.5%	1.6%	9.9%	1.8%	6.0%	1.8%	6.8%	7.2%	1.9%	9.0%	0.1%	2.5%	7.7%	6.9%	1.7%	3.0%	3.0%	7.1%	0.1%	20.6%
	QCEW	1.0%	0.5%	0.4%	5.2%	9.7%	4.6%	12.0%	4.3%	2.2%	4.6%	1.6%	6.8%	1.7%	6.7%	7.0%	13.3%	1.5%	9.7%	3.4%	3.5%	0.2%	–

Notes: This table reports the percentage distribution of workers in Revelio and employment in the QCEW across two-digit NAICS sectors in each year. The "All" row reports average annual shares over the available years. Revelio statistics use individual position records with valid end dates. *Sources:* Revelio Labs individual position data; BLS, Quarterly Census of Employment and Wages (QCEW).

Table 14: Distribution of Firm Size

	Revelio, December 2025		SUSB, 2022
Firm size (workers)	Full sample (%)	Nonmissing (%)	(%)
Fewer than 10	58.33	62.51	79.17
10–49	27.14	29.09	16.96
50–99	3.85	4.13	2.09
100–499	3.12	3.34	1.46
500–999	0.41	0.44	0.16
1,000+	0.46	0.49	0.17
Missing	6.69	–	–

Notes: This table reports the distribution of firm size in the Revelio estimation sample and in the Statistics of U.S. Businesses (SUSB). Revelio firm size is recorded employment in December 2025; SUSB firm size is enterprise employment in the pay period including March 12, 2022. The full Revelio sample contains 1,300,617 firms. The SUSB distribution covers 6,395,635 employer firms. The smallest SUSB category includes firms with zero employment in the reference pay period but positive employment during the year, whereas the corresponding Revelio category includes only firms with positive employment below 10. *Sources:* Revelio Labs; U.S. Census Bureau, 2022 SUSB. https://www.census.gov/data/tables/2022/econ/susb/2022-susb-annual.html

8.1.3 Worker Flows: Comparison with JOLTS

Figure 11 compares monthly hires and separations in the Revelio estimation sample with the JOLTS private-sector series over 2007–2025. Both series are measured in thousands of workers, with Revelio plotted on the left axis and JOLTS on the right. Revelio records far fewer hires and separations than JOLTS, reflecting its incomplete coverage of the workforce and the restrictions that define the estimation sample. Despite this difference in levels, the Revelio series track the broad trends and cyclical movements in the official data. Across the 228 monthly observations, the Pearson correlation with JOLTS is 0.514 for hires and 0.535 for separations.

The Revelio series also display recurring spikes in January, because workers who do not know the exact date of a job transition tend to report it as occurring at the start of the year. Such reporting errors reduce the monthly correlations with JOLTS. To limit their influence, the main analysis uses annual rather than monthly hiring and separation rates.

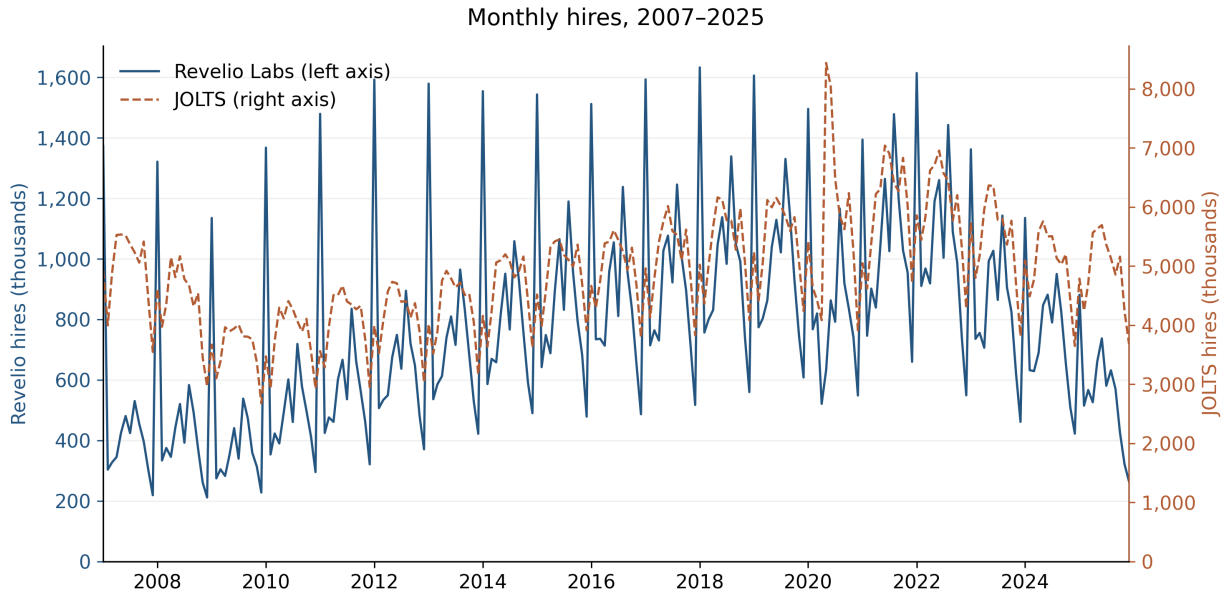
8.1.4 Firm Employment: Comparison with Compustat

To assess how accurately Revelio measures employment at the firm level, I compare firm size and net employment growth in Revelio with the corresponding measures in Compustat for firms matched across the two sources. In Compustat, firm size is total employment

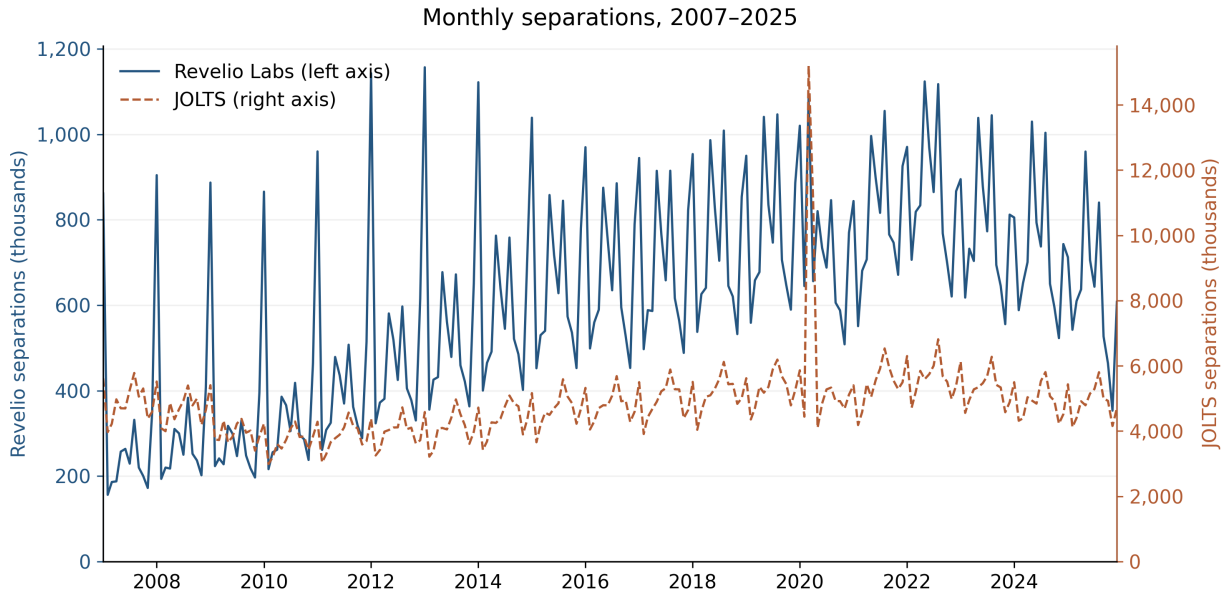
Table 15: Distribution of Firm Age

	Revelio, 2025		BDS, 2023
Firm age (years)	Full sample (%)	Nonmissing (%)	(%)
0–4	2.19	3.17	31.93
5+	66.81	96.83	68.07
5–9	10.18	14.75	–
10–19	20.54	29.77	–
20–29	13.34	19.33	–
30–49	13.08	18.95	–
50+	9.68	14.03	–
Missing	31.00	–	–
Median age (years)	20		–
Mean age (years)	28.9		–

Notes: This table reports the distribution of firm age in the Revelio estimation sample as of 2025 and in the Business Dynamics Statistics (BDS). Each firm is counted once. Revelio firm age equals 2025 minus the reported founding year; BDS firm age is based on administrative employment histories. The BDS is compared at the 0–4 and 5+ level; indented rows show the Revelio breakdown of firms aged 5 and older and are not additional categories. The median and mean are computed over Revelio firms with an observed founding year. The full Revelio sample contains 1,300,617 firms, of which 897,448 have an observed founding year; the nonmissing column excludes firms without one. BDS percentages are calculated from 5,593,727 firms across its published age categories. Firms classified by the BDS as left-censored are included in the 5+ category because they predate 1977, although their exact ages are unknown. Reference years, coverage, and age definitions differ across sources. *Sources:* Revelio Labs; U.S. Census Bureau, 2023 BDS. https://www.census.gov/data/datasets/time-series/econ/bds/bds-datasets.html



(a) Hires



(b) Separations

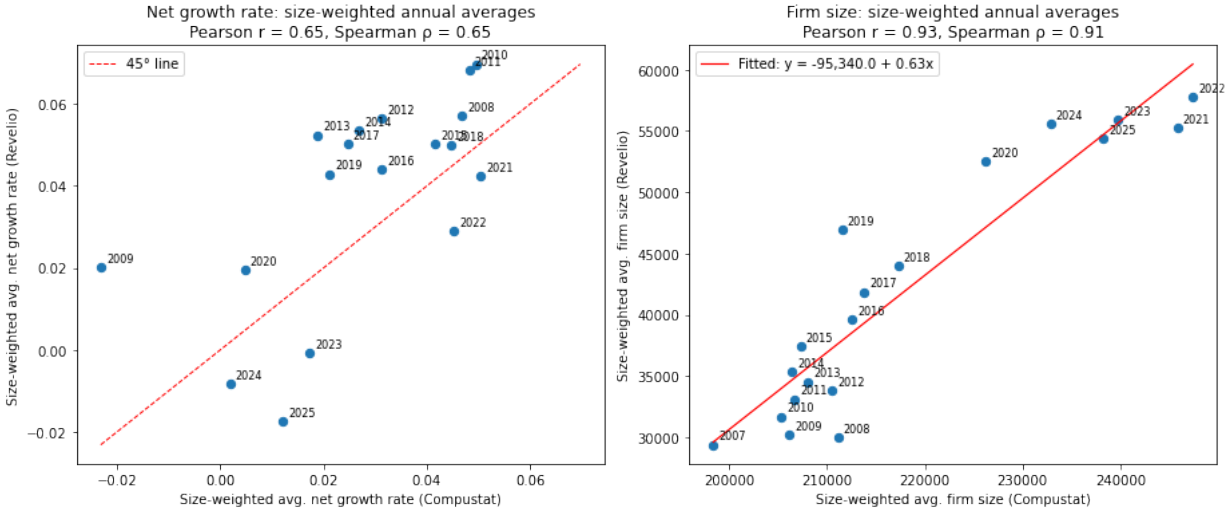
Notes: This figure compares monthly hires (panel a) and separations (panel b) in the Revelio estimation sample (left axis) with the corresponding JOLTS total private-sector series (right axis), from January 2007 through December 2025. All series are measured in thousands and are not seasonally adjusted. The JOLTS series are "Hires: Total Private" (`JTU1000HIL`) and "Total Separations: Total Private" (`JTU1000TSL`), published by the U.S. Bureau of Labor Statistics and retrieved from FRED. *Sources:* Revelio Labs; BLS, Job Openings and Labor Turnover Survey (JOLTS), via FRED. https://fred.stlouisfed.org/series/JTU1000HIL; https://fred.stlouisfed.org/series/JTU1000TSL

Figure 11: Monthly Hires and Separations: Revelio and JOLTS

as reported in firms' annual reports. In Revelio, firm size is the number of employees observed at the firm in a given year, constructed from workers' online professional profiles. In both sources, the net growth rate is the Davis–Haltiwanger symmetric growth rate, $(E_t - E_{t-1}) / [(E_t + E_{t-1})/2]$, where E_t is firm employment in year t.

Figure 12 plots annual averages of the two measures. For each year, I compute the size-weighted average of each measure across firms, weighting each source by its own measure of firm size. I restrict the sample to firm-years observed in both sources, so that the two series are averaged over the same set of firms. The two sources move closely together. For firm size, the Pearson correlation between the Revelio and Compustat annual averages is 0.93 (Spearman 0.91), and the two series rise together almost linearly over 2007–2025, with a fitted slope of 0.63. Revelio headcounts are lower than Compustat's in levels, consistent with incomplete coverage of the workforce in online profiles, but the stable relationship between the two indicates that this shortfall is systematic rather than erratic. For the net growth rate, the correlation is 0.65 (Spearman 0.65). Revelio reproduces the main cyclical movements in Compustat, including the weakness in 2009 and 2020 and the slowdown after 2022, although it shows somewhat higher growth than Compustat during the 2010s and lower growth in recent years.

The firm-year comparison is a more demanding test, because idiosyncratic measurement error no longer averages out across firms. The two sources nonetheless remain clearly related at this level. For firm size, the Pearson correlation between Revelio and Compustat is 0.67 across 86,071 firm-years (Spearman 0.68), and it is essentially unchanged at 0.68 when computed in logs, so the relationship is not driven by a small number of very large firms. The lower correlation relative to the annual averages is consistent with Revelio's coverage varying across firms, for example with the share of a firm's employees who maintain online profiles, even though this variation largely nets out in the aggregate. For the net growth rate, the correlation is 0.34 (Spearman 0.42) across 76,470 firm-years. The higher rank correlation suggests that the Pearson estimate is attenuated by a tail of extreme growth rates, as is common for firm-level growth measures, and that the two sources agree more closely on the ordering of firms by growth than on its exact magnitude. All firm-year correlations are significant at the 0.1 percent level. Overall, Revelio captures a substantial share of the variation in firm-level employment observed in Compustat, and its agreement with Compustat strengthens considerably once firm-level noise is averaged out.



Notes: This figure compares annual averages of firm-level employment measures in Revelio and Compustat. The sample consists of firms matched across the two sources over 2007–2025. For each measure, firm-years are included only if both sources report nonmissing values for that measure and for firm size. Firm size is total employment: in Compustat, as reported in firms' annual reports, and in Revelio, the number of employees observed at the firm based on workers' online professional profiles. The net growth rate is the Davis–Haltiwanger symmetric growth rate, $(E_t - E_{t-1})/\left[(E_t + E_{t-1})/2\right]$. Each point is the firm-size-weighted average across firms in a given year, with each source weighted by its own measure of firm size. Revelio averages are on the vertical axis, Compustat averages are on the horizontal axis, and points are labeled by year. In the left panel, the dashed line is the 45-degree line. In the right panel, the solid line is the OLS fit of the Revelio average on the Compustat average. Pearson and Spearman correlations are computed across years. *Sources:* Revelio Labs; S&P Global, Compustat North America; author's calculations.

Figure 12: Firm Size and Net Employment Growth: Revelio and Compustat

8.1.5 Employment Spells

Table 16 reports the distribution of the number of employment spells per worker in the estimation sample, which reflects both observed employment histories and the sample restrictions. Every worker in the sample has at least three recorded spells over 2007–2025. The mean number of spells per worker is 4.8, the median is 4, and the standard deviation is 1.92. About 90% of workers have seven or fewer spells, and 99% have 11 or fewer.

Table 16: Distribution of Employment Spells per Worker

Number of spells	Share (%)	Cumulative (%)
3	31.19	31.19
4	23.43	54.62
5	16.79	71.41
6	11.46	82.87
7	7.37	90.24
8	4.46	94.71
9	2.56	97.26
10	1.37	98.63
11	0.70	99.33
12	0.34	99.67
13	0.16	99.84
14	0.08	99.91
15	0.04	99.95
16	0.02	99.97
17	0.01	99.98
18	0.01	99.98
19	0.00	99.99
20	0.01	100.00

Notes: This table reports the distribution of the number of employment spells per worker in the Revelio estimation sample over 2007–2025. A share of 0.00 indicates a positive share below 0.005%. *Source:* Revelio Labs.

Table 17 reports the distribution of employment spell durations. About 18% of spells are right-censored, meaning they are still ongoing at the end of the observation window, so their completed durations are unknown. Among completed spells, the mean duration is 25.8 months and the median is 17 months, and 64.4% last two years or less.

8.2 Classification of job transition types

Internal transfers. I first check whether a subsequent position at the same employer begins within 31 days of the end date of the current position. If so, I classify the transition as

Table 17: Distribution of Employment Spell Duration

Duration (months)	All spells (%)	Completed spells (%)
$[0, 6]$	17.14	20.84
$(6, 12]$	14.88	18.16
$(12, 24]$	20.83	25.43
$(24, 36]$	11.83	14.44
$(36, 60]$	9.99	12.20
$(60, 120]$	5.84	7.13
> 120	1.47	1.80
Ongoing (right-censored)	18.02	–

Notes: This table reports the distribution of employment spell durations in the Revelio estimation sample. The first column includes all spells, with right-censored spells, those still ongoing at the end of the observation window, shown in a separate row. The second column restricts to completed spells. Durations of right-censored spells are not imputed. *Source:* Revelio Labs.

one of three internal transfer types based on the seniority level of the next position relative to the current one. A *downgrade* captures a move to a lower-seniority position within the same firm, signaling a deterioration in match quality that nonetheless remains above the worker's reservation threshold. A *lateral transfer* denotes a move to a different position at the same seniority level and carries no direct signal about the direction of match quality change. An *upgrade* captures a move to a higher-seniority position and signals an improvement in match quality. I then advance to the next position in the employment spell and repeat this classification until no subsequent qualifying position exists, at which point I treat the final position in the spell as a separation and apply the classification procedure described below.

Job-to-job moves within the same area. A transition is classified as a job-to-job move if a position at a different employer begins within 31 days of the end date of the current position, provided the new position extends beyond that end date. Among same-area job-to-job moves, I distinguish between two types. A *pull separation* is a move to a different employer within the same area at a strictly higher seniority level, indicating that the new match offers higher quality than the previous one. A *push separation* is a move to a different employer within the same area at an equal or lower seniority level. Although a downward move along the seniority ladder does not necessarily mean the previous match quality fell below the reservation threshold, it suggests the new match offers substantially higher utility for the worker to accept a lower rank.

Relocation. I classify a job-to-job move as a relocation if the new position is in a different geographic area and begins within two months of the end date of the current position.

Geographic area is defined at the MSA or CSA level.[9] Internal transfers across regions are not classified as relocation, since they reflect changes in match quality with the same employer rather than a geographic reallocation decision. I interpret relocation as primarily driven by shifts in personal or household preferences rather than a deterioration in the prior employment match.

Transition into education. A position is classified as transitioning into education if an education record begins within three months of the position's end date. Because enrollment decisions are typically planned in advance, this transition does not necessarily reflect declining match quality with the previous employer and is therefore treated as a separate category.

Transition into nonemployment. A position is classified as transitioning into nonemployment if none of the above conditions are met: there is no same-area position beginning within one month, no different-area position beginning within two months, and no education record beginning within three months. A transition into nonemployment signals that match quality with the previous employer has fallen below the quality threshold.

Internal transfers across the seniority ladder are associated with larger salary changes than job-to-job moves across firms. As shown in Figure 1, 19.3% of position changes are classified as internal transfers, with promotions (8.7%) and lateral transfers (7.3%) occurring more frequently than downgrades (3.3%). This distribution is consistent with the finding of Forsythe (2023) using CPS and SIPP data, who find that over 40% of moves are to lower ranked occupations. Although downgrades are relatively rare, they are associated with substantial salary losses, averaging $17,400 in annual salary — more than double the average decline of $8,125 observed for push separations. Promotions, by contrast, are associated with an average annual salary gain of $26,720, meaningfully larger than the $23,349 average gain for pull separations **??**. Table 18 shows these patterns are robust to controlling for year-month (Column 2) and sector fixed effects (Column 3), indicating that the pattern is not driven by aggregate macroeconomic conditions or by the concentration of different transition types in particular industries.

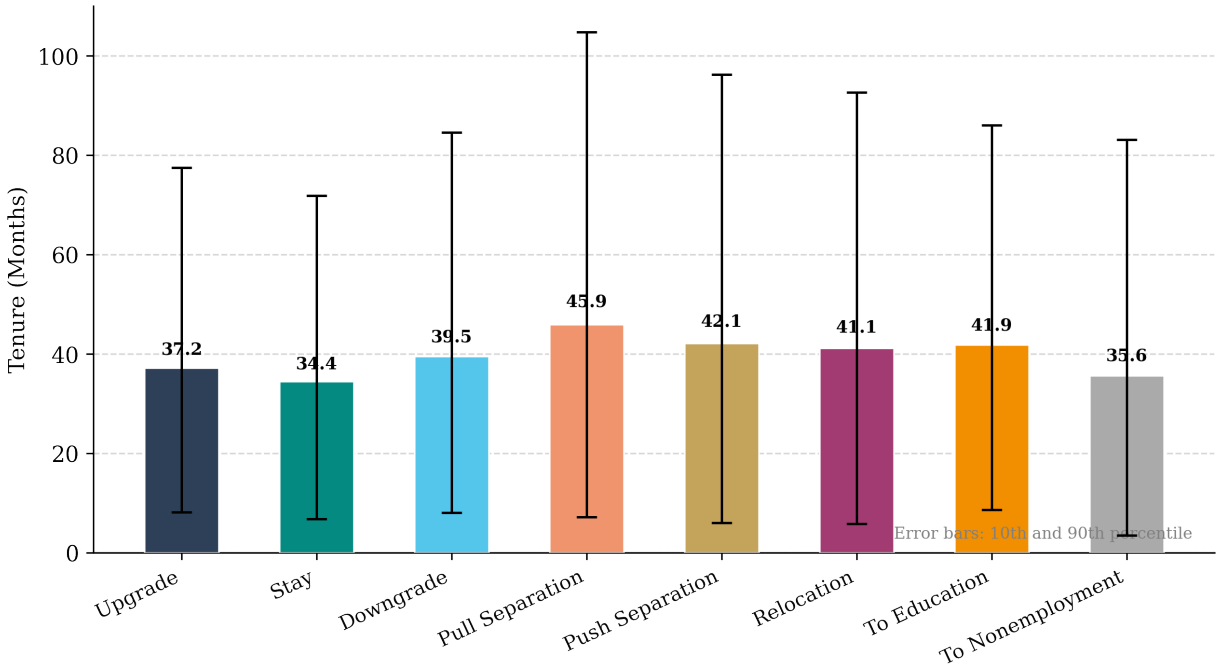
Turning to cyclical patterns, salary changes across all transition types decline during recessions, but internal promotions and push separations are most affected. Relative to normal times, salary growth associated with promotions falls by $1,789 during recessions, and push separations experience an additional decline of $1,766. Notably, during recessions the salary gains from promotions and pull separations become statistically in-

[9] We do not classify moves across MSAs within the same CSA as relocation, since CSAs represent regions with overlapping labor and commuting markets. The BLS defines CSAs as groups of metropolitan or micropolitan statistical areas with an employment interchange of at least 15%.

Table 18: Salary Change and Tenure by Separation Type

	(1) pct	(2) pct	(3) pct	(4) pct	(5) USD	(6) pct
Main effects						
Upgrade	0.345***	0.346***	0.352***	0.350***	21821.9***	0.314***
	(0.003)	(0.003)	(0.003)	(0.003)	(110.1)	(0.006)
Downgrade	−0.238***	−0.239***	−0.232***	−0.232***	−20299.8***	−0.207***
	(0.003)	(0.003)	(0.002)	(0.003)	(93.2)	(0.004)
Pull sep.	0.448***	0.443***	0.417***	0.416***	19501.9***	0.367***
	(0.004)	(0.004)	(0.003)	(0.004)	(82.3)	(0.007)
Push sep.	−0.085***	−0.089***	−0.096***	−0.098***	−9857.1***	−0.098***
	(0.003)	(0.003)	(0.003)	(0.003)	(106.9)	(0.004)
Relocation	0.120***	0.116***	0.102***	0.102***	473.7***	0.078***
	(0.004)	(0.004)	(0.005)	(0.005)	(101.8)	(0.012)
× *USREC*						
Upgrade × Rec.				0.030***	−458.9	
				(0.010)	(350.0)	
Downgrade × Rec.				−0.006	601.8*	
				(0.005)	(310.5)	
Pull sep. × Rec.				0.017**	58.7	
				(0.008)	(315.4)	
Push sep. × Rec.				0.031	−184.8	
				(0.026)	(352.0)	
Relocation × Rec.				0.006	720.0	
				(0.009)	(459.0)	
× *UNRATE*						
Upgrade × UR						0.007***
						(0.001)
Downgrade × UR						−0.004***
						(0.001)
Pull sep. × UR						0.009***
						(0.001)
Push sep. × UR						0.000
						(0.001)
Relocation × UR						0.004**
						(0.002)
Year-month FE	No	Yes	Yes	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes	Yes	Yes
Observations	80,005,113	80,005,113	71,065,892	71,065,892	71,065,892	71,060,757
R^2	0.001	0.001	0.001	0.001	0.109	0.001

Notes: The dependent variable in columns (1)–(3) is the salary change in USD, defined as the start salary of the next position minus the end salary of the current position. Column (4) uses a random sample. The baseline group is *Stay* (same employer, same seniority level). All specifications include a constant (not reported). Standard errors (in parentheses) are two-way clustered by worker and year-month. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) Job duration by separation types

Note: This figure shows the mean, 10th percentile, and 90th percentile of job duration for primary jobs, grouped by the type of subsequent separation. Source: Revelio Labs

distinguishable, suggesting that the advantage of internal advancement over cross-firm mobility compresses during downturns.

8.3 Testing persistence by employment gap

8.3.1 Cosine similarity

8.3.2 Vector autoregression

The rows of each matrix correspond to lagged components in states $1, \ldots, 7$, while the columns correspond to next-spell components in states $1, \ldots, 7$. The matrices $\widehat{A}_D^{\text{inn}}$ and $\widehat{A}_S^{\text{inn}}$ describe innovation persistence across and within firms, respectively. The matrices $\widehat{A}_D^{p}$ and $\widehat{A}_S^{p}$ provide the corresponding results for the latent-state probability vectors.

$$\widetilde{\text{inn}}_{i,n+1,k} = \sum_{j=1}^{7} A_{D,jk}\, \widetilde{\text{inn}}_{in,j} + \varepsilon_{i,n+1,k}, \qquad k = 1, \ldots, 7. \tag{22}$$

Table 19: Cosine Similarity by Pre-Employment-Gap Category

	Consecutive Spells	Random Baseline	Actual Minus Firm Random	Number of Pairs
Panel A: No gap transition				
Match innovations, different firm	0.1552	0.0375	0.1178	1,018,697
Match innovations, same firm	0.1190	-0.0170	0.1360	815,155
State probabilities, different firm	0.6004	0.3130	0.2874	1,018,697
State probabilities, same firm	0.6308	0.3162	0.3146	815,155
Panel B: Pre-employment gap of 1–6 months				
Match innovations, different firm	0.1654	0.0676	0.0977	250,588
Match innovations, same firm	0.0788	0.0625	0.0163	16,492
State probabilities, different firm	0.5589	0.3824	0.1764	250,588
State probabilities, same firm	0.5049	0.3704	0.1346	16,492
Panel C: Pre-employment gap of 7 or more months due to education				
Match innovations, different firm	0.0823	0.0654	0.0169	130,472
Match innovations, same firm	0.1138	0.0649	0.0488	136,483
State probabilities, different firm	0.2977	0.2953	0.0024	130,472
State probabilities, same firm	0.4063	0.2955	0.1108	136,483
Panel D: Other pre-employment gaps of 7 or more months				
Match innovations, different firm	0.1360	0.0518	0.0842	252,058
Match innovations, same firm	0.0261	0.0565	-0.0304	20,857
State probabilities, different firm	0.4782	0.3410	0.1371	252,058
State probabilities, same firm	0.4231	0.3427	0.0804	20,857

Notes: Each entry reports the mean cosine similarity between two seven-dimensional vectors. Actual consecutive spells compare adjacent spells belonging to the same worker. For different-firm transitions, the worker changes firms; for same-firm transitions, the worker changes positions within the same firm. The common-pool firm-random baseline keeps the source worker's previous-spell vector fixed and pairs it with a next-spell vector belonging to a different worker and firm in the same hire-quarter and pre-employment-gap cell. The donor pool is constructed from all eligible transitions before separating the same-firm and different-firm source samples. The final column reports the number of actual consecutive-spell pairs. Results use the filtered one-million-worker sample.

$$\widehat{A}_D^{inn} = \begin{bmatrix} 0.1475 & -0.0523 & 0.0349 & 0.0572 & -0.1408 & 0.0079 & -0.0544 \\ -0.0680 & 0.0190 & -0.0107 & 0.0037 & 0.1468 & -0.0591 & -0.0318 \\ 0.0436 & -0.0051 & 0.0554 & -0.0172 & -0.1716 & 0.1017 & -0.0068 \\ 0.0148 & 0.0671 & -0.0539 & -0.0232 & 0.0257 & -0.0348 & 0.0043 \\ -0.0629 & 0.0783 & 0.0037 & -0.0107 & 0.0692 & -0.0749 & -0.0027 \\ 0.0296 & -0.0588 & 0.0296 & 0.0421 & -0.1613 & 0.1640 & -0.0454 \\ -0.1047 & -0.0483 & -0.0589 & -0.0520 & 0.2320 & -0.1047 & 0.1367 \end{bmatrix}. \tag{23}$$

$$\widehat{A}_S^{\text{inn}} = \begin{bmatrix} -0.0245 & -0.0011 & 0.0267 & -0.0626 & 0.1147 & -0.0475 & -0.0056 \\ -0.0380 & -0.0221 & -0.0039 & -0.0192 & 0.0901 & 0.0313 & -0.0383 \\ 0.0613 & 0.0011 & 0.1008 & -0.0667 & -0.0886 & 0.0236 & -0.0315 \\ 0.0955 & -0.0059 & -0.0386 & 0.0216 & -0.0228 & 0.0061 & -0.0558 \\ 0.0241 & 0.0415 & -0.0305 & 0.1138 & -0.1387 & 0.0040 & -0.0142 \\ -0.0185 & 0.0156 & -0.0297 & 0.0796 & -0.1195 & 0.0882 & -0.0158 \\ -0.0999 & -0.0291 & -0.0247 & -0.0666 & 0.1648 & -0.1057 & 0.1611 \end{bmatrix}. \tag{24}$$

$$\widehat{A}_D^{p} = \begin{bmatrix} 3.6595 & 0.0834 & 8.6382 & -4.7580 & -6.4308 & -0.5466 & -0.6458 \\ 3.1977 & 0.5441 & 8.6277 & -4.6041 & -6.9447 & -0.1754 & -0.6454 \\ 2.5744 & 0.1206 & 9.0315 & -4.6568 & -5.6923 & -0.7137 & -0.6638 \\ 3.0725 & 0.0525 & 8.7802 & -3.5147 & -7.1097 & -0.6310 & -0.6499 \\ -19.3296 & -0.9756 & -52.6047 & 26.7694 & 39.6381 & 2.9162 & 3.5863 \\ 3.2377 & 0.1136 & 8.6760 & -4.6148 & -6.5269 & -0.2470 & -0.6387 \\ 3.5877 & 0.0614 & 8.8511 & -4.6210 & -6.9338 & -0.6026 & -0.3429 \end{bmatrix}. \tag{25}$$

$$\widehat{A}_S^{p} = \begin{bmatrix} 0.7910 & -0.0644 & -0.0787 & -0.6979 & 0.0302 & 0.0697 & -0.0499 \\ -0.0970 & 0.3711 & 0.0135 & -0.0756 & -0.5060 & 0.3520 & -0.0581 \\ -0.3148 & -0.0794 & 0.4291 & -0.0598 & 0.3088 & -0.2133 & -0.0705 \\ -0.1827 & -0.0809 & 0.1272 & 0.6756 & -0.3306 & -0.1494 & -0.0592 \\ -0.2995 & -0.0598 & -0.9240 & 0.3795 & 1.0890 & -0.1706 & -0.0147 \\ -0.0041 & -0.0236 & 0.0865 & -0.2341 & -0.0686 & 0.2903 & -0.0465 \\ 0.1070 & -0.0630 & 0.3464 & 0.0123 & -0.5229 & -0.1788 & 0.2989 \end{bmatrix}. \tag{26}$$

8.4 Firm-Level Outcomes

To examine how match quality relates to firm performance, I construct a panel of firm-quarter observations tracking firm-level employment and survival outcomes. *Firm employment* in a given quarter is defined as the number of workers with an active spell at

the firm during that quarter. *Employment growth* is the quarter-on-quarter log change in employment. *Firm survival* is an indicator equal to one if the firm has at least one active position recorded in the data in each quarter through the end of the observation window.

Because Revelio Labs data are derived from online profiles, coverage of firms is inherently incomplete and unequal. Large firms with a strong online presence are well represented, whereas small or private firms with few workers on professional networking platforms may have only a handful of observed positions—or none at all. Observed employment therefore understates true employment, and the degree of understatement varies across firms and over time. I account for this by including firm fixed effects in all firm-level regressions and by focusing on within-firm variation in employment growth and survival rather than cross-sectional levels.

Where relevant, I supplement the employment panel with information from Revelio Labs job postings. *Expected hires* is a forward-looking measure constructed from the stock of active job postings, scaled by historical offer-acceptance rates, and serves as a proxy for the firm's intended employment growth. This variable allows us to distinguish between realized employment changes and planned hiring activity.

9 Multi-channel Latent Markov Model Appendix

9.0.1 Estimation

We maximize the log-likelihood (**??**) using the *EM algorithm*. Direct maximization of (**??**) is computationally demanding because equation (7) sums over K^T latent-state histories. The EM algorithm alleviates this difficulty by alternating between two steps:

- *E-step*: Compute the posterior distribution of the latent-state history given the current parameter estimates, leveraging the Markovian structure through the Baum–Welch (forward–backward) algorithm. This reduces the computational complexity from $O(K^T)$ to $O(K^2T)$.
- *M-step*: Update all parameters by maximizing the expected complete-data log-likelihood given the posterior state probabilities from the E-step.

The algorithm iterates until the relative change in log-likelihood falls below 10^{-5}. Because the EM algorithm can converge to local optima, we estimate each model from five independent random restarts and retain the solution with the highest log-likelihood. Each restart is run as an independent task on a high-performance computing cluster and assigned a distinct seed to ensure reproducibility.

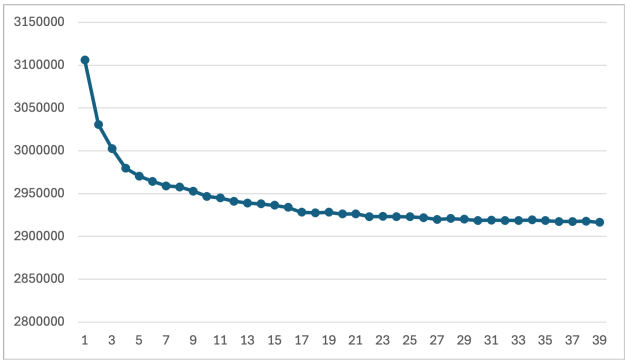


Figure 14: Akaike Information Criterion

Notes: The figure reports the Akaike Information Criterion (AIC) for latent Markov models estimated with different numbers of latent states K.

The model is estimated on a training sample of one million workers drawn from the Revelio Labs data. The remaining workers are used for out-of-sample validation, as described in Section **??**.

9.1 Model Selection

Table 20: Out-of-Sample Log-Likelihood per Worker

k	Training mean	Test mean	Train–test gap	Training workers	Test workers
7	−14.65269	−14.64467	−0.00802	100,000	900,000
9	−14.62349	−14.68503	0.06154	100,000	100,000
15	−14.54983	−14.61068	0.06085	100,000	100,000
28	−14.51930	**−14.58378**	0.06448	100,000	100,000

Notes: Log-likelihoods are reported as means per worker. A less negative test log-likelihood indicates better out-of-sample fit.

9.2 Emission Probabilities

9.3 Estimated Initial Distribution by Cohorts

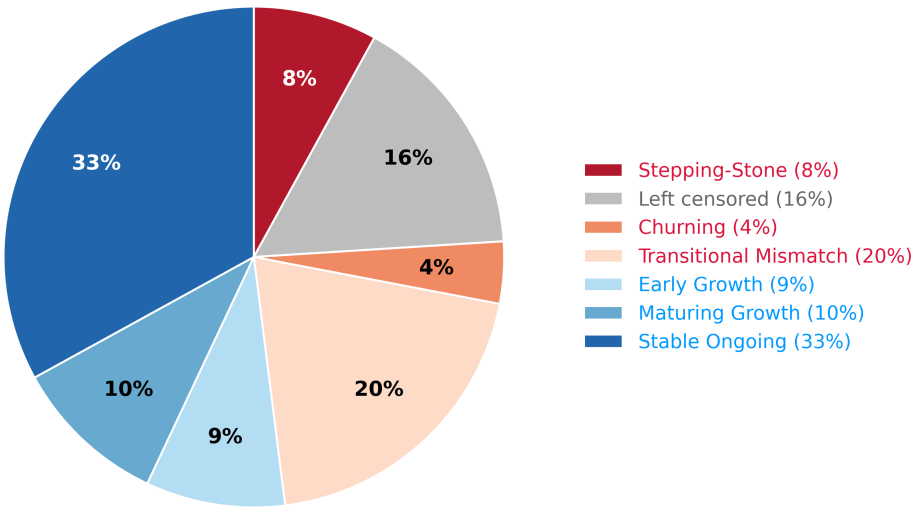
9.4 Standard Errors for Transition Probabilities.

We compute standard errors for the estimated transition probabilities using the delta method. The latent Markov model parameterizes each row of the transition matrix as a multinomial logit, so the transition probabilities are nonlinear functions of the estimated logit coefficients. Let $\widehat{\gamma}$ denote the vector of transition coefficients and let $\widehat{\mathbf{V}}_{\gamma}$ denote its

Table 21: Estimated Emission Probabilities

Initial distribution	**1**	**2**	**3**	**4**	**5**	**6**	**7**
	0.582	0.309	0.013	0.036	0.044	0.004	0.012
Emission 1: Employment duration	**1**	**2**	**3**	**4**	**5**	**6**	**7**
1–6 months	0.429	0.043	0.468	0.048	0.165	0.005	0.009
7–15 months	0.263	0.138	0.371	0.289	0.361	0.050	0.000
16–36 months	0.204	0.298	0.152	0.432	0.423	0.400	0.000
37 months or more	0.103	0.521	0.009	0.231	0.051	0.545	0.991
Emission 2: Job-exit type	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Education	0.367	0.105	0.094	0.000	0.034	0.017	0.000
Nonemployment	0.257	0.205	0.223	0.240	0.003	0.010	0.000
Downgrade	0.005	0.050	0.010	0.005	0.151	0.108	0.002
Same level	0.044	0.096	0.053	0.010	0.306	0.193	0.002
Upgrade	0.019	0.112	0.048	0.031	0.338	0.306	0.002
Pull separation	0.042	0.120	0.117	0.190	0.047	0.109	0.000
Push separation	0.150	0.203	0.255	0.334	0.087	0.154	0.000
Relocation	0.116	0.109	0.200	0.191	0.034	0.103	0.000
Ongoing	0.000	0.000	0.000	0.000	0.000	0.000	0.995

Notes: This table reports the estimated initial and emission probabilities from the seven-state multi-channel latent Markov model. The reported initial distribution is the weighted average of the cohort-specific initial distributions, where the weights are the cohort shares in the estimation sample. The complete set of cohort-specific initial distributions is reported in Appendix Table 22. The first emission is employment duration, and the second is job-exit type. Within each emission panel, a column reports the conditional distribution of the corresponding emission given the latent state. Entries may not sum exactly to one because of rounding.*Sources*: Revelio Labs and authors' calculations.



Notes: This figure shows the reported frequency distribution across the seven latent employment states. Each slice represents a state's share of the total, including the left-censored category. Percentages sum to 100%. *Sources*: Revelio Labs and author's calculations.

Figure 15: Distribution of Latent States

Table 22: Initial Distribution of Latent Match States by Birth Cohort

Birth cohort	S1	S2	S3	S4	S5	S6	S7
Pre-1982	0.0000	0.0260	0.0000	0.0000	0.0046	0.0000	0.9694
1982–1984	0.0007	0.2065	0.0000	0.0315	0.0040	0.0000	0.7572
1985–1987	0.0161	0.4330	0.0000	0.0585	0.0047	0.0000	0.4876
1988–1990	0.0257	0.6135	0.0000	0.0775	0.0054	0.0000	0.2778
1991–1993	0.0375	0.7282	0.0000	0.0867	0.0071	0.0002	0.1403
1994–1996	0.0395	0.8115	0.0000	0.0907	0.0091	0.0004	0.0487
1997–1999	0.0327	0.8629	0.0005	0.0706	0.0114	0.0087	0.0132
2000–2002	0.0332	0.8872	0.0063	0.0286	0.0165	0.0184	0.0098
2003–2005	0.0612	0.8249	0.0173	0.0157	0.0267	0.0304	0.0238
2006–2008	0.0965	0.6553	0.0189	0.0095	0.0326	0.0870	0.1002
Missing	0.0561	0.3511	0.0000	0.0110	0.0087	0.0000	0.5732
Overall	0.0363	0.5818	0.0039	0.0437	0.0119	0.0132	0.3092

Notes: The table reports the estimated initial distribution over the seven latent match states by worker birth cohort. Each row reports the probability that a worker's first observed position is assigned to each latent state. The final row reports the overall initial-state distribution in the estimation sample.

estimated covariance matrix, obtained from the inverse observed information matrix. For a given category, the corresponding transition probabilities can be written as

$$\widehat{\boldsymbol{\pi}} = g(\widehat{\gamma}; \mathbf{x}),$$

where $\mathbf{x}$ contains the covariates of the model. I approximate the covariance matrix of these probabilities using

$$\widehat{\mathbf{V}}_{\pi} = \mathbf{G}\widehat{\mathbf{V}}_{\gamma}\mathbf{G}', \qquad \mathbf{G} = \left.\frac{\partial g(\gamma; \mathbf{x})}{\partial \gamma'}\right|_{\gamma=\widehat{\gamma}}.$$

The reported standard error of each transition probability is the square root of the corresponding diagonal element of $\widehat{\mathbf{V}}_{\pi}$. By explicitly incorporating the Jacobian of the multinomial logit transformation, this procedure properly propagates the covariance among the estimated coefficients into the covariance of the transition probabilities, while respecting the adding-up constraint that each row of the transition matrix must sum to one.

9.5 Spell-level regression coefficients

Table 23: Worker age odds ratios by latent state

	S2	S6	S4	S3
≤20	60.277	7.782	1.680	0.501
	(0.843)	(0.064)	(0.031)	(0.007)
20–25	13.398	3.494	1.819	0.630
	(0.138)	(0.020)	(0.019)	(0.004)
25–30	2.424	1.612	1.288	0.852
	(0.020)	(0.009)	(0.008)	(0.004)
35–45	0.638	0.726	0.750	0.993
	(0.007)	(0.005)	(0.006)	(0.006)
45–55	0.581	0.645	0.515	0.815
	(0.009)	(0.006)	(0.007)	(0.007)
55+	0.636	0.686	0.436	0.657
	(0.018)	(0.012)	(0.009)	(0.010)
Missing	11.713	1.808	0.798	0.604
	(0.198)	(0.024)	(0.015)	(0.010)

Table 24: Gender odds ratios by latent state

	S2	S6	S4	S3
F	1.065	1.064	1.091	0.987
	(0.007)	(0.005)	(0.007)	(0.005)
Missing	1.112	1.068	1.188	1.017
	(0.036)	(0.024)	(0.035)	(0.025)

Table 25: Education odds ratios by latent state

	S2	S6	S4	S3
High School	0.931	0.982	0.565	0.529
	(0.014)	(0.013)	(0.010)	(0.009)
Associate	0.534	1.035	0.627	0.756
	(0.007)	(0.010)	(0.009)	(0.008)
MBA	1.218	1.348	1.376	1.103
	(0.014)	(0.010)	(0.017)	(0.010)
Master	1.789	1.412	1.019	0.992
	(0.013)	(0.009)	(0.010)	(0.008)
Doctor	1.810	1.500	0.891	1.074
	(0.032)	(0.022)	(0.014)	(0.012)
Missing	0.821	1.291	1.301	1.052
	(0.092)	(0.117)	(0.198)	(0.142)

Table 26: Ethnicity odds ratios by latent state

	S2	S6	S4	S3
API	1.286	0.961	0.844	0.926
	(0.019)	(0.009)	(0.011)	(0.014)
Black	1.120	1.055	0.827	0.828
	(0.008)	(0.006)	(0.007)	(0.006)
Hispanic	1.154	1.050	0.879	0.858
	(0.009)	(0.007)	(0.009)	(0.007)
Multiple	1.252	1.045	0.907	0.882
	(0.043)	(0.031)	(0.039)	(0.034)
Native	1.187	1.078	0.821	0.874
	(0.083)	(0.064)	(0.072)	(0.072)
Missing	1.287	1.087	0.919	0.944
	(0.139)	(0.095)	(0.132)	(0.123)

Table 27: Firm size odds ratios by latent state

	S2	S6	S4	S3
11–30	0.931	1.015	1.399	1.053
	(0.007)	(0.006)	(0.014)	(0.009)
31–50	0.878	1.006	1.685	1.112
	(0.008)	(0.007)	(0.020)	(0.011)
51–200	0.840	1.002	1.973	1.167
	(0.007)	(0.006)	(0.022)	(0.011)
200+	0.870	1.057	2.811	1.410
	(0.010)	(0.008)	(0.037)	(0.015)
Missing	1.797	1.210	1.458	1.271
	(0.057)	(0.027)	(0.053)	(0.038)

Table 28: Firm age odds ratios by latent state

	S2	S6	S4	S3
1–3 year	1.240	1.242	1.268	0.885
	(0.027)	(0.020)	(0.029)	(0.017)
3–5 year	1.243	1.275	1.376	0.885
	(0.028)	(0.022)	(0.035)	(0.018)
6–9 year	1.217	1.264	1.388	0.915
	(0.026)	(0.020)	(0.034)	(0.018)
10–14 year	1.214	1.247	1.371	0.992
	(0.026)	(0.020)	(0.034)	(0.020)
15–30 year	1.229	1.228	1.304	1.060
	(0.027)	(0.020)	(0.031)	(0.021)
31–50 year	1.263	1.228	1.360	1.218
	(0.030)	(0.021)	(0.039)	(0.029)
50+ year	1.409	1.285	1.598	1.490
	(0.035)	(0.023)	(0.044)	(0.033)
Missing	1.296	1.241	1.299	1.098
	(0.028)	(0.020)	(0.030)	(0.021)

Table 29: Sector odds ratios by latent state

	S2	S6	S4	S3
Admin and Waste Management and Remed Serv [NAICS 56]	1.093	1.239	0.803	0.755
	(0.032)	(0.022)	(0.046)	(0.042)
Agriculture, Forestry, Fishing and Hunting [NAICS 11]	1.252	1.126	1.022	1.036
	(0.091)	(0.045)	(0.081)	(0.085)
Arts, Entert, and Recr [NAICS 71]	1.898	1.400	0.792	0.763
	(0.058)	(0.023)	(0.034)	(0.021)
Construction [NAICS 23]	1.016	1.010	0.842	0.948
	(0.065)	(0.058)	(0.040)	(0.027)
Educational Services [NAICS 61]	2.073	1.636	0.693	0.819
	(0.059)	(0.027)	(0.021)	(0.019)
Finance and Insurance [NAICS 52]	0.902	0.951	0.954	1.043
	(0.027)	(0.016)	(0.032)	(0.024)
Health Care and Social Assistance [NAICS 62]	1.169	1.164	0.656	0.768
	(0.046)	(0.022)	(0.018)	(0.017)
Information [NAICS 51]	1.012	1.074	1.036	0.922
	(0.032)	(0.017)	(0.031)	(0.027)
Mng of Companies and Enterprises [NAICS 55]	0.931	0.968	0.873	0.925
	(0.056)	(0.038)	(0.089)	(0.050)
Mining, Quar and Oil and Gas Extrac [NAICS 21]	1.055	1.024	1.305	1.079
	(0.054)	(0.027)	(0.176)	(0.101)
Prof, Scien, and Tech Services [NAICS 54]	1.011	1.044	0.899	0.911
	(0.035)	(0.020)	(0.023)	(0.021)
Real Estate and Rental and Leasing [NAICS 53]	0.920	0.935	0.725	0.786
	(0.030)	(0.020)	(0.078)	(0.023)
Retail Trade [NAICS 44/45]	1.530	1.272	0.912	0.783
	(0.052)	(0.025)	(0.060)	(0.032)
Trans and Warehousing [NAICS 48/49]	1.022	1.089	0.890	0.917
	(0.031)	(0.018)	(0.041)	(0.036)
Utilities [NAICS 22]	0.922	0.999	1.168	1.354
	(0.040)	(0.030)	(0.082)	(0.095)
Wholesale Trade [NAICS 42]	1.077	1.044	0.909	0.944
	(0.048)	(0.025)	(0.038)	(0.030)
Missing	1.449	1.200	1.174	1.126
	(0.138)	(0.088)	(0.136)	(0.119)

9.6 Persistence of Match Innovations within Workers

The match innovation is constructed to remove the component of the realized match state that can be predicted from worker characteristics and prior employment history. This section evaluates how well the construction achieves that goal by measuring how much of the persistence in latent match states across a worker's consecutive spells is removed by the prediction procedure.

Cosine similarity. Using a random sample of one million workers, I compute the cosine similarity between the seven-dimensional vectors associated with consecutive job spells of the same worker.[10] I do this separately for transitions across firms and for transitions between positions within the same firm, and compare the observed similarity with a firm-random baseline. For each actual transition, the baseline replaces the subsequent-spell vector with the vector of a different worker at a different firm, drawn from the same hire-quarter and pre-employment-gap cell. Because transitions within and across firms are distributed differently across these cells, the baseline can differ between the two samples. An excess similarity of zero would mean that consecutive spells of the same worker are no more alike than observably comparable spells of unrelated workers and firms, i.e., that the innovation fully eliminates worker-level persistence in the latent-state distribution.

Table 30 shows that the prediction procedure removes a large share of this persistence. For workers moving across firms, mean cosine similarity falls from 0.552 for state probabilities to 0.148 for match innovations. For workers changing positions within the same firm (with no gap between spells), it falls from 0.631 to 0.119. Relative to the firm-random baseline, excess similarity falls from 0.225 to 0.102 across firms and from 0.315 to 0.136 within firms, reductions of roughly 55 and 57 percent, respectively. The innovation construction therefore substantially reduces the persistence in latent-state distributions, although the remaining excess similarity across firms is still statistically different from zero.

Residual similarity is expected for two reasons. First, the innovation is not purged of firm-specific effects, so persistence should be greater when a worker stays at the same firm. Consistent with this, excess similarity of match innovations is larger for position changes within the same firm (0.136) than for job-to-job moves across firms (0.102). Second, realized matches are a selected sample: a match is observed only when the match-specific shock exceeds an acceptance threshold, and endogenous separations add further

[10] For two seven-dimensional vectors $\mathbf{x}$ and $\mathbf{y}$, cosine similarity is defined as $\cos(\mathbf{x}, \mathbf{y}) = \mathbf{x}'\mathbf{y}/(\|\mathbf{x}\|\|\mathbf{y}\|)$. I calculate this measure directly from the raw state-probability or innovation vectors without standardization.

selection (Borovičková and Shimer, 2024). Even if potential innovations were independent across a worker's matches, realized innovations could remain weakly correlated.

Table 19 repeats the comparison separately by pre-employment-gap category. The innovation construction substantially reduces persistence in most categories, and especially for transitions with no gap between spells.

Table 30: Cosine Similarity of State Probabilities and Match Innovations

	Consecutive Spells	Random Baseline	Consecutive minus Random	Number of Pairs
Match innovations				
Different firm	0.148	0.046	0.102	1,651,815
Same firm (no gap)	0.119	-0.017	0.136	815,155
State probabilities				
Different firm	0.552	0.326	0.225	1,651,815
Same firm (no gap)	0.631	0.316	0.315	815,155

Notes: The table reports the mean cosine similarity of state-probability and match-innovation vectors across consecutive job spells of the same worker, using a random sample of one million workers. Each entry is the mean cosine similarity between two seven-dimensional vectors. Consecutive spells compare adjacent spells belonging to the same worker. In different-firm transitions the worker changes firms; in same-firm transitions the worker changes positions within the same firm with no intervening gap. The firm-random baseline keeps the worker's previous-spell vector fixed and pairs it with a next-spell vector belonging to a different worker and firm in the same hire-quarter and pre-employment-gap cell. Randomized next-spell vectors are drawn from the common pool of all eligible transitions before the same-firm and different-firm samples are separated. The comparisons use 1,651,563 valid different-firm randomized pairs and 815,155 valid same-firm randomized pairs. Results use the filtered one-million-worker sample.

Vector autoregression. As a second check, I estimate an unrestricted vector autoregression in which each component of the previous spell's distribution may predict each component of the subsequent spell's distribution. Tables 32 and 31 report the coefficient matrices for state probabilities and match innovations, respectively. Consistent with the cosine-similarity results, the predictive relationships are much weaker for match innovations than for state probabilities. The innovation coefficients are generally small in magnitude, indicating that the prediction procedure removes most of the state-specific dependence across consecutive spells. Some dependence remains, consistent with the positive excess cosine similarity documented above.

Table 31: Persistence of Match Innovations across Consecutive Employment Spells

Panel A: Different-firm transitions

	Subsequent spell						
Previous spell	inn_1	inn_2	inn_3	inn_4	inn_5	inn_6	inn_7
inn_1	0.148	-0.052	0.035	0.057	-0.141	0.008	-0.054
inn_2	-0.068	0.019	-0.011	0.004	0.147	-0.059	-0.032
inn_3	0.044	-0.005	0.055	-0.017	-0.172	0.102	-0.007
inn_4	0.015	0.067	-0.054	-0.023	0.026	-0.035	0.004
inn_5	-0.063	0.078	0.004	-0.011	0.069	-0.075	-0.003
inn_6	0.030	-0.059	0.030	0.042	-0.161	0.164	-0.045
inn_7	-0.105	-0.048	-0.059	-0.052	0.232	-0.105	0.137

Panel B: Same-firm transitions

	Subsequent spell						
Previous spell	inn_1	inn_2	inn_3	inn_4	inn_5	inn_6	inn_7
inn_1	-0.024	-0.001	0.027	-0.063	0.115	-0.047	-0.006
inn_2	-0.038	-0.022	-0.004	-0.019	0.090	0.031	-0.038
inn_3	0.061	0.001	0.101	-0.067	-0.089	0.024	-0.031
inn_4	0.095	-0.006	-0.039	0.022	-0.023	0.006	-0.056
inn_5	0.024	0.041	-0.030	0.114	-0.139	0.004	-0.014
inn_6	-0.018	0.016	-0.030	0.080	-0.119	0.088	-0.016
inn_7	-0.100	-0.029	-0.025	-0.067	0.165	-0.106	0.161

Notes: The table reports the unrestricted persistence matrix for seven-dimensional match-innovation vectors across consecutive employment spells. Rows index the innovation associated with the previous spell and columns index the innovation associated with the subsequent spell. Estimation accounts for the linear dependence among the seven innovation components using six orthonormal Helmert coordinates. Each innovation component is demeaned by the subsequent spell's hire-quarter, pre-employment-gap category, and same-firm status prior to estimation. Panel A uses 1,651,815 transitions in which the worker changes firms. Panel B uses 988,987 transitions between positions within the same firm.

Table 32: Persistence of Latent-State Probabilities across Consecutive Employment Spells

Panel A: Different-firm transitions

	Subsequent spell						
Previous spell	p_1	p_2	p_3	p_4	p_5	p_6	p_7
p_1	3.660	0.083	8.638	-4.758	-6.431	-0.547	-0.646
p_2	3.198	0.544	8.628	-4.604	-6.945	-0.175	-0.645
p_3	2.574	0.121	9.032	-4.657	-5.692	-0.714	-0.664
p_4	3.073	0.053	8.780	-3.515	-7.110	-0.631	-0.650
p_5	-19.330	-0.976	-52.605	26.769	39.638	2.916	3.586
p_6	3.238	0.114	8.676	-4.615	-6.527	-0.247	-0.639
p_7	3.588	0.061	8.851	-4.621	-6.934	-0.603	-0.343

Panel B: Same-firm transitions

	Subsequent spell						
Previous spell	p_1	p_2	p_3	p_4	p_5	p_6	p_7
p_1	0.791	-0.064	-0.079	-0.698	0.030	0.070	-0.050
p_2	-0.097	0.371	0.014	-0.076	-0.506	0.352	-0.058
p_3	-0.315	-0.079	0.429	-0.060	0.309	-0.213	-0.071
p_4	-0.183	-0.081	0.127	0.676	-0.331	-0.149	-0.059
p_5	-0.300	-0.060	-0.924	0.380	1.089	-0.171	-0.015
p_6	-0.004	-0.024	0.087	-0.234	-0.069	0.290	-0.047
p_7	0.107	-0.063	0.346	0.012	-0.523	-0.179	0.299

Notes: The table reports the unrestricted persistence matrix for seven-dimensional posterior latent-state probability vectors across consecutive employment spells. Rows index the state probability associated with the previous spell and columns index the state probability associated with the subsequent spell. Because the seven probabilities sum to one, estimation is conducted using six orthonormal Helmert coordinates, with the resulting coefficients mapped back to the seven-state representation shown in the table. Each probability component is demeaned by the subsequent spell's hire-quarter, pre-employment-gap category, and same-firm status prior to estimation. Panel A uses 1,651,815 different-firm transitions and Panel B uses 988,987 same-firm transitions.

9.7 Frequency of Match Innovation Shifts Among Incumbent Workers and Completed Spells

Appendix Tables 33 and 34 report the corresponding frequencies of changes in firms' average match innovations among incumbent workers and completed spells, respectively.

For the stock of incumbent workers, changes of at least 10 percentage points are less frequent than among new hires. The most common pairing is a decline in Transitional Mismatch innovations alongside an increase in Stable Ongoing innovations, occurring in 5.8% of consecutive-year firm observations in the full sample and 3.2% among Compustat firms. The reverse pairing occurs in 3.2% and 1.2% of observations, respectively. Declines in Churning innovations paired with increases in Early Growth or Maturing Growth are comparatively uncommon, each occurring in less than 1% of observations in both samples.

Among completed spells, the most frequent pairing in the full sample is a decline in Churning innovations alongside an increase in Transitional Mismatch innovations, occurring in 10.5% of consecutive-year firm observations. Declines in Early Growth innovations paired with increases in Transitional Mismatch are also relatively frequent, occurring in 9.0% of observations. This latter pairing is the most common among Compustat firms, at 5.0%. These changes describe the composition of spells ending at the firm, rather than its current workforce: increases in a state's completed-spell innovation indicate greater-than-predicted exposure to that state among positions that have ended.

9.8 Summary states of match innovation exposures

Table 35 reports summary statistics for the firm-year innovations. The first two columns use the full Revelio sample, and the last two restrict to firms matched to Compustat. Standard deviations vary substantially across states—from 0.093 for Early Growth in the full sample to 0.112 for Churning. The dispersion is generally, though not uniformly, smaller among Compustat firms.

9.9 Lagged firm outcomes (Full sample)

9.10 Lagged firm outcomes (Compustat)

Table 33: Changes in Worker-Stock State Innovations of at Least 10 Percentage Points

	Increasing state						
Decreasing state	L-censored	Stepping	Churning	Transitional	Early Gr.	Mature Gr.	Ongoing
Panel A: Full sample (N = 13,591,261)							
L-censored	0.0000	0.0091	0.0034	0.0066	0.0022	0.0018	0.0079
Step.	0.0077	0.0000	0.0024	0.0081	0.0039	0.0024	0.0072
Churn.	0.0037	0.0038	0.0000	0.0106	0.0043	0.0064	0.0246
Trans.	0.0058	0.0089	0.0093	0.0000	0.0098	0.0070	0.0579
Early Gr.	0.0024	0.0068	0.0060	0.0138	0.0000	0.0031	0.0160
Mature Gr.	0.0015	0.0033	0.0070	0.0087	0.0025	0.0000	0.0132
Ongoing	0.0036	0.0062	0.0155	0.0323	0.0090	0.0052	0.0000
Panel B: Compustat sample (N = 80,293)							
L-censored	0.0000	0.0032	0.0011	0.0028	0.0011	0.0012	0.0027
Step.	0.0030	0.0000	0.0006	0.0023	0.0014	0.0010	0.0016
Churn.	0.0013	0.0009	0.0000	0.0029	0.0016	0.0021	0.0081
Trans.	0.0027	0.0024	0.0026	0.0000	0.0062	0.0050	0.0316
Early Gr.	0.0014	0.0021	0.0019	0.0078	0.0000	0.0014	0.0097
Mature Gr.	0.0010	0.0012	0.0023	0.0057	0.0013	0.0000	0.0059
Ongoing	0.0013	0.0017	0.0042	0.0115	0.0044	0.0021	0.0000

Notes: This table reports changes in average match innovations among employment spells active in the fourth quarter of each year. Each entry gives the fraction of consecutive-year firm-year pairs in which the innovation for the row state decreases by at least 10 percentage points and that for the column state increases by at least 10 percentage points. The sample includes firms with observed innovations for all seven states in both years. Each pair receives equal weight, and N denotes the total number of pairs used as the denominator. Panel B restricts the sample to firms matched to Compustat in the starting year. A pair may satisfy the conditions for multiple cells, so row sums need not equal one.

Table 34: Changes in Completed-Spell State Innovations of at Least 10 Percentage Points

	Increasing state						
Decreasing state	L-censored	Stepping	Churning	Transitional	Early Gr.	Mature Gr.	Ongoing
Panel A: Full sample (N = 4,210,865)							
L-censored	0.0000	0.0407	0.0241	0.0262	0.0140	0.0097	0.0020
Step.	0.0564	0.0000	0.0406	0.0865	0.0438	0.0264	0.0121
Churn.	0.0302	0.0361	0.0000	0.1052	0.0542	0.0535	0.0364
Trans.	0.0297	0.0711	0.0794	0.0000	0.0683	0.0418	0.0283
Early Gr.	0.0166	0.0469	0.0578	0.0901	0.0000	0.0180	0.0154
Mature Gr.	0.0102	0.0237	0.0435	0.0421	0.0167	0.0000	0.0050
Ongoing	0.0048	0.0217	0.0636	0.0687	0.0376	0.0151	0.0000
Panel B: Compustat sample (N = 72,780)							
L-censored	0.0000	0.0098	0.0059	0.0077	0.0058	0.0035	0.0006
Step.	0.0231	0.0000	0.0069	0.0235	0.0163	0.0086	0.0019
Churn.	0.0081	0.0060	0.0000	0.0260	0.0213	0.0194	0.0071
Trans.	0.0100	0.0155	0.0184	0.0000	0.0374	0.0240	0.0070
Early Gr.	0.0082	0.0133	0.0184	0.0502	0.0000	0.0093	0.0046
Mature Gr.	0.0042	0.0068	0.0139	0.0202	0.0079	0.0000	0.0018
Ongoing	0.0011	0.0040	0.0156	0.0238	0.0180	0.0074	0.0000

Notes: This table reports changes in average match innovations among employment spells ending in each calendar year. Each entry gives the fraction of consecutive-year firm-year pairs in which the innovation for the row state decreases by at least 10 percentage points and that for the column state increases by at least 10 percentage points. The sample includes firms with observed innovations for all seven states in both years. Each pair receives equal weight, and N denotes the total number of pairs used as the denominator. Panel B restricts the sample to firms matched to Compustat in the starting year. A pair may satisfy the conditions for multiple cells, so row sums need not equal one.

Table 35: Summary Statistics of Firm-Year Exposure to Match Innovations

	Full Sample		Compustat Firms	
	Stock	New Positions	Stock	New Positions
inn_1 (Stepping-stone)	−0.041 (0.101)	0.007 (0.127)	−0.037 (0.045)	−0.004 (0.059)
inn_2 (Stable-Stagnant)	0.023 (0.102)	0.004 (0.086)	0.093 (0.192)	−0.016 (0.140)
inn_3 (Churning)	−0.087 (0.112)	−0.002 (0.154)	−0.064 (0.053)	−0.004 (0.075)
inn_4 (Transitional)	−0.027 (0.215)	0.014 (0.184)	−0.004 (0.113)	0.014 (0.102)
inn_5 (Early Gr)	−0.079 (0.093)	−0.048 (0.113)	−0.048 (0.066)	0.002 (0.085)
inn_6 (Maturing Gr)	−0.009 (0.101)	−0.015 (0.082)	0.032 (0.068)	0.004 (0.058)
inn_7 (Stable Ongoing)	0.220 (0.375)	0.042 (0.295)	0.028 (0.055)	0.004 (0.048)
N (firm-years)	17,053,441	10,083,811	87,745	81,513

Notes: The table reports summary statistics for the seven match innovations aggregated to the firm-year level. Each cell reports the mean, with the standard deviation in parentheses. The stock measure averages spell-level innovations across employment spells active at the firm in the fourth quarter of each year. The new-hire measure averages spell-level innovations across employment spells beginning at the firm during the year. Columns (1)–(2) report statistics for the full Revelio sample, while columns (3)–(4) restrict the sample to firm-years matched to Compustat. *Source: Revelio Labs and Compustat.*

Table 36: New-Hire Match Innovations and Firm Outcomes (Full Sample, Lag 2)

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
inn_1 (Stepping-st)	−0.1031*	−0.0045*	0.1074*	0.0135*	−0.1055*	−0.0049*	0.1090*	0.0136*
	(0.0010)	(0.0010)	(0.0013)	(0.0004)	(0.0010)	(0.0010)	(0.0013)	(0.0004)
inn_3 (Churning)	−0.1292*	−0.0212*	0.1189*	0.0022*	−0.1298*	−0.0218*	0.1198*	0.0023*
	(0.0009)	(0.0009)	(0.0011)	(0.0003)	(0.0009)	(0.0009)	(0.0011)	(0.0003)
inn_5 (Early Gr)	−0.0488*	0.0120*	0.0725*	0.0109*	−0.0429*	0.0115*	0.0631*	0.0104*
	(0.0009)	(0.0009)	(0.0011)	(0.0003)	(0.0009)	(0.0009)	(0.0012)	(0.0003)
inn_6 (Maturing Gr)	−0.2076*	−0.0370*	0.1812*	0.0132*	−0.2134*	−0.0371*	0.1874*	0.0137*
	(0.0011)	(0.0013)	(0.0016)	(0.0003)	(0.0011)	(0.0013)	(0.0016)	(0.0003)
inn_7 (Stable Ongoing)	−0.1232*	−0.0170*	0.1147*	0.0085*	−0.1247*	−0.0169*	0.1162*	0.0087*
	(0.0005)	(0.0006)	(0.0007)	(0.0002)	(0.0005)	(0.0006)	(0.0007)	(0.0002)
pred_1 (Stepping-st)					−0.0051*	0.0291*	0.0115*	−0.0029*
					(0.0013)	(0.0013)	(0.0016)	(0.0004)
pred_3 (Churning)					−0.0056*	0.0306*	0.0036*	−0.0096*
					(0.0016)	(0.0017)	(0.0021)	(0.0006)
pred_5 (Early Gr)					0.0619*	0.0234*	−0.0770*	−0.0053*
					(0.0017)	(0.0018)	(0.0023)	(0.0006)
pred_6 (Maturing Gr)					−0.0588*	0.0681*	0.0646*	−0.0101*
					(0.0036)	(0.0038)	(0.0048)	(0.0012)
pred_7 (Stable Ongoing)					−0.0424*	0.0393*	0.0522*	−0.0000
					(0.0015)	(0.0015)	(0.0019)	(0.0004)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0159	0.0019	0.0067	0.0038	0.0168	0.0021	0.0075	0.0040
N	6,902,126	6,902,126	6,930,601	6,989,518	6,902,126	6,902,126	6,930,601	6,989,518

Notes: Estimates relate lagged new-hire match-state innovations to firm outcomes in the full sample. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival. All models include firm and 6-digit NAICS industry-by-year fixed effects. Separation and hiring models control for their lagged outcome; growth and survival models control for lagged employment growth. All lags are 2 years. Columns (5)–(8) also include lagged predicted state probabilities. State 4 (Transitional) is the reference for innovations and probabilities; state 2 (Stable/Stagnant) coefficients are estimated but not displayed. R^2 is within R^2; N counts firm-years. Standard errors are in parentheses. Coefficients and standard errors are rounded to four decimal places.

* $p < 0.10$, based on unrounded reported p-values. *Source: Revelio Labs.*

Table 37: New-Hire Match Innovations and Firm Outcomes (Full Sample, Lag 3)

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
inn_1 (Stepping-st)	−0.0267*	−0.0078*	0.0301*	0.0083*	−0.0274*	−0.0087*	0.0297*	0.0082*
	(0.0010)	(0.0010)	(0.0012)	(0.0003)	(0.0010)	(0.0010)	(0.0012)	(0.0003)
inn_3 (Churning)	−0.0933*	−0.0321*	0.0758*	0.0033*	−0.0939*	−0.0324*	0.0766*	0.0034*
	(0.0010)	(0.0009)	(0.0011)	(0.0003)	(0.0009)	(0.0009)	(0.0011)	(0.0003)
inn_5 (Early Gr)	−0.0047*	0.0014	0.0098*	0.0083*	0.0001	−0.0005	0.0009	0.0078*
	(0.0009)	(0.0009)	(0.0011)	(0.0003)	(0.0009)	(0.0009)	(0.0012)	(0.0003)
inn_6 (Maturing Gr)	−0.0802*	−0.0231*	0.0756*	0.0085*	−0.0826*	−0.0217*	0.0796*	0.0087*
	(0.0011)	(0.0012)	(0.0015)	(0.0003)	(0.0011)	(0.0012)	(0.0015)	(0.0003)
inn_7 (Stable Ongoing)	−0.0598*	−0.0160*	0.0538*	0.0066*	−0.0602*	−0.0157*	0.0545*	0.0068*
	(0.0005)	(0.0006)	(0.0007)	(0.0002)	(0.0005)	(0.0006)	(0.0007)	(0.0002)
pred_1 (Stepping-st)					−0.0027*	0.0322*	0.0213*	−0.0016*
					(0.0013)	(0.0013)	(0.0016)	(0.0004)
pred_3 (Churning)					−0.0064*	0.0318*	0.0194*	−0.0067*
					(0.0017)	(0.0017)	(0.0021)	(0.0006)
pred_5 (Early Gr)					0.0445*	0.0159*	−0.0546*	−0.0032*
					(0.0018)	(0.0018)	(0.0022)	(0.0006)
pred_6 (Maturing Gr)					−0.0261*	0.0853*	0.0726*	−0.0083*
					(0.0036)	(0.0037)	(0.0047)	(0.0012)
pred_7 (Stable Ongoing)					−0.0132*	0.0507*	0.0461*	0.0005
					(0.0015)	(0.0015)	(0.0019)	(0.0004)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0081	0.0005	0.0036	0.0018	0.0083	0.0008	0.0040	0.0019
N	6,318,902	6,318,902	6,344,316	6,394,277	6,318,902	6,318,902	6,344,316	6,394,277

Notes: Estimates relate lagged new-hire match-state innovations to firm outcomes in the full sample. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival. All models include firm and 6-digit NAICS industry-by-year fixed effects. Separation and hiring models control for their lagged outcome; growth and survival models control for lagged employment growth. All lags are 3 years. Columns (5)–(8) also include lagged predicted state probabilities. State 4 (Transitional) is the reference for innovations and probabilities; state 2 (Stable/Stagnant) coefficients are estimated but not displayed. R^2 is within R^2; N counts firm-years. Standard errors are in parentheses. Coefficients and standard errors are rounded to four decimal places.

* $p < 0.10$, based on unrounded reported p-values. *Source: Revelio Labs.*

Table 38: Match Innovations and Firm Outcomes II (Compustat Firms, Lag 2)

	Emp. growth (1)	Survival (2)	Log sales (3)	Emp. growth (4)	Survival (5)	Log sales (6)
inn_1 (Stepping-st)	0.0957*	0.0649*	0.1339	0.0944*	0.0681*	0.1565
	(0.0382)	(0.0189)	(0.0941)	(0.0389)	(0.0191)	(0.0964)
inn_3 (Churning)	0.0884*	0.0049	−0.3118*	0.0912*	0.0032	−0.3023*
	(0.0335)	(0.0192)	(0.0907)	(0.0336)	(0.0192)	(0.0908)
inn_5 (Early Gr)	0.1835*	0.0637*	0.5261*	0.1830*	0.0650*	0.5248*
	(0.0279)	(0.0148)	(0.0833)	(0.0285)	(0.0148)	(0.0844)
inn_6 (Maturing Gr)	0.2144*	0.0766*	0.2072*	0.2166*	0.0750*	0.2033*
	(0.0356)	(0.0168)	(0.0984)	(0.0356)	(0.0166)	(0.0990)
inn_7 (Stable Ongoing)	0.1692*	0.0671*	0.1641*	0.1694*	0.0676*	0.1697*
	(0.0202)	(0.0108)	(0.0579)	(0.0203)	(0.0108)	(0.0588)
pred_1 (Stepping-st)				−0.0136	−0.0053	−0.1039
				(0.0420)	(0.0242)	(0.1072)
pred_3 (Churning)				−0.0395	−0.0138	−0.0970
				(0.0623)	(0.0373)	(0.1534)
pred_5 (Early Gr)				−0.0583	0.0539*	−0.3123*
				(0.0545)	(0.0301)	(0.1654)
pred_6 (Maturing Gr)				0.0077	−0.0784	−0.3203
				(0.1141)	(0.0624)	(0.2942)
pred_7 (Stable Ongoing)				−0.0167	−0.0182	0.0112
				(0.0614)	(0.0325)	(0.1422)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	Yes	Yes	Yes
Within R^2	0.0078	0.0040	0.2199	0.0079	0.0042	0.2203
N	54,192	59,475	48,759	54,192	59,475	48,759

Notes: Estimates relate lagged new-hire match-state innovations to Compustat firm outcomes. Outcomes are employment growth following Davis et al. (2013), firm survival, and log sales. All models include firm and 6-digit NAICS industry-by-year fixed effects. Growth and survival models control for lagged employment growth; sales models control for lagged log sales. Innovations, predicted probabilities, and outcome controls are lagged 2 years. Columns (4)–(6) additionally include predicted state probabilities. State 4 (Transitional) is the reference for innovations and probabilities; state 2 (Stable/Stagnant) coefficients are estimated but not displayed. R^2 is within R^2; N counts firm-years. Standard errors are in parentheses. Coefficients and standard errors are rounded to four decimal places.
* $p < 0.10$, based on unrounded reported p-values. *Source: Revelio Labs and Compustat.*

Table 39: Match Innovations and Firm Outcomes II (Compustat Firms, Lag 3)

	Emp. growth (1)	Survival (2)	Log sales (3)	Emp. growth (4)	Survival (5)	Log sales (6)
inn_1 (Stepping-st)	0.0900*	0.0255	0.0763	0.0940*	0.0273	0.0962
	(0.0369)	(0.0187)	(0.1025)	(0.0372)	(0.0189)	(0.1046)
inn_3 (Churning)	0.1171*	−0.0233	−0.3126*	0.1238*	−0.0227	−0.3044*
	(0.0381)	(0.0188)	(0.0965)	(0.0382)	(0.0188)	(0.0969)
inn_5 (Early Gr)	0.0564*	0.0659*	0.6062*	0.0573*	0.0672*	0.6026*
	(0.0289)	(0.0154)	(0.0954)	(0.0291)	(0.0154)	(0.0969)
inn_6 (Maturing Gr)	0.1531*	0.0250	0.1980*	0.1566*	0.0247	0.1946*
	(0.0349)	(0.0166)	(0.1106)	(0.0348)	(0.0166)	(0.1102)
inn_7 (Stable Ongoing)	0.0953*	0.0529*	0.2003*	0.0976*	0.0536*	0.2052*
	(0.0221)	(0.0116)	(0.0654)	(0.0224)	(0.0116)	(0.0665)
pred_1 (Stepping-st)				−0.0342	−0.0259	−0.1900
				(0.0410)	(0.0254)	(0.1238)
pred_3 (Churning)				−0.0912	−0.0069	−0.1821
				(0.0631)	(0.0383)	(0.1703)
pred_5 (Early Gr)				−0.1385*	−0.0078	−0.2464
				(0.0522)	(0.0308)	(0.1905)
pred_6 (Maturing Gr)				−0.0329	−0.0500	−0.5940*
				(0.1062)	(0.0662)	(0.3435)
pred_7 (Stable Ongoing)				−0.0247	−0.0115	−0.0661
				(0.0561)	(0.0389)	(0.1670)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	Yes	Yes	Yes
Within R^2	0.0050	0.0023	0.1088	0.0055	0.0024	0.1092
N	48,530	53,657	43,728	48,530	53,657	43,728

Notes: Estimates relate lagged new-hire match-state innovations to Compustat firm outcomes. Outcomes are employment growth following Davis et al. (2013), firm survival, and log sales. All models include firm and 6-digit NAICS industry-by-year fixed effects. Growth and survival models control for lagged employment growth; sales models control for lagged log sales. Innovations, predicted probabilities, and outcome controls are lagged 3 years. Columns (4)–(6) additionally include predicted state probabilities. State 4 (Transitional) is the reference for innovations and probabilities; state 2 (Stable/Stagnant) coefficients are estimated but not displayed. R^2 is within R^2; N counts firm-years. Standard errors are in parentheses. Coefficients and standard errors are rounded to four decimal places.
* $p < 0.10$, based on unrounded reported p-values. *Source: Revelio Labs and Compustat.*

Table 40: New-Hire Match Innovations and Firm Outcomes I (Compustat Firms, Lag 2)

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
inn_1 (Stepping-st)	−0.1662*	0.0205	0.2082*	0.0017	−0.1727*	0.0212	0.2137*	0.0017
	(0.0157)	(0.0171)	(0.0236)	(0.0033)	(0.0160)	(0.0173)	(0.0239)	(0.0034)
inn_3 (Churning)	−0.1658*	−0.0438*	0.1364*	−0.0013	−0.1664*	−0.0421*	0.1378*	−0.0011
	(0.0142)	(0.0164)	(0.0221)	(0.0036)	(0.0142)	(0.0164)	(0.0220)	(0.0036)
inn_5 (Early Gr)	−0.1011*	0.0892*	0.2103*	0.0017	−0.1015*	0.0875*	0.2087*	0.0019
	(0.0110)	(0.0123)	(0.0171)	(0.0014)	(0.0110)	(0.0124)	(0.0172)	(0.0015)
inn_6 (Maturing Gr)	−0.3125*	0.0177	0.3514*	−0.0009	−0.3126*	0.0174	0.3515*	−0.0004
	(0.0143)	(0.0169)	(0.0228)	(0.0019)	(0.0142)	(0.0168)	(0.0227)	(0.0018)
inn_7 (Stable Ongoing)	−0.1994*	0.0404*	0.2540*	0.0009	−0.2013*	0.0404*	0.2555*	0.0010
	(0.0084)	(0.0094)	(0.0128)	(0.0012)	(0.0084)	(0.0095)	(0.0128)	(0.0012)
pred_1 (Stepping-st)					−0.0033	−0.0140	−0.0132	−0.0050
					(0.0174)	(0.0196)	(0.0274)	(0.0037)
pred_3 (Churning)					0.0040	0.0057	−0.0048	−0.0107*
					(0.0271)	(0.0281)	(0.0401)	(0.0049)
pred_5 (Early Gr)					0.0454*	−0.0748*	−0.1357*	−0.0003
					(0.0208)	(0.0234)	(0.0322)	(0.0048)
pred_6 (Maturing Gr)					−0.0062	−0.0186	−0.0055	−0.0094
					(0.0457)	(0.0519)	(0.0727)	(0.0072)
pred_7 (Stable Ongoing)					−0.0649*	0.0149	0.0732*	−0.0042
					(0.0222)	(0.0251)	(0.0342)	(0.0032)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0365	0.0101	0.0268	0.0010	0.0377	0.0110	0.0283	0.0015
N	69,014	69,014	69,026	69,056	69,014	69,014	69,026	69,056

Notes: Estimates relate lagged new-hire match-state innovations to firm outcomes in the Compustat-matched sample. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival. All models include firm and 6-digit NAICS industry-by-year fixed effects. Separation and hiring models control for their lagged outcome; growth and survival models control for lagged employment growth. All lags are 2 years. Columns (5)–(8) also include lagged predicted state probabilities. State 4 (Transitional) is the reference for innovations and probabilities; state 2 (Stable/Stagnant) coefficients are estimated but not displayed. R^2 is within R^2; N counts firm-years. Standard errors are in parentheses. Coefficients and standard errors are rounded to four decimal places. * $p < 0.10$, based on unrounded reported p-values. *Source: Revelio Labs; Compustat sample.*

Table 41: New-Hire Match Innovations and Firm Outcomes I (Compustat Firms, Lag 3)

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
inn_1 (Stepping-st)	−0.0700*	0.0043	0.0833*	0.0002	−0.0716*	0.0063	0.0888*	0.0003
	(0.0152)	(0.0165)	(0.0237)	(0.0023)	(0.0155)	(0.0167)	(0.0239)	(0.0023)
inn_3 (Churning)	−0.1290*	−0.0039	0.1402*	−0.0006	−0.1298*	−0.0015	0.1427*	−0.0006
	(0.0146)	(0.0134)	(0.0204)	(0.0027)	(0.0145)	(0.0135)	(0.0204)	(0.0027)
inn_5 (Early Gr)	−0.0033	0.0270*	0.0373*	0.0030	−0.0014	0.0245*	0.0331*	0.0028
	(0.0106)	(0.0118)	(0.0167)	(0.0020)	(0.0106)	(0.0119)	(0.0168)	(0.0019)
inn_6 (Maturing Gr)	−0.1661*	0.0291*	0.2047*	0.0019	−0.1663*	0.0291*	0.2045*	0.0017
	(0.0131)	(0.0151)	(0.0208)	(0.0017)	(0.0130)	(0.0150)	(0.0206)	(0.0016)
inn_7 (Stable Ongoing)	−0.1018*	0.0390*	0.1568*	0.0018	−0.1023*	0.0395*	0.1584*	0.0018
	(0.0082)	(0.0089)	(0.0126)	(0.0012)	(0.0082)	(0.0089)	(0.0126)	(0.0012)
pred_1 (Stepping-st)					−0.0193	0.0095	0.0098	−0.0054
					(0.0167)	(0.0195)	(0.0269)	(0.0034)
pred_3 (Churning)					−0.0197	−0.0053	−0.0067	−0.0025
					(0.0240)	(0.0273)	(0.0379)	(0.0041)
pred_5 (Early Gr)					0.0677*	−0.1010*	−0.1910*	−0.0065
					(0.0212)	(0.0232)	(0.0334)	(0.0045)
pred_6 (Maturing Gr)					−0.0536	0.0149	0.0105	−0.0155
					(0.0449)	(0.0516)	(0.0704)	(0.0112)
pred_7 (Stable Ongoing)					−0.0516*	0.0201	0.0631*	−0.0067
					(0.0227)	(0.0263)	(0.0353)	(0.0045)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0126	0.0020	0.0091	0.0014	0.0136	0.0035	0.0116	0.0020
N	64,971	64,971	64,986	65,014	64,971	64,971	64,986	65,014

Notes: Estimates relate lagged new-hire match-state innovations to firm outcomes in the Compustat-matched sample. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival. All models include firm and 6-digit NAICS industry-by-year fixed effects. Separation and hiring models control for their lagged outcome; growth and survival models control for lagged employment growth. All lags are 3 years. Columns (5)–(8) also include lagged predicted state probabilities. State 4 (Transitional) is the reference for innovations and probabilities; state 2 (Stable/Stagnant) coefficients are estimated but not displayed. R^2 is within R^2; N counts firm-years. Standard errors are in parentheses. Coefficients and standard errors are rounded to four decimal places. * $p < 0.10$, based on unrounded reported p-values. *Source: Revelio Labs; Compustat sample.*

9.11 Results from completed spells

Table 42 examines whether match innovations revealed through recently completed positions predict subsequent firm performance. This specification addresses a concern with assigning match innovations to new hires or incumbent workers: the duration and transition type used to infer match quality may not yet be realized when those exposures are measured. Restricting exposure to spells completed in year $t-1$ ensures that these spell characteristics are determined before firm outcomes are measured in year t. Combined with filtered decoding, which excludes information from later spells conditional on the estimated model parameters, this approach separates the realization of match outcomes from the subsequent firm outcomes they predict. The regressions further condition on lagged firm outcomes, firm fixed effects, and industry-by-year fixed effects.

Unlike the main analysis, which constructs exposures from new hires or the stock of incumbent workers, this specification captures the composition of positions that have recently ended. The economic interpretation depends on how those positions end. Completion of Stepping-stone and Churning spells typically reflects workers departing from the firm, so their innovations describe matches that the firm has recently lost. By contrast, completion of Early Growth and Maturing Growth spells is more likely to reflect promotion or another internal transition, with the worker remaining at the firm. These innovations may therefore capture successful matches whose workers continue to contribute to the firm through new positions. However, the exposure reflects match quality in the completed position, not in the worker's subsequent position.

For clarity, the table suppresses the Stable Ongoing coefficients because this state's interpretation is less direct among completed positions. Its innovation and predicted probability remain included in the regressions, since completed spells can still receive positive posterior probability for this latent state.

Each coefficient is interpreted relative to Transitional Mismatch. The comparisons below consider a 10-percentage-point increase in a state's average innovation, offset by an equal decrease in the Transitional Mismatch innovation, holding other innovations and regression controls fixed. All specifications include firm and industry-by-year fixed effects and lagged firm outcomes; columns (5)–(8) additionally control for predicted state probabilities.

Completed Stepping-stone and Churning matches are associated with higher subsequent separation rates. A 10-percentage-point shift toward Stepping-stone is associated with a 0.278–0.281 percentage-point increase in next year's separation rate, while the corresponding increase for Churning is 0.180–0.235 percentage points. They indicate that firms with more separation-prone completed matches also experience higher subsequent

turnover. The two growth-oriented states show different separation patterns. A shift toward Early Growth is associated with a 0.416–0.421 percentage-point increase in subsequent separations, whereas a shift toward Maturing Growth is associated with a 0.061–0.095 percentage-point reduction. By contrast, the Maturing Growth estimates are consistent with greater subsequent workforce retention, although the magnitude is small.

Hiring and net employment growth are positively associated with innovations toward each of the four displayed states relative to Transitional Mismatch. Maturing Growth has the largest point estimates: a 10-percentage-point shift is associated with a 0.343–0.344 percentage-point increase in the hiring rate and a 0.916–0.934 percentage-point increase in employment growth. The corresponding increases for Early Growth are 0.130–0.134 and 0.481–0.495 percentage points. Reduction in the firm's composition of Stepping-stone and Churning matches also predict higher hiring and growth, although their employment-growth associations are roughly half as large as those for Maturing Growth, and their hiring associations are substantially smaller. This pattern is consistent with Maturing Growth capturing high-quality matches whose completion, potentially through internal advancement, is followed by continued firm expansion in the subsequent year.

A 10-percentage-point shift toward Stepping-stone, Churning, or Early Growth in completed spells is associated with an increase in next year's survival probability of approximately 0.060, 0.059–0.081, and 0.094–0.099 percentage points, respectively. Maturing Growth has a much smaller association with survival: approximately 0.009 percentage points without predicted exposures, falling to essentially zero and becoming statistically insignificant when those exposures are included.

Tables 43 and 44 restrict the sample to Compustat firms. Tables 43 shows a particularly strong association between completed Maturing Growth matches and subsequent firm expansion. A 10-percentage-point shift toward Maturing Growth among completed spells is associated with a 0.626–0.640 percentage-point reduction in separations, a 0.594–0.652 percentage-point increase in hiring, and a 1.330–1.384 percentage-point increase in employment growth. These estimates retain the pattern observed in the full sample, with larger point estimates for Compustat firms. They are consistent with successful matches that culminate in internal advancement being followed by both greater retention and additional hiring.

Early Growth exhibits a different pattern: the same shift is associated with increases of 0.377–0.389 percentage points in separations and 0.200–0.209 percentage points in hiring, but its employment-growth association is not statistically significant. Thus, completed Early Growth matches predict greater subsequent workforce turnover without a clear increase in net employment. Stepping-stone innovations are associated with a smaller

Table 42: Completed-Spell Match Innovations and Firm Outcomes (Full Sample, Lag 1)

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
Stepping-st	0.0278*	0.0073*	0.0567*	0.0060*	0.0281*	0.0070*	0.0564*	0.0060*
	(0.0011)	(0.0011)	(0.0013)	(0.0004)	(0.0011)	(0.0011)	(0.0013)	(0.0004)
Churning	0.0235*	0.0050*	0.0466*	0.0059*	0.0180*	0.0056*	0.0549*	0.0081*
	(0.0009)	(0.0010)	(0.0012)	(0.0004)	(0.0009)	(0.0010)	(0.0012)	(0.0004)
Early Gr	0.0416*	0.0130*	0.0495*	0.0099*	0.0421*	0.0134*	0.0481*	0.0094*
	(0.0011)	(0.0009)	(0.0011)	(0.0003)	(0.0011)	(0.0009)	(0.0011)	(0.0003)
Maturing Gr	−0.0061*	0.0343*	0.0916*	0.0009*	−0.0095*	0.0344*	0.0934*	0.0000
	(0.0013)	(0.0013)	(0.0016)	(0.0005)	(0.0013)	(0.0013)	(0.0017)	(0.0005)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0071	0.0066	0.0141	0.0092	0.0073	0.0067	0.0146	0.0100
N	6,366,605	6,366,605	6,400,141	6,495,741	6,366,605	6,366,605	6,400,141	6,495,741

Notes: This table reports associations between firm outcomes in year t and average filtered match-state innovations among spells completed in year $t-1$, including internal promotions and firm departures. Both exposure vectors use equal weights over the same completed spells. The sample is the full eligible firm sample, without a Compustat restriction. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival, measured using Revelio Labs data. Separation and hiring regressions control for their respective lagged outcomes; growth and survival regressions control for lagged employment growth. Columns (5)–(8) also include lagged predicted state probabilities. All models include firm and 6-digit NAICS industry-by-year fixed effects, with industry measured in $t-1$. All lags are one year. Transitional Mismatch is the reference for innovations and probabilities. Within R^2 is reported; firm-clustered standard errors are in parentheses. * $p < 0.10$. *Source: Revelio Labs.*

increase in employment growth of 0.352–0.370 percentage points, significant at the 10 percent level, but neither hiring nor separations is significantly associated with these innovations. The Churning estimates are statistically insignificant across all four outcomes.

Table II provides further evidence of the positive relationship between Maturing Growth and subsequent expansion using Compustat employment. A 10-percentage-point shift toward this state is associated with a 1.549–1.626 percentage-point increase in employment growth, the largest point estimate among the displayed states. Higher exposure in completed spells in Early Growth and Churning also predict higher Compustat employment growth, with corresponding increases of 0.447–0.476 and 0.641–0.725 percentage points. The Stepping-stone estimates are positive but statistically insignificant.

The sales results distinguish growth-oriented matches from Stepping-stone and Churning matches. A 10-percentage-point shift from Transitional Mismatch toward Early Growth among completed spells is associated with approximately 3.0–3.1% higher subsequent sales, while the corresponding association for Maturing Growth is 1.8–2.2%. Both associations are statistically significant across specifications. By contrast, the estimates for Stepping-stone and Churning are smaller and statistically insignificant. Thus, even when exposure is measured using completed positions, innovations toward growth-oriented matches predict higher subsequent sales, consistent with the main analysis based on new hires and incumbent workers.

Survival results are less consistent across the two data sources. In Table I, Stepping-stone innovations are associated with a 0.088–0.091 percentage-point reduction in Revelio-based survival, while Maturing Growth has a small negative association that is significant only when predicted exposures are included. In Table II, all four displayed states are positively associated with Compustat-based survival: a 10-percentage-point shift corresponds to increases of 0.425–0.455 percentage points for Stepping-stone, 0.327–0.391 for Churning, 0.589–0.591 for Early Growth, and 0.660–0.726 for Maturing Growth.

Taken together, the most consistent finding is that innovations toward Maturing Growth among completed spells predict subsequent employment expansion across both data sources. Innovations toward Early Growth and Maturing Growth also predict higher Compustat sales, suggesting that completed growth-oriented positions capture productive employment relationships whose relevance to firm performance extends beyond the end of the position.

Table 43: Completed-Spell Match Innovations and Firm Outcomes I (Compustat Firms)

	Sep. (1)	Hires (2)	Growth (3)	Surv. (4)	Sep. (5)	Hires (6)	Growth (7)	Surv. (8)
Stepping-st	0.0026	0.0152	0.0370*	−0.0088*	0.0028	0.0129	0.0352*	−0.0091*
	(0.0141)	(0.0142)	(0.0202)	(0.0038)	(0.0141)	(0.0142)	(0.0203)	(0.0039)
Churning	−0.0061	−0.0162	0.0150	−0.0047	−0.0071	−0.0182	0.0154	−0.0059
	(0.0128)	(0.0137)	(0.0192)	(0.0043)	(0.0130)	(0.0140)	(0.0196)	(0.0044)
Early Gr	0.0389*	0.0209*	0.0138	−0.0031	0.0377*	0.0200*	0.0152	−0.0035
	(0.0094)	(0.0104)	(0.0145)	(0.0021)	(0.0096)	(0.0106)	(0.0148)	(0.0022)
Maturing Gr	−0.0626*	0.0652*	0.1384*	−0.0046	−0.0640*	0.0594*	0.1330*	−0.0052*
	(0.0132)	(0.0138)	(0.0194)	(0.0029)	(0.0139)	(0.0145)	(0.0204)	(0.0029)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	No	Yes	Yes	Yes	Yes
Within R^2	0.0082	0.0662	0.0334	0.0048	0.0085	0.0666	0.0337	0.0059
N	70,367	70,367	70,400	70,461	70,367	70,367	70,400	70,461

Notes: This table reports associations between firm outcomes in year t and average exposure to match innovations among spells completed in year $t-1$, including internal transitions and job departures. Both exposure vectors use equal weights over the same completed spells. The sample comprises firms matched to Compustat in $t-1$. Outcomes are annual separation and hiring rates, employment growth following Davis et al. (2013), and firm survival, measured using Revelio Labs data. Separation and hiring regressions control for their respective lagged outcomes; growth and survival regressions control for lagged employment growth. Columns (5)–(8) also include lagged predicted state probabilities. All models include firm and 6-digit NAICS industry-by-year fixed effects, with industry measured in $t-1$. All lags are one year. Transitional Mismatch is the reference for innovations and probabilities. Firm-clustered standard errors are in parentheses. * $p < 0.10$. *Source: Revelio Labs, Compustat.*

Table 44: Completed-Spell Match Innovations and Firm Outcomes II (Compustat Firms)

	Emp. growth (1)	Survival (2)	Log sales (3)	Emp. growth (4)	Survival (5)	Log sales (6)
Stepping-st	0.0477	0.0425*	0.0904	0.0475	0.0455*	0.0925
	(0.0299)	(0.0158)	(0.0633)	(0.0299)	(0.0158)	(0.0637)
Churning	0.0725*	0.0391*	0.0568	0.0641*	0.0327*	0.0200
	(0.0304)	(0.0155)	(0.0664)	(0.0314)	(0.0161)	(0.0678)
Early Gr	0.0476*	0.0591*	0.3129*	0.0447*	0.0589*	0.3042*
	(0.0235)	(0.0120)	(0.0550)	(0.0238)	(0.0122)	(0.0554)
Maturing Gr	0.1549*	0.0726*	0.2187*	0.1626*	0.0660*	0.1830*
	(0.0324)	(0.0181)	(0.0738)	(0.0345)	(0.0185)	(0.0810)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Ind.-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Predicted	No	No	No	Yes	Yes	Yes
Within R^2	0.0015	0.0058	0.4703	0.0018	0.0062	0.4706
N	57,960	63,345	51,999	57,960	63,345	51,999

Notes: This table reports associations between firm outcomes in year t and average exposure to match innovations among spells completed in year $t-1$, including internal transitions and job departures. Both exposure vectors use equal weights over the same completed spells. The sample comprises firms matched to Compustat in $t-1$. Outcomes are employment growth following Davis et al. (2013), survival, and log sales, measured using Compustat data. Growth and survival regressions control for lagged Compustat employment growth; the sales regression controls for lagged log sales. Columns (4)–(6) also include lagged predicted state probabilities. All models include firm and 6-digit NAICS industry-by-year fixed effects, with industry measured in $t-1$. All lags are one year. Transitional Mismatch is the reference for innovations and probabilities. Within R^2 is reported; firm-clustered standard errors are in parentheses. * $p < 0.10$. *Source: Revelio Labs, Compustat.*

9.11.1 Estimating firm-level TFP

I estimate firm-level total factor productivity (TFP) from Compustat Fundamentals Annual following İmrohoroğlu and Tüzel (2014), who adapt the Olley and Pakes (1992) estimator to Compustat data. I assume a Cobb–Douglas value-added production function,

$$y_{jt} = \beta_0 + \beta_l \, l_{jt} + \beta_k \, k_{jt} + \omega_{jt} + \epsilon_{jt}, \tag{27}$$

where y_{jt} is log real value added, l_{jt} is log employment, k_{jt} is log beginning-of-period capital, ω_{jt} is firm productivity, and ϵ_{jt} captures transitory shocks. Because Compustat does not consistently report labor costs, I impute each firm's wage bill as its employment (EMP) times the national private-sector annual wage per employee in its NAICS industry from the BLS Quarterly Census of Employment and Wages, matched at the finest available level from six to two digits. Materials equal total expenses (SALE $-$ OIBDP) minus this wage bill, and value added equals sales minus materials, deflated by the GDP deflator. Capital is gross property, plant, and equipment (PPEGT), deflated by the nonresidential fixed investment deflator at the average vintage of the capital stock, computed as accumulated over current depreciation (DPACT/DP) and smoothed over three years, and lagged one year. Investment is capital expenditures (CAPX) deflated by the same investment deflator. Where Compustat reports staff expenses (XLR), the imputed wage bill closely tracks it: the correlation in logs is 0.95 and the median ratio of imputed to reported labor costs is 0.99.

I estimate the production function by Olley and Pakes (1992), pooling firms in all non-financial sectors and including 2-digit NAICS industry-by-year dummies. The first stage regresses log value added on log employment and a second-order polynomial in log investment and log capital, which proxies for productivity. The second stage recovers the capital coefficient by nonlinear least squares from the law of motion for productivity, controlling for the probability of remaining in the sample. The estimation sample contains 53,412 firm-years and 6,901 firms with positive sales, employment, capital, depreciation, investment, SG&A, materials, and value added. The estimated output elasticities are 0.652 (s.e. 0.013) for labor and 0.199 (s.e. 0.072) for capital, with standard errors from a firm-level bootstrap, implying returns to scale of 0.85. Firm TFP is $\hat{\omega}_{jt} + \epsilon_{jt} = y_{jt} - \hat{\beta}_l \, l_{jt} - \hat{\beta}_k \, k_{jt}$, which has an annual autoregressive coefficient of 0.84. The approach departs from İmrohoroğlu and Tüzel (2014) in three respects, all driven by data coverage: wages come from the QCEW rather than the NBER-CES Manufacturing Database, deflators are aggregate rather than industry-specific, and the sample covers all non-financial sectors rather than manufacturing only. Restricting the sample to manu-

facturing yields elasticities of 0.78 for labor and 0.10 for capital, compared with 0.83 and 0.17 in İmrohoroğlu and Tüzel (2014). Because output is measured by deflated revenue, the resulting TFP is a revenue-based measure and also reflects differences in prices and markups across firms.

Results with 6-digit industry-by-year fixed effects. Table 45 repeats the regressions in Table 7, replacing 2-digit with 6-digit NAICS industry-by-year fixed effects. The results are very similar. In the level regressions, the coefficients on Early Growth, Mature Growth, and Stable Ongoing remain large and significant at the 1 percent level; with size and age controls, a 10 percentage point shift in the stock's innovation exposure from Transitional Mismatch is associated with 13.0, 15.3, and 10.5 percent higher TFP, respectively, compared with 14.7, 15.2, and 9.2 percent with 2-digit fixed effects.

In the growth regressions, the estimates are nearly unchanged: with size and age controls, the same shift in new hires' innovation exposure is associated with 1.6, 3.2, and 2.2 percentage points faster TFP growth toward Early Growth, Mature Growth, and Stable Ongoing, respectively, compared with 1.6, 3.2, and 2.0 percentage points with 2-digit fixed effects. Overall, the finer industry controls leave both the ranking and the magnitude of the estimates essentially unchanged.

Table 45: Match Innovations and Firm Productivity: 6-Digit Industry-Year Fixed Effects (Compustat, Olley–Pakes TFP)

	Log TFP$_t$ Stock innovations, $t-1$		Δ log TFP$_{t,t+1}$ New-hire innovations, t	
	(1)	(2)	(3)	(4)
Stepping Stone	1.2057***	0.6124*	0.1915*	0.1991*
	(0.3701)	(0.3389)	(0.1110)	(0.1111)
Churning	0.7775*	0.6452*	0.2132*	0.2035*
	(0.4036)	(0.3788)	(0.1139)	(0.1136)
Early Growth	2.4323***	1.2178***	0.1801**	0.1647*
	(0.2804)	(0.2444)	(0.0870)	(0.0893)
Mature Growth	2.3039***	1.4221***	0.3324***	0.3166***
	(0.3099)	(0.2453)	(0.1118)	(0.1131)
Stable Ongoing	1.3915***	0.9960***	0.2201***	0.2170***
	(0.1628)	(0.1500)	(0.0680)	(0.0681)
NAICS6-year FE	Yes	Yes	Yes	Yes
Size & age bins	No	Yes	No	Yes
Predicted	Yes	Yes	Yes	Yes
Within R^2	0.0900	0.1988	0.0028	0.0048
Firms	3,376	3,376	2,993	2,993
N	29,469	29,469	25,243	25,243

Notes: This table reports the conditional correlation between firm-level total factor productivity (TFP) and average filtered match-state innovations. TFP is estimated from Compustat following **?**, using the **?** estimator with investment as the proxy: value added is sales minus materials, where materials equal total expenses net of an imputed wage bill (Compustat employment times the BLS QCEW national private-sector wage per employee in the firm's NAICS industry); capital is gross property, plant, and equipment deflated at its average vintage and measured at the beginning of the year. The outcome is log TFP; all models include 6-digit NAICS industry-by-year fixed effects. In columns (1)–(2), the outcome is log TFP in year t and the exposures are innovations of the stock of all workers in year $t-1$. In columns (3)–(4), the outcome is the change in log TFP of the same firm from t to $t+1$ and the exposures are innovations of workers hired in year t. All models control for predicted state probabilities of the same population measured in the exposure year. Columns (2) and (4) add employment-size and firm-age bins. Industry (NAICS), employment-size bins, and firm-age bins are from Revelio Labs and measured in the exposure year. Firms are linked to Compustat through GVKEY, ticker, or CUSIP using Compustat membership in the exposure year; firm-years matched to multiple parents are excluded. Transitional Mismatch is the reference state for innovations and probabilities. Within R^2 is reported; firm-clustered standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Source: Compustat, Revelio Labs, BLS QCEW.*